

Exercicis de classe

MÈTODES NUMÈRICS II

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MÈTODES ITERATIUS PER A SISTEMES LINEALS

Exercici 1.1. Let $\|\cdot\|$ be a norm on $\mathbb{R}^n$.

1. If A is a real and non singular n -by- n matrix, show that $\|x\|_A = \|Ax\|$ is a norm on $\mathbb{R}^n$.
2. Find the matrix norm induced by the previous norm.
3. Let A a positive definite matrix. Prove that $\|x\| = \sqrt{x^T A x}$ is a norm (hint: use the Cholesky decomposition).

Resolució. Hem de provar que $\|x\|_A = \|Ax\|$ és norma a $\mathbb{R}^n$. Efectivament:

1. Si $\|x\|_A = 0 = \|Ax\|$ tenim que $Ax = 0$. Com A és no singular, $x = 0$. Si $x = 0$, aleshores $Ax = 0$ i $\|Ax\| = 0 = \|x\|_A$.
2. Sigui $c \in \mathbb{R}$ i $x \in \mathbb{R}^n$. Aleshores:

$$\|cx\|_A = \|A(cx)\| = \|c \cdot Ax\| = |c| \cdot \|Ax\| = |c| \cdot \|x\|_A.$$

3. Siguin $x, y \in \mathbb{R}^n$ i $\|x + y\|_A$. Volem veure la desigualtat triangular:

$$\|x + y\|_A = \|A(x + y)\| = \|Ax + Ay\| \leq \|Ax\| + \|Ay\| = \|x\|_A + \|y\|_A.$$

Així doncs, $\|\cdot\|_A$ és una norma. Per a trobar la norma matricial induïda per la norma anterior:

$$\|A\| = \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \sup_{\|y\|=1} \|Ay\|.$$

Aplicant-ho a l'exercici:

$$\|B\|_A = \sup_{\|y\|=1} \|By\|_A = \sup_{\|Ay\|=1} \|A(By)\| = \sup_{\|x\|=1} \|ABA^{-1}x\| = \|ABA^{-1}\|.$$

Sigui A una matriu definida positiva. Hem de provar que $\|x\| = \sqrt{x^T A x}$ és una norma. Recordem que A és definida positiva si, i només si, $x^T A x > 0$ per a tot $x \in \mathbb{R}^n$. Per la descomposició de Cholesky, existeix $G = \{g_{ij}\}_{ij}$ tal que $g_{ij} > 0$ i G és una matriu triangular superior amb $A = G^T G$.

$$\|x\| = \sqrt{x^T A x} = \sqrt{x^T G^T G x} = \sqrt{(Gx)^T \cdot (Gx)} = \sqrt{y^T \cdot y} = \|y\|_2,$$

que és la norma euclidiana (i ja sabem que és norma). ■

Exercici 1.2. Si $x \in \mathbb{R}^n$, prova que $\lim_p \|x\|_p = \|x\|_\infty$.

Resolució. És evident que el cas $x = 0$ és cert. Prenem $x \neq 0$ i treballem amb la norma $\mathcal{L}_p$:

$$\|x\|_p = (|x_1|^p + \dots + |x_n|^p)^{\frac{1}{p}} = \frac{\|x\|_\infty}{\|x\|_\infty} (|x_1|^p + \dots + |x_n|^p)^{\frac{1}{p}} = \|x\|_\infty \cdot \left(\frac{|x_1|^p}{\|x\|_\infty^p} + \dots + \frac{|x_n|^p}{\|x\|_\infty^p} \right)^{\frac{1}{p}}.$$

Per definició, $\|x\|_\infty = \sup_i |x_i|$. Evidentment, $|x_i| \leq \sup_i |x_i| = \|x\|_\infty$; per tant, $\frac{|x_i|}{\|x\|_\infty} \leq 1$ i $\left(\frac{|x_i|}{\|x\|_\infty} \right)^p \leq 1$. Ens queda:

$$1^{\frac{1}{p}} \leq \left(\sum_i \left(\frac{|x_i|}{\|x\|_\infty} \right)^p \right)^{\frac{1}{p}} \leq n^{\frac{1}{p}}.$$

Multiplicant per $\|x\|_\infty > 0$ a tot arreu:

$$\|x\|_\infty \leq \|x\|_p \leq \|x\|_\infty \cdot n^{\frac{1}{p}} \xrightarrow{p \rightarrow \infty} \|x\|_\infty.$$

Per tant, $\|x\|_p \rightarrow \|x\|_\infty$ quan $p \rightarrow \infty$, com volíem veure. ■

Exercici 1.3. Si $A \in \mathcal{M}_{n \times n}$, proveu que:

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}|, \quad \|A\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}|.$$

Demostració. Sigui $(Ax)_i$ la i -èsima fila d' Ax . Aleshores, definint $\sum_{i=1}^n |a_{ij}| \leq \alpha := \max_i \sum_{i=1}^n |a_{ij}|$:

$$\begin{aligned} \|Ax\|_1 &= |(Ax)_1| + \dots + |(Ax)_n| = |a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n| + \dots + |a_{n1}x_1 + \dots + a_{nn}x_n| \\ &= \left| \sum_{j=1}^n a_{1j}x_j \right| + \dots + \left| \sum_{j=1}^n a_{nj}x_j \right| = \sum_{i=1}^n \left| \sum_{j=1}^n a_{ij}x_j \right| \leq \sum_{i=1}^n \sum_{j=1}^n |a_{ij}| \cdot |x_j| = \sum_{j=1}^n \sum_{i=1}^n |a_{ij}| \cdot |x_j| \\ &\leq \sum_{j=1}^n \alpha \cdot |x_j| = \alpha \cdot \sum_{j=1}^n |x_j| = \alpha \cdot \|x\|_1 \implies \begin{cases} \|Ax\|_1 \leq \alpha \cdot \|x\|_1 \\ \|Ax\|_1 \leq \|A\|_1 \cdot \|x\|_1 \end{cases} \end{aligned}$$

Per tant:

$$\|A\|_1 = \sup_{\|x\|_1 \neq 0} \frac{\|Ax\|_1}{\|x\|_1} \leq \alpha.$$

Sigui i tal que $\sum_j |a_{ij}| = \alpha$. Si prenem $x = e_i = (0, \dots, 0, 1, 0, \dots, 0)$, $\|x\|_1 = 1$:

$$\|Ax\|_1 = \|Ae_i\|_1 = \sum_{j=1}^n |a_{ij}| = \alpha, \quad \|Ae_i\|_1 \leq \|A\|_1 \cdot \|e_i\|_1 = \|A\|_1.$$

Per tant, $\|A\|_1 \geq \alpha$. Com ja tenim $\|A\|_1 \leq \alpha$, ens queda $\|A\|_1 = \alpha$. ■

Exercici 1.4. Sigui $A \in \mathcal{M}_{n \times n}(\mathbb{R})$. Si P i Q són matrius reals $n \times n$ i ortogonals, demostreu que $\|A\|_2 = \|PAQ\|_2$.

Demostració. P és ortogonal si, i només si, $P^T P = P P^T = Id$; és a dir, si $P^T = P^{-1}$. La norma $\|A\|_2 = \sqrt{\rho(A^T A)}$, on ρ és el radi espectral de la matriu; és a dir, $\rho(A^T A) = \max_i |\lambda_i|$, on $\lambda_1, \dots, \lambda_n$ són els VAPs d' $A^T A$. Ens preguntem què val $\rho((PAQ)^T (PAQ))$. Fent aquest producte de matrius:

$$(PAQ)^T \cdot (PAQ) = Q^T A^T P^T P A Q = Q^T A^T A Q = Q^{-1} A^T A Q,$$

i com $Q^{-1} A^T A Q$ té els mateixos VAPs que $A^T A$,

$$\rho((PAQ)^T (PAQ)) = \rho(Q^{-1} A^T A Q) = \rho(A^T A) \implies \|PAQ\|_2 = \|A\|_2. \quad \blacksquare$$

Exercici 1.5. Sigui $A \in M_{n \times n}(\mathbb{R})$. Si A és regular, proveu que $A^T A$ és simètrica i definida positiva. És cert si A és singular?

Demostració. $M = A^T A$ és simètrica si, i només si, $M = M^T$. Per tant,

$$M^T = (A^T A)^T = A^T (A^T)^T = A^T A = M.$$

Pel que fa a la segona part, M és definida positiva si, i només si, $x^T M x > 0$ per a tot $x \neq 0$. Per tant:

$$x^T M x = x^T A^T A x = (Ax)^T A x = \|Ax\|_2^2 \geq 0.$$

Finalment, $\|Ax\|_2 = 0 \iff Ax = 0$ i, suposant A regular, $x = 0$. Per tant, per a tot $x \neq 0$ tenim que $x^T M x = \|Ax\|_2^2 > 0$ i M és definida positiva. Ens queda distingir el cas en què A és singular. Aleshores, existeix $v \neq 0$ tal que $Av = 0$, pel que no es compleix que $\|Ax\|_2 = 0 \iff x = 0$. ■

Exercici 1.6. Si A és una matriu $n \times n$, proveu que $\|A\|_2^2 \leq \|A\|_1 \|A\|_\infty$.

Demostració. Tenim que $\|A\|_2^2 = \rho(A^T A) = |\lambda|$, on $\lambda = \max(\text{Spec}(A))$. Per tant, existeix $v \neq 0$ tal que $A^T A v = \lambda v$.

$$\begin{cases} \|A^T A v\|_1 = \|\lambda v\|_1 = |\lambda| \cdot \|v\|_1 \\ \|A^T A v\|_1 \leq \|A^T\|_1 \cdot \|A\|_1 \cdot \|v\|_1 \end{cases} \implies \|A\|_2^2 = |\lambda| \leq \|A\|_\infty \cdot \|A\|_1, \quad \|v\|_1 \neq 0.$$

Per tant, es compleix $|\lambda| \leq \|A\|_\infty \cdot \|A\|_1$, com volíem. ■

Exercici 1.7. Sigui A una matriu simètrica:

1. Proveu que A és definida positiva si, i només si, els seus valors propis són estrictament positius.
2. Supposem que A és estrictament diagonal dominant (per files o columnes), amb elements positius. Proveu que A és definida positiva.

Demostració.

1. Sigui $x \neq 0$ a $\mathbb{R}^n$. A és simètrica si, i només si, existeix Q ortogonal tal que $Q A Q^{-1} = A$. Recordem que si Q és ortogonal, aleshores $Q^T = Q^{-1}$. Sigui $y = Q^T x \neq 0$. Aleshores, $y = Q^{-1} x$ i $x = Q y$. Així doncs:

$$x^T A x = (Q y)^T A (Q y) = y^T Q^T A Q y = y^T D y = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2, \quad \forall \lambda_i > 0.$$

Per tant, $x^T A x > 0$ per a tot $x \neq 0$ A és definida positiva. Agafem $v_i \neq 0$ el VEP de VAP λ_i , pel que $A v_i = \lambda_i v_i$.

$$v_i^T (A v_i) = v_i^T (\lambda_i v_i) = \lambda_i v_i^T v_i = \lambda_i \underbrace{\|v_i\|_2^2}_{>0} \implies \lambda_i = \frac{v_i^T A v_i}{\|v_i\|_2^2} > 0.$$

2. A és diagonal dominant per columnes si, i només si, $a_{ii} > \sum_{j=1, j \neq i}^n |a_{ij}|$. Sigui λ un VAP amb VAP $v \neq 0$, tal que $v = (v_1, \dots, v_i, \dots, v_n)^T$, amb $v_i = \max_j |v_j|$. Definim:

$$v' = \frac{v}{v_i} = \left(\frac{v_1}{v_i}, \frac{v_2}{v_i}, \dots, 1, \dots, \frac{v_n}{v_i} \right)^T, \quad v'_i = 1 \text{ i } |v'_j| \leq 1, \quad i \neq j.$$

v' és, també, VEP d' A : $A v' = \lambda v'$. En particular, la component i -èsima és:

$$\lambda v'_i = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} v'_1 \\ \vdots \\ v'_n \end{pmatrix} = a_{i1} v'_1 + \cdots + a_{ii} \cdot 1 + \cdots + a_{in} v'_n = a_{ii} + \sum_{\substack{j=1 \\ j \neq i}}^n a_{ij} v'_j.$$

Sabem que per a tot $\alpha \in \mathbb{R}$, $-|\alpha| \leq \alpha \leq |\alpha|$. Prenem, en aquest cas, $\alpha = \lambda - a_{ii}$. Per tant:

$$\lambda - a_{ii} \geq - \left| \sum_{\substack{j=1 \\ j \neq i}}^n a_{ij} v'_j \right| \geq - \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}| \cdot |v'_j| \geq - \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}|.$$

Per tant, $\lambda > a_{ii} - \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}| > 0$ i pel primer apartat, A és definida positiva. ■

Exercici 1.8. Per a qualsevol $A \in \mathcal{M}_{n \times n}(\mathbb{R})$ definim:

$$F(A) = \left(\sum_{i=1}^n \sum_{j=1}^n a_{ij}^2 \right)^{\frac{1}{2}}.$$

1. Proveu que $F(A)$ és una norma a l'espai de matrius reals $\mathcal{M}_{n \times n}(\mathbb{R})$. $F(A)$ és coneguda com la norma de Fröbenius d' A i es denota usualment per $\|A\|_F$.
2. Existeix una norma $\|\cdot\|_f$ on $\mathbb{R}^n$ tal que $\|A\|_F = \max_{\|x\|_f=1} \|Ax\|_f$? (Pista: considereu $A = I$).
3. Proveu que $\|A\|_2 \leq \|A\|_F \leq \sqrt{n} \|A\|_2$. (Pista: $\|A\|_F^2 = \text{tr}(A^T A)$).

4. *Discutiu si la següent desigualtat és sempre certa:* $\|Ax\|_2 \leq \|A\|_F \cdot \|x\|_2$.

Demostració.

1. Hem de veure que $F(A)$ és norma. Notem que:

$$F(A)^2 = \sum_i \sum_j a_{ij}^2 = a_{11}^2 + a_{12}^2 + \cdots + a_{1n}^2 + a_{21}^2 + \cdots + a_{nn}^2 = \|x\|_2^2,$$

$$x = (a_{11}, a_{12}, \dots, a_{1n}, a_{21}, \dots, a_{2n}, \dots, a_{n1}, \dots, a_{nn}) \in \mathbb{R}^{n \times n}.$$

Ara, $\|x\|_2^2 = x^T x$ i $\|Ax\|_2^2 = x^T A^T A x$, pel que:

$$F(A) \equiv \|A\|_F = \sup_{\|x\|_2=1} \|Ax\|_2 \implies F(A) \text{ és una norma.}$$

Si $A = (a_1, \dots, a_n)$ i $B = (b_1, \dots, b_n)$, tenim:

$$\begin{aligned} A^T B &= \begin{pmatrix} a_1 b_1 & \cdots & a_1 b_n \\ \vdots & \ddots & \vdots \\ a_n b_1 & \cdots & a_n b_n \end{pmatrix} \implies \|AB\|_F^2 = \sum_i \sum_j (a_i b_j)^2 \leq \sum_i \sum_j \|a_i\|^2 \|b_j\|^2 \\ &\leq \sum_i \|a_i\|_2^2 \cdot \sum_j \|b_j\|_2^2 = \|A\|_F^2 \cdot \|B\|_F^2. \end{aligned}$$

2. Volem provar que $\|A\|_F = \max_{\|x\|_f=1} \|Ax\|_f$. Considerem $A = I$. Per tant, procedim a calcular $\|I\|_F$:

$$\|I\|_F = \max_{\|x\|_f=1} \|Ix\|_f = \max_{\|x\|_f=1} \|x\|_f = 1 \implies \|I\|_F^2 = \sum_i \sum_j a_{ij}^2 = n \iff \|I\|_F = n^{\frac{1}{2}}.$$

Per tant, si $n = 1$ sí que $\|I\|_F = 1$, però per a $n > 1$ aquesta igualtat no és certa; és a dir, acabem de trobar un contraexemple.

3. Se'ns aconsella usar que $\|A\|_F^2 = \text{tr}(A^T A)$. Calculem $A^T A$ manualment:

$$\begin{pmatrix} a_{11} & \cdots & a_{n1} \\ \vdots & \ddots & \vdots \\ a_{1n} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \implies \text{tr}(A^T A) = \begin{pmatrix} a_{11}^2 + a_{21}^2 + \cdots + a_{n1}^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & a_{1n}^2 + a_{2n}^2 + \cdots + a_{nn}^2 \end{pmatrix}$$

Sigui $\|A\|_2^2 = \rho(A^T A) = \max_i |\lambda_i|$, λ_i VAPs d' $A^T A$.

- Pel que fa a la primera desigualtat, i tenint en compte que $\sum_j a_{ij} \cdot x_j = a_i \cdot x$:

$$\begin{aligned}\|A\|_2^2 &= \sup_{\|x\|_2=1} \|Ax\|_2^2 = \sup_{\|x\|_2=1} \sum_i \left(\sum_j a_{ij} \cdot x_j \right)^2 \stackrel{\text{C-Sch}}{\leq} \sup_{\|x\|_2=1} \sum_i \|a_i\|_2^2 \cdot \underbrace{\|x\|_2^2}_1 \\ &= \sup_{\|x\|_2=1} \sum_i \sum_j a_{ij}^2 = \|A\|_F^2.\end{aligned}$$

- Pel que fa a la segona desigualtat, és molt més senzill:

$$\|A\|_F^2 = \lambda_1 + \dots + \lambda_n \leq n \cdot \rho(A^T A) = n \|A\|_2^2 \iff \|A\|_F^2 \leq \sqrt{n} \|A\|_2.$$

4. $\|Ax\|_2 \leq \|A\|_F \cdot \|x\|_2$ si, i només si, $\|Ax\|_2 \leq \|A\|_2 \cdot \|x\|_2 \leq \|A\|_F \cdot \|x\|_2$. La primera desigualtat l'hem vist abans, i la segona és perquè $\|A\|_2 \leq \|A\|_F$, que també ho hem vist abans. ■

Observació 1.9. Sigui $Ax = b$ un sistema lineal i $A = D + L + U$, per a matrius D (diagonal), L (triangular inferior) i U (triangular superior). Hem vist dos mètodes iteratius lineals fins ara:

1. Jacobi: $x^{(k+1)} = -D^{-1}(L + U)x^{(k)} + D^{-1}b$.
2. Gauss-Seidel: $x^{(k+1)} = -(L + D)^{-1}Ux^{(k)} + (L + D)^{-1}b$, on $B_{GS} = -(L + D)^{-1}U$.

$$\|e^{(k+1)}\|_2 = \frac{\|B\|_2}{1 - \|B\|_2} \|x^{(k+1)} - x^{(k)}\|_2 \leq \frac{\|B\|_2^{k+1}}{1 - \|B\|_2} \|x^{(1)} - x^{(0)}\|_2.$$

Exercici 1.10. Consider a linear system $Ax = b$, where:

$$A = \begin{pmatrix} 4 & -1 & -1 & 0 \\ -1 & 4 & 0 & -1 \\ -1 & 0 & 4 & -1 \\ 0 & -1 & -1 & 4 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$$

Solve it using the Jacobi and Gauss-Seidel methods, with an absolute error (in Euclidean norm) below 10^{-9} . Which method requires fewer iterations?

Demostració. Tenim que:

$$D^{-1} = \begin{pmatrix} \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{1}{4} \end{pmatrix}, \quad D^{-1}(L + U) = \begin{pmatrix} 0 & -\frac{1}{4} & -\frac{1}{4} & 0 \\ -\frac{1}{4} & 0 & 0 & -\frac{1}{4} \\ -\frac{1}{4} & 0 & 0 & -\frac{1}{4} \\ 0 & -\frac{1}{4} & -\frac{1}{4} & 0 \end{pmatrix} \implies B_J = \begin{pmatrix} 0 & \frac{1}{4} & \frac{1}{4} & 0 \\ \frac{1}{4} & 0 & 0 & -\frac{1}{4} \\ \frac{1}{4} & 0 & 0 & -\frac{1}{4} \\ 0 & \frac{1}{4} & \frac{1}{4} & 0 \end{pmatrix}$$

Obtenim que $\|B_J\|_1 = \|B_J\|_\infty = \frac{1}{2} < 1$. Per l'exercici 6:

$$\|B_J\|_2 \stackrel{\text{ex. 6}}{\leq} \sqrt{\|B_J\|_\infty + \|B_J\|_1} \leq \frac{1}{2}.$$

Prenent D i usant $\frac{1}{1-\frac{1}{2^k}}$, podem imposar que la cota d'error sigui 10^{-9} : $\|D^{-1}b\|_2 < 10^{-9} \iff k = 30$. Sigui $(x^{(0)})^T = (0, 0, 0, 0)$. Aleshores:

$$\|e_a^{(k)}\|_2 \leq \frac{\|B_J\|_2}{1 - \|B_J\|_2} \cdot \|x^{(1)} - x^{(0)}\|_2 \leq \frac{\frac{1}{2^k}}{\frac{1}{2}} \cdot \sqrt{\frac{3}{8}} < 10^{-9},$$

ja que:

$$x^{(k+1)} = B_J x^{(k)} + D^{-1}b \implies x^{(1)} = D^{-1}b = \begin{pmatrix} \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{1}{4} \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{4} \\ \frac{1}{2} \\ 0 \\ \frac{1}{4} \end{pmatrix}, \quad \|x^{(1)}\| = \sqrt{\frac{3}{8}}.$$

Per tant,

$$\ln\left(\frac{1}{2^k} \sqrt{\frac{3}{8}} \cdot 2\right) < \ln(10^{-9}) \iff k > \frac{\ln(10^{-9}) - \ln\left(\sqrt{\frac{3}{8}}\right)}{\ln \frac{1}{2}} = 30,558 \approx 31.$$

Ara fem el mateix, però pel mètode de Gauss-Seidel. Sigui $B_{GS} = -(L + D)^{-1}U$ tal que:

$$B_{GS} = \begin{pmatrix} 0 & \frac{1}{4} & \frac{1}{4} & 0 \\ 0 & \frac{1}{16} & \frac{1}{16} & \frac{1}{4} \\ 0 & \frac{1}{16} & \frac{1}{16} & \frac{1}{4} \\ 0 & \frac{1}{32} & \frac{1}{32} & \frac{1}{8} \end{pmatrix} \xrightarrow[\|B_{GS}\|_\infty = \frac{5}{2}]{\|B_{GS}\|_1 = \frac{5}{8}} \|B_{GS}\|_2 \leq \sqrt{\|B_{GS}\|_1 \cdot \|B_{GS}\|_\infty} \leq \frac{\sqrt{5}}{4}.$$

Fets aquests càlculs, podem aplicar l'algorisme:

$$\|e_a\|_2 = \frac{\|B_{GS}\|_2^k}{1 - \|B_{GS}\|_2} \cdot \|x^{(1)}\|_2 < 10^{-9},$$

i si fixem $(x^{(0)})^T = (0, 0, 0, 0)$ com abans:

$$x^{(1)} = (L + D)^{-1}b = \begin{pmatrix} \frac{1}{4} \\ \frac{9}{16} \\ \frac{1}{16} \\ \frac{13}{32} \end{pmatrix}, \quad k = 17. \quad \blacksquare$$

Exercici I.II. Given a linear system $Ax = b$, prove that if A is strictly diagonally dominant by columns, then the Jacobi method is convergent.

Demostració. Recordem que A és diagonal dominant per columnes si, i només si, $|a_j| > \sum_{i \neq j} |a_{ij}|$. Prenent el sistema $Ax = b$, volem veure què passa amb $A^T x = b$ (si A^T és diagonal dominant per files, aleshores Jacobi convergeix). Sigui D, L, U matrius diagonal, triangular inferior i superior, pel que $A^T = D + L^T +$

U^T . Per tant,

$$B_J(A) = -D^{-1}(L + U) \implies B_J(A^T) = -D^{-1}(L^T + U^T),$$

i com Jacobi convergeix tenim que $\rho(B_J(A^T)) < 1$.

$$\begin{aligned} 1 > \rho(B_J(A^T)) &= \rho(D^{-1}(L^T + U^T)) = \rho((D^{-1}(L^T + U^T))^T) \\ &= \rho((L^T + U^T)^T \cdot (D^{-1})^T) = \rho((L + U)D^{-1}) \\ &\stackrel{1}{=} \rho(D^{-1}(L + U)) = \rho(-D^{-1}(L + U)) = \rho(B_J(A)) < 1. \end{aligned}$$

Per tant, Jacobi convergeix per A . ■

Resolució alternativa. Sigui $\|x\|_D = \|Dx\|_1$ la matriu induïda i $\|A\|_D = \|DAD^{-1}\|_1$. Aleshores:

$$\|B_J\|_D = \|D(D^{-1}(L + U))D^{-1}\|_1 = \|(L + U)D^{-1}\|_1 = \max_{1 \leq i \leq n} \sum_{j=1}^n \frac{|a_{ij}|}{|a_{jj}|} < 1,$$

on:

$$L + U = \begin{pmatrix} 0 & a_{12} & \cdots & a_{1n} \\ a_{21} & 0 & \ddots & a_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & 0 \end{pmatrix}, \quad D^{-1} = \begin{pmatrix} \frac{1}{a_{11}} & 0 & \cdots & 0 \\ 0 & \frac{1}{a_{22}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{a_{nn}} \end{pmatrix}, \quad (L + U)D^{-1} = \begin{pmatrix} 0 & \frac{a_{12}}{a_{22}} & \cdots & \frac{a_{1n}}{a_{nn}} \\ \frac{a_{21}}{a_{11}} & 0 & \ddots & \frac{a_{2n}}{a_{nn}} \\ \vdots & \ddots & \ddots & \vdots \\ \frac{a_{n1}}{a_{11}} & \frac{a_{n2}}{a_{22}} & \cdots & 0 \end{pmatrix}$$

Acabem el proper dia (pista: no l'hem acabat mai). ■

Exercici I.12. Show that the Jacobi method converges for 2-by-2 symmetric positive definite systems.

Demostració. Sigui $A = (a_{ij})_{1 \leq i, j \leq 2}$; A és simètrica si, i només si $a_{12} = a_{21}$. A més, és definida positiva si, i només si $a_{11}, a_{22} > 0$ i $a_{11}a_{22} - a_{12}^2 > 0$. Sigui $x \neq 0$, plantegem:

$$x^T Ax = (x_1, x_2)A(x_1, x_2)^T = a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2 > 0.$$

- $x_2 = 0, x_1 \neq 0 \iff x^T Ax = a_{11}x_1^2 > 0 \iff a_{11} > 0.$
- $x_1 = 0, x_2 \neq 0 \iff x^T Ax = a_{22}x_2^2 > 0 \iff a_{22} > 0.$

Pel que fa als VAPs d' A . Per l'exercici 7, $\lambda_1, \lambda_2 > 0$:

$$\begin{vmatrix} a_{11} - \lambda & a_{12} \\ a_{12} & a_{22} - \lambda \end{vmatrix} = \lambda^2 - (a_{11} + a_{22})\lambda + a_{11}a_{22} - a_{12}^2 = 0.$$

¹ Hem usat que si tenim A, B, C, D tal que $AB = C$ i $BA = D$, aleshores $BAB = BC \iff D = BCB^{-1} = Q^{-1}Q = Q^{-1}CQ$, pel que C i D tenen els mateixos VAPs.

Imposem que $\lambda_1 \cdot \lambda_2 = a_{11}a_{22} - a_{12}^2 > 0$. Per tant:

$$B_J = -D^{-1}(L + U) = -\begin{pmatrix} a_{11}^{-1} & 0 \\ 0 & a_{22}^{-1} \end{pmatrix} \cdot \begin{pmatrix} 0 & a_{12} \\ a_{12} & 0 \end{pmatrix} = -\begin{pmatrix} 0 & \frac{a_{12}}{a_{11}} \\ \frac{a_{12}}{a_{22}} & 0 \end{pmatrix}.$$

Per tant, plantegem trobar els VAPs:

$$\begin{vmatrix} -\lambda & \frac{a_{12}}{a_{11}} \\ \frac{a_{12}}{a_{22}} & -\lambda \end{vmatrix} = \lambda^2 - \frac{a_{12}^2}{a_{11}a_{22}} = 0 \iff \lambda_1, \lambda_2 = \pm \sqrt{\frac{a_{12}^2}{a_{11}a_{22}}} = \pm \frac{|a_{12}|}{\sqrt{a_{11}a_{22}}} \iff \rho(B_J) = \frac{|a_{12}|}{\sqrt{a_{11}a_{22}}} < 1,$$

ja que $a_{11}a_{22} - a_{12}^2 > 0$. ■

Exercici 1.13. Let A be a real $n \times n$ matrix and $b \in \mathbb{R}^n$. We write $A = D + L + U$, where D is diagonal, L is strictly lower triangular and U is strictly upper triangular. Moreover, we suppose that $a_{ii} \neq 0, i = 1, \dots, n$. We want to solve the system $Ax = b$ by using the following iterative method:

$$x^{(k+1)} = B_\omega x^{(k)} + c_\omega, \quad \omega > 0,$$

where

$$B_\omega = (1 - \omega)I - \omega D^{-1}(L + U), \quad c_\omega = \omega D^{-1}b.$$

1. Prove that if $(x^{(k)})_{k \geq 0}$ is convergent, then its limit $\bar{x}$ satisfies $A\bar{x} = b$. Moreover, λ is an eigenvalue of B_J (iteration matrix of the Jacobi method) iff $1 - \omega + \omega\lambda$ is an eigenvalue of B_ω . Conclude that if the Jacobi method is convergent, the defined iterative method converges if $0 < \omega \leq 1$.
2. Suppose that the Jacobi method is convergent and B_J has only real eigenvalues:

$$\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$$

Prove that the optimal value of ω is $\omega_{opt} = \frac{2}{2 - \lambda_1 - \lambda_n}$. Which is the value of $\rho(B_{\omega_{opt}})$?

Demostració.

1. Sigui $x^{(k)}$ tal que $x^{(k)} \rightarrow \bar{x}$ quan $k \rightarrow \infty$. Definim el mètode iteratiu $x^{(k+1)} = B_\omega x^{(k)} + c_\omega$ i $\bar{x} = B_\omega \bar{x} + c_\omega$. Aleshores:

$$\begin{aligned} \bar{x} &= (1 - \omega)\bar{x} - \omega D^{-1}(L + U)\bar{x} + \omega D^{-1}b \iff (I - (1 - \omega)I + \omega D^{-1}(L + U))\bar{x} = \omega D^{-1}b \\ &\iff \omega \bar{x} + \omega D^{-1}(L + U)\bar{x} = \omega D^{-1}b \xrightarrow{\omega \neq 0} \bar{x} + D^{-1}(L + U)\bar{x} = D^{-1}b \\ &\iff (D + L + U)\bar{x} = b. \end{aligned}$$

Recordem que $B_J = -D^{-1}(L + U)$. Si λ és el VAP de B_J si, i només si, existeix $x \neq 0$ tal que x és un VEP i $B_J x = \lambda x \iff -D^{-1}(L + U)x = \lambda x$.

$$B_\omega x = (1 - \omega)x - \omega D^{-1}(L + U)x = (1 - \omega + \lambda\omega)x \implies 1 - \omega + \lambda\omega \text{ és VAP de } B_\omega.$$

Sigui λ de la forma $\lambda = r(\cos \varphi + i \sin \varphi) \iff \rho(B_J) < 1 \iff r < 1$.

$$\begin{aligned} |\mu|^2 &= |1 - w + w\lambda|^2 = |(1 - w + rw \cos \varphi) + iwr \sin \varphi|^2 = (1 - w + rw \cos \varphi)^2 + w^2 r^2 \sin^2 \varphi \\ &= (1 - w)^2 + 2(1 - w)rw \cos \varphi + w^2 r^2 \cos^2 \varphi + w^2 r^2 \sin^2 \varphi \\ &= (1 - w)^2 + 2w(1 - w)r \cos \varphi + w^2 r^2 \leq (1 - w)^2 + 2w(1 - w)r + w^2 r^2 \\ &= (1 - w + wr)^2 < 1 \iff |\mu| < 1, \end{aligned}$$

pel que B_w convergeix.

2. Suposem que el mètode de Jacobi és convergent i B_J solament té VEPs reals tals que $\lambda_1 \leq \dots \leq \lambda_n$, que seran de la forma $1 - w + w\lambda_1 \leq \dots \leq 1 - w + w\lambda_n$. Pel que fa al radi espectral $\rho(B_w)$:

- és $1 - w + w\lambda_n$ si $1 - w + w\lambda_1 > 0 \iff w \leq \frac{1}{1 - \lambda_1}$;
- és $1 - w + w\lambda_1$ si $1 - w + w\lambda_n > 0 \iff w \leq \frac{1}{1 - \lambda_n}$;
- és $\max\{1 - w + w\lambda_n, 1 - w + w\lambda_1\}$ en cas contrari.

El punt d'intersecció és $1 - w + w\lambda_n = -1 + w - w\lambda_1$ pel que $w_{opt} = \frac{2}{2 - \lambda_1 - \lambda_n}$:

$$\rho(B_{w_{opt}}) = 1 - (\lambda_n - 1) \frac{2}{2 - \lambda_1 - \lambda_n} = \frac{\lambda_n - \lambda_1}{2 - \lambda_1 - \lambda_n}. \quad \blacksquare$$

Exercici I.14. We want to solve the linear system $Ax = b$, with matrix:

$$A = \begin{pmatrix} 1 & a & a \\ a & 1 & a \\ a & a & 1 \end{pmatrix}, \quad a \in \mathbb{R}.$$

1. For which values of a the matrix A is positive definite?
2. For which values of a the Jacobi method applied to the system is convergent.
3. Prove that if $|a| \geq 1$, then the Gauss-Seidel method is divergent.

Demostració.

1. El criteri de Sylvester ens diu que A és definida positiva si, i només si, tots els menors principals són positius. Per tant, $A_1 = 1 > 0$ i:

$$A_2 = \det \begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix} = 1 - a^2 > 0 \iff |a| < 1, \quad A_3 = \det(A) = (a - 1)^2(2a + 1) > 0 \iff a > -\frac{1}{2},$$

$a \neq 1$. Per tant, A és definida positiva si, i només si, $a \in (-\frac{1}{2}, 1)$.

2. Sigui $B_J = D^{-1}(L + U)$. El mètode de Jacobi convergeix si $\rho(B_J) < 1$. Calculem $B_J = -a \cdot M$:

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad L = \begin{pmatrix} 0 & a & a \\ 0 & 0 & a \\ 0 & 0 & 0 \end{pmatrix}, \quad U = \begin{pmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ a & a & 0 \end{pmatrix}, \quad B_J = -a \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}. \quad (1.1)$$

Sigui ara el polinomi característic de M , $p_M(\lambda)$. Si λ és un VAP d' M tindrem que $-a\lambda$ serà un VAP de B_J , i:

$$p_H(\lambda) = \lambda^3 - 3\lambda - 2 = (\lambda + 1)^2(\lambda - 2) \implies \rho(M) = \max\{|-1|, |2|\} = 2.$$

Amb el que hem dit abans és clar que $\rho(B_J) = |a| \cdot \rho(M)$, pel que $\rho(B_J) = 2 \cdot |a| < 1 \iff |a| < \frac{1}{2}$. Per tant, el mètode de Jacobi és convergent per a aquests valors d' a .

3. Fem el mateix amb el mètode de Gauss-Seidel. Aquí, $B_{GS} = -(L + D)^{-1} \cdot U$ i prenent L, D, U com a (1.1) ens queda:

$$B_{GS} = -(L + D)^{-1} \cdot U = \begin{pmatrix} 0 & -a & -a \\ 0 & a^2 & a^2 - a \\ 0 & a^2 - a^3 & 2a^2 - a^3 \end{pmatrix} = a \begin{pmatrix} 0 & -1 & -1 \\ 0 & a & a - 1 \\ 0 & a - a^2 & 2a - a^2 \end{pmatrix}.$$

El polinomi característic de B_{GS} és $p_{B_{GS}}(\lambda) = -\lambda(\lambda^2 - a^2(3-a)\lambda + a^3)$, pel que els VAPS $\lambda = \lambda_1, \lambda_2, \lambda_3$ tals que $p_{B_{GS}}(\lambda) = 0$ són:

$$\lambda_1 = 0, \quad \begin{cases} \lambda_1 + \lambda_2 = a^2(3-a), \\ \lambda_1\lambda_2 = a^3. \end{cases}$$

D'aquesta última equació en deduïm que $|\lambda_1\lambda_2| \geq 1$, pel que com a mínim $|\lambda_1| \geq 1$ o bé $|\lambda_2| \geq 1$. Per tant, $\rho(B_{GS}) = \max_i\{|\lambda_i|\} \geq 1$ i el mètode de Gauss-Seidel divergeix. ■

Exercici 1.15. Consider the system $Ax = b$, 4×4 with matrix:

$$A = \begin{pmatrix} 2 & 0 & -1 & -1 \\ 0 & 2 & 0 & -1 \\ -1 & 0 & 2 & 0 \\ -1 & -1 & 0 & 2 \end{pmatrix}.$$

1. Prove that the methods of Jacobi and Gauss-Seidel converge. Which is faster?
2. Obtain an estimate of the number of iterates of the method of Jacobi needed to reduce the initial error by a factor of 10^{-8} .
3. Write the iteration matrix of the SOR method in the case $\omega = 3/2$. Give all the non-zero components as irreducible fractions.
4. Is it possible to get the optimal value of ω ?

Demostració.

1. $B_J = -D^{-1} \cdot (L + U)$, pel que seleccionem D, L, U i fem el càlcul:

$$D = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}, L = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ -1 & -1 & 0 & 0 \end{pmatrix}, U = \begin{pmatrix} 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, B_J = \frac{1}{2} \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

Si considerem $B_J = \frac{1}{2}M$, podem trobar els valors propis de M amb $p_M(\lambda)$ i calcular $\rho(B_J) = \frac{1}{2}\rho(M)$:

$$\begin{aligned} p_M(\lambda) = \lambda^4 - 3\lambda^2 + 1 = 0 &\iff \lambda_{1,2,3,4} = \pm \sqrt{\frac{3 \pm \sqrt{5}}{2}} \implies \rho(M) = \sqrt{\frac{3 + \sqrt{5}}{2}} \\ &\implies \rho(B_J) = \sqrt{\frac{3 + \sqrt{5}}{8}} < 1. \end{aligned}$$

Per tant, el mètode de Jacobi és convergent. Pel que fa al mètode de Gauss-Seidel, amb $B_{GS} = -(L + D)^{-1}U$:

$$B_{GS} = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ -1 & 0 & 2 & 0 \\ -1 & -1 & 0 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & 0 & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{2} \end{pmatrix} \cdot U = \frac{1}{4} \begin{pmatrix} 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix} = \frac{1}{4}N.$$

Amb el polinomi característic $p_N(\lambda) = \lambda^2(\lambda^2 - 3\lambda + 1)$, trobem que els VAPs de N són $\lambda_{1,2} = 0$ i $\lambda_{3,4} = \frac{3 \pm \sqrt{5}}{2}$. Per tant, $\rho(N) = \frac{3 + \sqrt{5}}{2}$ i $\rho(B_{GS}) = \frac{1}{4} \cdot \rho(N) = \frac{1}{4} \cdot \frac{3 + \sqrt{5}}{2} = \frac{3 + \sqrt{5}}{8}$. Com $\rho(N) > \rho(B_{GS})$, Gauss-Seidel convergeix més ràpidament que Jacobi.

2. Sigui $x^{(k)}$ la k -èsima iteració del mètode de Jacobi, x la solució exacta del sistema lineal del que $e^{(k)} = x^{(k)} - x$ n'és el k -èsim iterat de l'error.

$$e^{(k+1)} = B_J e^{(k)} \iff \|e^{(k)}\| \leq \|B_J\|^n \cdot \|e^{(0)}\|.$$

Notem que $\|B_J\|_1 = \|B_J\|_\infty = 1$, ja que $A^T = A$ i podem posar:

$$\|B_J\|_2^2 = \rho(B_J^T \cdot B_J) = \rho(B_J^2) = \rho(B_J)^2 \implies \|B_J\|_2 = \rho(B_J) \approx 0.809.$$

Ara, $\rho(B_J)^k < 10^{-8}$, pel que $k \geq -\frac{8}{\log_{10}(\rho(B_J))} \approx 87$.

3. El mètode SOR depèn d'un factor de relaxació w , que en aquest cas $w = \frac{3}{2}$. Escrivim la matriu d'iteració

$$B_w = (D + wL)^{-1}((I - w)D - wU):$$

$$\begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ -w & 0 & 2 & 0 \\ -w & -w & 0 & 2 \end{pmatrix}^{-1} \cdot \begin{pmatrix} 2(I-w) & 0 & w & w \\ 0 & 2(I-w) & 0 & w \\ 0 & 0 & 2(I-w) & 0 \\ 0 & 0 & 0 & 2(I-w) \end{pmatrix} \xrightarrow{w=\frac{3}{2}} \begin{pmatrix} -\frac{1}{2} & 0 & \frac{3}{4} & \frac{3}{4} \\ 0 & -\frac{1}{2} & 0 & \frac{3}{4} \\ -\frac{3}{8} & 0 & \frac{1}{16} & \frac{9}{16} \\ -\frac{3}{8} & -\frac{3}{8} & \frac{9}{16} & \frac{5}{8} \end{pmatrix}.$$

4. A és una matriu banda; per tant, tridiagonal a blocs i simètrica. És definida positiva pel criteri de Sylvester. Efectivament, es pot comprovar que tots els menors principals són positius. De fet, per l'estructura de la matriu solament n'hem de calcular un:

$$\begin{vmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{vmatrix} = 6 > 0.$$

Pel teorema d'Ostrowski-Reich, si A és simètrica i definida positiva, llavors el mètode SOR convergeix per a tot $w \in (0, 2)$. Que convergeixi no vol dir que sigui el paràmetre òptim, i el podem trobar amb la següent igualtat²:

$$w_{\text{opt}} = \frac{2}{1 - \sqrt{1 - \rho(B_{GS})}} = \frac{4\sqrt{2}}{\sqrt{8} + \sqrt{5} - \sqrt{5}} \approx 1.26 \implies \rho(B_{w_{\text{opt}}}) \approx 0.26. \quad \blacksquare$$

Exercici 1.16. Consider the linear system given in Exercise 9.

1. Is matrix A positive definite?
2. Let B_J be the iteration matrix of the Jacobi method for this system. Knowing that $\rho(B_J) = \cos \frac{\pi}{3}$, can you predict the optimal relaxation factor ω^* for the SOR method?
3. Apply the SOR method with the optimal relaxation factor ω^* , and compare the speed of convergence of the SOR method with the one obtained in Exercise 9.

Demostració.

1. Com A és diagonal dominant per files, simètrica i amb elements positius, aleshores A és definida positiva (cf. 1.7).
2. Com A és una matriu real, simètrica, definida positiva i tridiagonal a blocs, $\rho(B_{GS}) = \rho(B_J)^2 = \frac{1}{4}$ i el valor òptim del paràmetre de relaxació és:

$$\tilde{w} = \frac{2}{1 + \sqrt{1 - \rho(B_{GS})}} = \frac{2}{1 - \frac{\sqrt{3}}{2}} \approx 1.0718.$$

² El resultat de teoria és: *Si A una matriu real, simètrica, definida positiva i tridiagonal a blocs. Llavors, $\rho(B_{GS}) = \rho(B_J)^2$ i el valor òptim del paràmetre de relaxació $\tilde{w} = \frac{2}{1 + \sqrt{1 - \rho(B_{GS})}}$, $\rho(B_{GS}) < 1$.*

3. Sigui w^* el factor de relaxació, de tal manera que $B_{w^*} = (D + w^*L)^{-1}((1 - w^*)D - w^*U)$. Aleshores:

$$\|B_{w^*}\|_\infty = \|B_{w^*}\| = 0.60770, \quad \|B_{w^*}\|_2 \leq \sqrt{\|B_{w^*}\|_1 \cdot \|B_{w^*}\|_\infty} = 0.6077,$$

$$\|e^{(k)}\|_2 \leq \frac{\|B_{w^*}\|_2^k}{1 - \|B_{w^*}\|_2} \cdot \|x^{(1)}\|_2 < 10^{-8}.$$

Per SOR, necessitaríem 10 iteracions; per GS, 15; per Jacobi, 28. ■

Exercici 1.17. Let $Ax = b$ be a linear system of dimension n , where A is a symmetric matrix such that $a_{ii} = d > 0$ for all $i = 1, \dots, n$.

1. Assume that the method of Jacobi is convergent (for any initial point) and prove that the method of Gauss-Seidel is also convergent.

From this point on, we will work with the following n -dimensional matrix:

$$\begin{pmatrix} 2 & 1 & 0 & \cdots & \cdots & 0 \\ 1 & 2 & 1 & 0 & & \vdots \\ 0 & 1 & 2 & 1 & \ddots & \vdots \\ \vdots & 0 & 1 & \ddots & \ddots & 0 \\ \vdots & & \ddots & \ddots & 2 & 1 \\ 0 & \cdots & \cdots & 0 & 1 & 2 \end{pmatrix}$$

2. Show that the method of Gauss-Seidel is convergent. In this particular example, the convergence of Gauss-Seidel implies the convergence of Jacobi? If both are convergent, which one converges faster?
3. Consider now the linear system $(A + 2I)x = b$, where $\|b\|_\infty = 1$. If $x^{(0)} = 0$, how many iterates of the Jacobi method are needed to approximate the solution with an error (in sup norm) below 10^{-8} ?

Demostració.

1. Sigui $A = L + D + U$, tal que $D = d \cdot I$. Provem la següent cadena d'equivalències: els VAPs d' A són positius $\iff A$ és definida positiva $\iff$ Gauss-Seidel convergeix.

Jacobi convergeix si, i només si $\rho(B_J) < 1$; per tant, si $B_J = -D^{-1}(L + U)$ és simètrica aleshores els VAPs de B_J són reals. λ és un VAP de $B_J \iff \exists v \neq 0$ tal que $B_J v = \lambda v$:

$$-\frac{1}{d}(L + U)v = \lambda v \iff \begin{cases} -(L + U)v = d\lambda v \\ (L + U + D)v = -d\lambda v + Dv \\ Av = d(1 - \lambda)v \end{cases}$$

Per tant, $|\lambda| < 1$ i com els VAPs són positius, $1 - \lambda > 0$ i $d(1 - \lambda) > 0$ (VEP d' A) és positiu.

2. Optem per usar el criteri del Sylvester: A és definida positiva si, i només si, tots els menors principals

són positius. Sigui $a_m = \det |A_m|$:

$$A_m = \begin{pmatrix} 2 & 1 & 0 & \cdots & \cdots & 0 \\ 1 & 2 & 1 & 0 & & \vdots \\ 0 & 1 & 2 & 1 & & \vdots \\ \vdots & 0 & 1 & \ddots & \ddots & 0 \\ \vdots & & & \ddots & 2 & 1 \\ 0 & \cdots & \cdots & 0 & 1 & 2 \end{pmatrix}, \quad a_m = 2 \cdot \begin{vmatrix} 2 & 1 & 0 & \cdots & \cdots & 0 \\ 1 & 2 & 1 & 0 & & \vdots \\ 0 & 1 & 2 & 1 & & \vdots \\ \vdots & 0 & 1 & \ddots & \ddots & 0 \\ \vdots & & & \ddots & 2 & 1 \\ 0 & \cdots & \cdots & 0 & 1 & 2 \end{vmatrix} - \begin{vmatrix} 1 & 0 & 0 & \cdots & \cdots & 0 \\ 1 & 2 & 1 & 0 & & \vdots \\ 0 & 1 & 2 & 1 & & \vdots \\ \vdots & 0 & 1 & \ddots & \ddots & 0 \\ \vdots & & & \ddots & 2 & 1 \\ 0 & \cdots & \cdots & 0 & 1 & 2 \end{vmatrix}.$$

Per tant, $a_m = 2a_{m-1} - a_{m-2}$. Per inducció sobre m que $a_m > 0$ i que $a_m > a_{m-1}$, per a tot $m \geq 2$. Efectivament, per a $m = 2$ tenim $a_2 = 3 \geq 2 = a_1$. Suposem-ho cert per k (i.e. $a_k > a_{k-1}$) i provem-ho per a $k+1$:

$$a_{k+1} = 2a_k - a_{k-1} > 2a_k - a_k = a_k.$$

Per tant, A és definida positiva i el mètode de Gauss-Seidel és convergent. Podríem trobar que A és definida positiva a partir de la mateixa definició (i.e. $x^T A x > 0$ per a tot $x \neq 0$). Sigui $x = (x_1, \dots, x_n)$. Aleshores:

$$\begin{aligned} x^T A x &= (x_1, \dots, x_n) A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (x_1, \dots, x_n) \begin{pmatrix} 2x_1 + x_2 \\ x_1 + 2x_2 + x_3 \\ \vdots \\ x_{n-1} + 2x_n \end{pmatrix} \\ &= x_1(2x_1 + x_2) + x_2(x_1 + 2x_2 + x_3) + \cdots + x_n(x_{n-1} + 2x_n) \\ &= x_1^2 + (x_1^2 + 2x_1x_2 + x_2^2) + (x_2^2 + 2x_2x_3 + x_3^2) + \cdots + (x_{n-1}^2 + 2x_{n-1}x_n + x_n^2) + x_n^2 \\ &= x_1^2 + \sum_{i=1}^{n-1} (x_i + x_{i+1})^2 + x_n^2 > 0. \end{aligned}$$

Ara, com A és definida positiva, tridiagonal a blocs, simètrica i real, el mètode de Gauss-Seidel convergeix. Usant que també es compleix $\rho(B_{GS}) = \rho(B_J)^2 < 1$, aleshores Jacobi convergeix. De fet, Gauss-Seidel convergeix molt més ràpid que Jacobi.

3. Siguin $B = A + 2I$ i $B_J = -D^{-1}(L + U) = -\frac{L+U}{4}$:

$$B = \begin{pmatrix} 4 & 1 & 0 & \cdots & \cdots & 0 \\ 1 & 4 & 1 & 0 & & \vdots \\ 0 & 1 & 4 & 1 & & \vdots \\ \vdots & 0 & 1 & \ddots & \ddots & 0 \\ \vdots & & & \ddots & 4 & 1 \\ 0 & \cdots & \cdots & 0 & 1 & 4 \end{pmatrix}, \quad B_J = \begin{pmatrix} 0 & -\frac{1}{4} & 0 & \cdots & \cdots & 0 \\ -\frac{1}{4} & 0 & -\frac{1}{4} & 0 & & \vdots \\ 0 & -\frac{1}{4} & 0 & -\frac{1}{4} & & \vdots \\ \vdots & 0 & -\frac{1}{4} & \ddots & \ddots & 0 \\ \vdots & & & \ddots & 0 & -\frac{1}{4} \\ 0 & \cdots & \cdots & 0 & -\frac{1}{4} & 0 \end{pmatrix},$$

Pel que $\|B_J\|_\infty = \frac{1}{2} = \beta$. Prenent $x^{(0)} = 0$ com a condició inicial, ens queda que

$$\|x^{(k)} - \bar{x}\|_\infty = \|e^{(k)}\|_\infty \leq \frac{\beta^k}{1 - \beta} \|x^{(1)} - x^{(0)}\|_\infty = \frac{\beta^k}{1 - \beta} \|x^{(1)}\|_\infty,$$

però com $x^{(1)} = B_J x^{(0)} + D^{-1}b = D^{-1}b$, aleshores $\|x^{(1)}\|_\infty \leq \|D^{-1}\|_\infty \cdot \|b\|_\infty = \frac{1}{4} \cdot 1 = \frac{1}{4}$.

$$\|e^{(k)}\| \leq \frac{\beta^k}{1 - \beta} \cdot \|x^{(1)}\|_\infty < 10^{-8} \iff k > 25.575 \approx 26. \quad \blacksquare$$

Exercici I.18. Consider a linear system $Ax = b$, where A is symmetric.

1. Assume that there exist two values $\mu \in \mathbb{R} \setminus \{0\}$ and $\varepsilon > 0$ such that $\text{Spec}(A) \subset \{z \in \mathbb{C} \mid |z - \mu| \leq \varepsilon\}$. Prove that there exists a value $\varepsilon_0 > 0$ such that, if $\varepsilon < \varepsilon_0$, then A is strictly diagonally dominant (either by rows or by columns).
2. Assume that $\text{Spec}(A) \subset \{z \in \mathbb{C} \mid |z - 1| \leq \varepsilon\} \cup \{z \in \mathbb{C} \mid |z + 1| \leq \varepsilon\}$. If ε is small enough, then derive an iterative method to solve the system and prove its convergence (hint: use A^2).

Demostració.

1. Sigui $\text{Spec}(A) \in \{|z - \mu| < \varepsilon\}$. Provem que existeix $\varepsilon_0 > 0$ tal que $\forall \varepsilon < \varepsilon_0$, A és estrictament diagonal dominant. Com A és simètrica, existeix Q una matriu ortogonal tal que $A = Q^T D Q$, on $D = \text{diag}(d_1, \dots, d_n)$. Per tant, $|d_i - \mu| < \varepsilon$ i $d_i = \mu + \delta_i$ (i.e. $|\delta_i| \leq \varepsilon$). Sigui $D = \mu I + \tilde{D}$; aleshores,

$$A = \mu I + Q^T \tilde{D} Q = \mu I + \begin{pmatrix} q_1^T \tilde{D} q_1 & \cdots & q_1^T \tilde{D} q_n \\ \vdots & \ddots & \vdots \\ q_n^T \tilde{D} q_1 & \cdots & q_n^T \tilde{D} q_n \end{pmatrix}.$$

Per Cauchy-Schwarz, $|q_i^T \tilde{D} q_j| \leq \|q_i^T\|_2 \cdot \|\tilde{D} q_j\|_2 \leq \|\tilde{D}\|_2 \cdot \|q_j\|_2 \leq \varepsilon$. A més a més, $\|D\|_2 \leq \varepsilon$. Fixem-nos ara en els elements diagonals (i.e. $i = j$):

$$|a_{ii}| = |\mu + q_i^T \tilde{D} q_i| \geq |\mu| - |q_i^T \tilde{D} q_i| \geq |\mu| - \varepsilon.$$

Com hem vist abans, $|q_i^T \tilde{D} q_j| \leq \varepsilon$, per a tot i, j , pel que pels elements no diagonals $\sum_{i \neq j} |q_i^T \tilde{D} q_j| \leq (n - 1)\varepsilon$. Així doncs:

$$|\mu| - \varepsilon > (n - 1)\varepsilon \iff \varepsilon < \frac{|\mu|}{n} =: \varepsilon_0.$$

2. $Ax = b \iff A^2 x = Ab$. Aleshores, λ és VAP d' A si, i només si λ^2 VAPs d' A .

- Si $|\lambda - 1| < \varepsilon$, aleshores $\lambda^2 - 1 = (\lambda - 1)^2 + 2(\lambda - 1)$, pel que $|\lambda^2 - 1| \leq |\lambda - 1|^2 + 2|\lambda - 1| < \varepsilon^2 + 2\varepsilon$.
- Si $|\lambda + 1| < \varepsilon$, aleshores $\lambda^2 - 1 = (\lambda + 1)^2 - 2(\lambda + 1)$, pel que $|\lambda^2 - 1| < \varepsilon^2 + 2\varepsilon$.

Per l'exercici 17(a), $\mu_1, \varepsilon_0 = \varepsilon^2 + \varepsilon$ i A és diagonalment dominant. Així doncs, Gauss-Seidel és convergent. ■

Exercici 1.19. Let $\|\cdot\|$ be a matrix norm induced by a vector norm, and let A be a n -by- n real matrix such that $a_{ii} = 1$ for all $1 \leq i \leq n$.

1. Let us write $A = I - B$, and assume that $\|B\| < 1$. Prove that A is invertible and that $A^{-1} = I + B + B^2 + B^3 + \dots$. Derive a bound for $\|A^{-1}\|$ in terms of $\|B\|$ (hint: show that $A(I + B + B^2 + \dots + B^k) = I - B^{k+1}$).
2. Let us write $A = L + I + U$, where L and U denote the lower and upper parts of A . Assume that $\|L\| = 1$ and $\|U\| = u$. If u is small enough, prove that the Gauss-Seidel method (applied to $Ax = b$) is convergent (hint: prove that L is nilpotent).

Demostració.

1. Si $A = I - B$, μ VAP de B , existeix $v \neq 0$ tal que $Bv = \mu v$, $\|v\| = 1$:

$$\|Bv\| \leq \max_{\|v\|=1} \|Bv\| = \|B\| < 1 \iff \|\mu v\| = |\mu| \cdot \|v\| = |\mu| \iff |\mu| < 1.$$

Com $|\mu| < 1$, aleshores $1 - \mu \neq 0$ i $1 - \mu$ és VAP d' A . Per tant, existeix A^{-1} la inversa d' A .

$$\begin{aligned} A(I + B + B^2 + B^3 + \dots + B^k) &= (I - B) \cdot (I + B + \dots + B^k) \\ &= I - B + B - B^2 + \dots + (-B^k) + B^k - B^{k+1} \\ &= I - B^{k+1} \implies I + B + B^2 + \dots + B^k = A^{-1} - A^{-1}B^{k+1}. \end{aligned}$$

Per tant, $I + B + B^2 + B^3 + \dots + B^k \xrightarrow{k \rightarrow \infty} A^{-1} - 0$. Això és ja que $\|A^{-1}B^{k+1}\| \leq \|A^{-1}\| \cdot \|B^{k+1}\| \leq \|A^{-1}\| \rightarrow 0$:

$$\|A^{-1}\| = \|I + B + B^2 + \dots\| \leq \|I\| + \underbrace{\|B\|}_{\leq 1} + \underbrace{\|B\|^2}_{\leq 1} = \frac{1}{1 - \|B\|}.$$

2. Provem que L és nilpotent, és a dir, que existeix $k \in \mathbb{N}$ tal que $L^k = 0$. Sigui $v \in \mathbb{R}^n$:

$$Lv = \begin{pmatrix} 0 & & & 0 \\ * & 0 & & \\ * & * & 0 & \\ \vdots & \vdots & & \ddots \\ * & * & * & \dots & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} 0 \\ v_2^{(1)} \\ \vdots \\ v_n^{(1)} \end{pmatrix} = v^{(1)} \implies L^n v = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} = x^n = 0.$$

Ara ja tenim que L és nilpotent. Posem $B_{GS} = -(L + I)^{-1}U$ i $\|B_{GS}\| \leq \|(I + L)^{-1}\| \cdot \|U\|$.

$$(I + L)^{-1} \leq \underbrace{\|I\|}_1 + \underbrace{\|L\|}_1 + \|L\|^2 + \dots + \underbrace{\|L\|^{n-1}}_1 = n.$$

Si L no fos nilpotent, la suma tendria a infinit. Per tant:

$$\|B_{GS}\| \leq n\mu < 1 \iff \mu < \frac{1}{n} \text{ i Gauss-Seidel convergeix.} \quad \blacksquare$$

Exercici 1.20. Let us consider a linear system $Ax = b$ of dimension n , where:

$$A = \begin{pmatrix} 8 & -1 & 0 & \cdots & 0 & 0 \\ 3 & 8 & -1 & 0 & & 0 \\ 0 & 3 & 8 & -1 & & \vdots \\ \vdots & 0 & 3 & \ddots & \ddots & 0 \\ 0 & & & \ddots & 8 & -1 \\ 0 & 0 & \cdots & 0 & 3 & 8 \end{pmatrix}.$$

We want to solve it using the Gauss-Seidel iteration, $x^{(k+1)} = B_{GS}x^{(k)} + c$.

1. Find a bound for $\|B_{GS}\|_\infty$. Is the Gauss-Seidel method convergent? If it is convergent, does it converge faster than the method of Jacobi?

From now on, we assume that $b = e_n$.

2. Give a bound, in sup norm, for the solution of the linear system.
3. If we choose $x_0 = 0$, compute how many Gauss-Seidel iterations are needed to approximate the solution with an error below 10^{-12} in sup norm.
4. Show that, if the m first components of $x^{(k-1)}$ are zero, then the $m-1$ first component of $x^{(k)}$ are also zero. If we denote by ℓ the amount of iterations found in last part, how many components of $x^{(\ell)}$ are zero?

Demostració.

1. Usant que $B_{GS} = (L + D)^{-1}U$, tenim que:

$$\begin{aligned} \|(L + D)^{-1}U\|_\infty &\leq \|(L + D)^{-1}\|_\infty \cdot \|U\|_\infty \stackrel{3}{=} \|(D(D^{-1}L + I))^{-1}\|_\infty \stackrel{4}{=} \|(I + D^{-1}L)^{-1} \cdot D^{-1}\|_\infty \\ &\leq \|(I + D^{-1}L)^{-1}\|_\infty + \|D^{-1}\|_\infty = \frac{1}{8} \|I + (D^{-1}L) + (D^{-1}L)^2 + \cdots\|_\infty \\ &\leq \frac{1}{8} (\|I\|_\infty + \|D^{-1}L\|_\infty + \|D^{-1}L\|_\infty^2 + \cdots) = \frac{1}{8} \cdot \frac{1}{1 - \|D^{-1}L\|_\infty} = \frac{1}{5} < 1. \end{aligned}$$

Per tant, el mètode de Gauss-Seidel convergeix. En particular:

$$D^{-1}L = \begin{pmatrix} \frac{1}{8} & & \\ & \ddots & \\ & & \frac{1}{8} \end{pmatrix} \cdot \begin{pmatrix} 0 & & & 0 \\ 3 & 0 & & \\ & 3 & 0 & \\ & & \ddots & \ddots \\ 0 & & & 3 & 0 \end{pmatrix} = \begin{pmatrix} 0 & & & 0 \\ \frac{3}{8} & 0 & & \\ & \frac{3}{8} & 0 & \\ & & \ddots & \ddots \\ 0 & & & \frac{3}{8} & 0 \end{pmatrix} \iff \|D^{-1}L\|_\infty = \frac{3}{8}.$$

³ Hem usat que $A(B + C) = AB + AC$.

⁴ Hem usat que $(AB)^{-1} = B^{-1}A^{-1}$.

Com $\|B_{GS}\| < 1$, aleshores el mètode de Gauss-Seidel convergeix i:

$$D^{-1}U = \begin{pmatrix} 0 & -\frac{1}{8} & & 0 \\ & 0 & -\frac{1}{8} & \\ & & 0 & \ddots \\ & & & \ddots & -\frac{1}{8} \\ 0 & & & & 0 \end{pmatrix} \Rightarrow \|B_J\|_\infty = \|D^{-1}(L+U)\|_\infty = \left\| \begin{pmatrix} 0 & -\frac{1}{8} & & 0 \\ \frac{3}{8} & 0 & -\frac{1}{8} & \\ & \frac{3}{8} & 0 & \ddots \\ & & \ddots & \ddots & -\frac{1}{8} \\ 0 & & & \frac{3}{8} & 0 \end{pmatrix} \right\|,$$

i el resultat és $\frac{3}{8} + \frac{1}{8} = \frac{1}{2}$. Per tant, B_J també convergeix (més lentament que B_{GS}).

2. Sabem que $x^{(k+1)} = B_{GS}x^{(k)} + c$ tendeix a $\bar{x} = B_{GS}\bar{x} + c$ quan $k \rightarrow \infty$. Aleshores:

$$\|x\|_\infty \leq \|B_{GS}\|_\infty \cdot \|\bar{x}\|_\infty + \|c\|_\infty \iff \|\bar{x}\|_\infty \leq \frac{\|c\|_\infty}{1 - \|B_{GS}\|_\infty} \leq \frac{\frac{1}{8}}{1 - \frac{1}{5}} = \frac{5}{32},$$

ja que $x^{(0)} = (0, \dots, 0)^T$ i $x^{(1)} = c = (L+D)^{-1}b$, pel que:

$$\|c\|_\infty = \|(L+D)^{-1}e_n\|_\infty = \left\| \begin{pmatrix} \frac{1}{8} & 0 & & 0 \\ * & \frac{1}{8} & 0 & \\ * & * & \frac{1}{8} & \ddots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ * & * & \dots & * & \frac{1}{8} \end{pmatrix} \cdot \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} \right\| = \left\| \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \frac{1}{8} \end{pmatrix} \right\| \iff \|c\|_\infty = \frac{1}{8}.$$

3. Definim $e^{(k)} = x^{(k)} - \bar{x}$, $\bar{x}$ la solució exacta, $x^{(k)}$ la k -èsima iteració del mètode de Gauss-Seidel. Escollim $x^{(0)} = 0$, pel que $x^{(1)} = c$ (ja hem usat aquest fet a l'apartat anterior). Per tant, $\|x^{(1)} - x^{(0)}\|_\infty = \|c\|_\infty$ i:

$$\|x^{(k)} - \bar{x}\|_\infty = \|e^{(k)}\|_\infty \leq \frac{\|B_{GS}\|_\infty^k}{1 - \|B_{GS}\|_\infty} \cdot \|x^{(1)} - x^{(0)}\|_\infty \leq \frac{1}{8} \cdot \frac{\frac{1}{5^k}}{1 - \frac{1}{5}} < 10^{-12},$$

pel que $k \geq \frac{12 \ln 10 + \ln 5 - \ln 32}{\ln 5} = 16.01$ i prenem $k = 17$.

4. Sigui $x^{(k-1)} = (0, \dots, 0, x_{m+1}, \dots, x_n)^T$. Utilitzem el mètode de Gauss-Seidel, a partir de la iteració $x^{(k)} = B_{GS}x^{(k-1)} + c$. Tenim $B_{GS}x^{(k-1)} = -(L+D)^{-1}Ux^{(k-1)}$. Fem els càlculs pertinents:

$$- \begin{pmatrix} 8 & 0 & & 0 \\ 3 & 8 & 0 & \\ & 3 & \ddots & \ddots \\ 0 & & \ddots & \ddots & 0 \\ & & & 3 & 8 \end{pmatrix}^{-1} \begin{pmatrix} 0 & -1 & & 0 \\ 0 & 0 & -1 & \\ & 0 & \ddots & \ddots \\ 0 & & \ddots & \ddots & -1 \\ & & & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ \vdots \\ 0 \\ x_{m+1} \\ \vdots \\ x_n \end{pmatrix} = - \begin{pmatrix} \frac{1}{8} & 0 & & 0 \\ * & \frac{1}{8} & 0 & \\ * & * & \ddots & \ddots \\ \vdots & & \ddots & \ddots & 0 \\ * & \dots & \dots & * & \frac{1}{8} \end{pmatrix} \begin{pmatrix} 0 \\ \vdots \\ 0 \\ x_{m+1} \\ \vdots \\ x_n \end{pmatrix}$$

Recordem que, anteriorment, hem definit $b = e_n = (0, \dots, 0, 1)^T$ i hem trobat que $c = (0, \dots, 0, \frac{1}{8})^T$.

Per tant,

$$x^{(k)} = - \begin{pmatrix} \frac{1}{8} & 0 & & 0 \\ * & \frac{1}{8} & 0 & \\ * & * & \ddots & \ddots \\ \vdots & & \ddots & \ddots & 0 \\ * & \dots & \dots & * & \frac{1}{8} \end{pmatrix} \cdot \begin{pmatrix} 0 \\ \vdots \\ 0 \\ x_{m+1} \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \frac{1}{8} \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ -\frac{1}{8}x_{m+1} \end{pmatrix}.$$

Per tant, per a $\ell = 17$ tenim que $x^{(17)}$ té les primeres $n - 17$ files nul·les. ■

Exercici I.21. We consider the linear system $Ax = b$, where:

$$A = \begin{pmatrix} I & -I & & & \\ & I & -I & & \\ & & \ddots & \ddots & \\ & & & I & -I \\ a & & & & I \end{pmatrix}$$

is an $n \times n$ matrix and b is a real vector of dimension n . The elements corresponding to the blanks are zero.

1. Prove that the previous system has a unique solution if $a \neq -1$.
2. Study the convergence of the Jacobi method depending on the real parameter a .
3. Study the convergence of the Gauss-Seidel method as a function of the real parameter a .
4. If for a value of a , both Jacobi and Gauss-Seidel methods converge, is it possible to say which of them converges faster?

Demostració.

1. Simplement comprovem que la matriu tingui rang màxim, calculant-ne el determinant:

$$\det(A) = \begin{vmatrix} I & -I & & & \\ & I & -I & & \\ & & \ddots & \ddots & \\ & & & I & -I \\ 0 & & & & I \end{vmatrix} + a \cdot (-1)^{n+1} \cdot \begin{vmatrix} -I & 0 & & & \\ I & -I & 0 & & \\ & I & \ddots & \ddots & \\ & & \ddots & -I & 0 \\ 0 & & & I & -I \end{vmatrix} = 1 + a \cdot (-1)^{n+1} \cdot (-1)^{n-1} = 1 + a.$$

Per tant, $1 + a \neq 0 \iff a \neq -1$, i existeix una única solució si $a \neq -1$.

2. Prenem la matriu d'iteració $B_J = -D^{-1}(L + U) = -(L + U)$ (ja que $D = I$), i:

$$B_J = -(L + U) = \begin{pmatrix} 0 & I & & \\ & 0 & I & \\ & & \ddots & \ddots \\ & & & 0 & I \\ -a & & & & 0 \end{pmatrix} \Rightarrow \|B_J\|_\infty = \max\{|-1|, |a|\} \geq 1.$$

El polinomi característic és $p_{B_J}(\lambda) = (-1)^n(\lambda^n + a)$, ja que:

$$\det(B_J - \lambda I) = -\lambda \begin{vmatrix} -\lambda & I & & \\ & -\lambda & I & \\ & & \ddots & \ddots \\ & & & \ddots & I \\ & & & & -\lambda \end{vmatrix} - a \cdot (-1)^{n+1} \begin{vmatrix} I & & & \\ -\lambda & I & & \\ & -\lambda & I & \\ & & \ddots & \ddots \\ & & & -\lambda & I \end{vmatrix},$$

i $(-\lambda) \cdot \lambda^{n-1} - a \cdot (-1)^{n+1} = (-1)^n(\lambda^n + a)$. Per tant, $\rho(B_J) = |a|^{\frac{1}{n}} < 1$ si $|a| < 1$. Per tant, Jacobi convergeix.

3. La matriu d'iteració de Gauss-Seidel és $B_{GS} = -(L + D)^{-1}U$. Com $D = I$ i $(I + L)^{-1} = I - L^5$, tenim que $B_{GS} = -(I - L) \cdot U$. Fem els càlculs pertinents per trobar els VAPs:

$$B_{GS} = \begin{pmatrix} 0 & I & & \\ & 0 & \ddots & \\ & & \ddots & I \\ 0 & -a & & 0 \end{pmatrix} \Rightarrow \det(B_{GS} - \lambda I) = (-\lambda) \begin{vmatrix} -\lambda & -I & & \\ & \ddots & \ddots & \\ & & \ddots & -I \\ -a & & & -\lambda \end{vmatrix}.$$

Notem que:

$$\begin{vmatrix} -\lambda & -I & & \\ & \ddots & \ddots & \\ & & \ddots & -I \\ -a & & & -\lambda \end{vmatrix} = (-\lambda) \begin{vmatrix} -\lambda & -I & & \\ & \ddots & \ddots & \\ & & \ddots & -I \\ & & & -\lambda \end{vmatrix} - a \cdot (-1)^{n+1} \begin{vmatrix} -I & & & \\ -\lambda & -I & & \\ & -\lambda & -I & \\ & & \ddots & \ddots \\ & & & -\lambda & -I \end{vmatrix},$$

pel que $p_{B_{GS}}(\lambda) = -\lambda((-\lambda) \cdot (-\lambda)^{n-2} - a(-1)^n(-1)^{n-2})$, i $p_{B_{GS}}(\lambda) = -\lambda((-\lambda)^{n-1} - a)$. Per tant, $\rho(B_{GS}) = |a|^{\frac{1}{n-1}}$, que és menor que 1 si $|a| < 1$.

4. Fixem-nos que $\rho(B_J) = |a|^{\frac{1}{n}}$, mentre que $\rho(B_{GS}) = |a|^{\frac{1}{n-1}}$. Si $a = 0$, els dos són igual de ràpids; així, posem $a \neq 0$, $|a| < 1$. Aleshores, $|a|^{\frac{1}{n-1}} < |a|^{\frac{1}{n}}$, pel que $\rho(B_{GS}) < \rho(B_J)$ i Gauss-Seidel convergeix

⁵ Això ho hem deduït perquè $L^2 = 0$, pel que $(I + L)(I - L) = I - L^2 = I$, i ens queda que $(I + L)^{-1} = I - L$.

més ràpidament. ■

Exercici 1.22. We want to solve a system $Ax = b$, for $b \in \mathbb{R}^3$,

$$A = \begin{pmatrix} 1 & -2 & \alpha \\ -1 & 1 & -1 \\ -\alpha & -2 & 1 \end{pmatrix},$$

and $\alpha \in \mathbb{R}$. We write $A = L + D + U$, where L is strictly upper triangular, D is diagonal and U is strictly lower triangular.

1. Suppose that $\alpha = 2$.

1.1. Prove that the Jacobi method is convergent and for any initial vector $x^{(0)}$ the sequence of iterates $x^{(k)}$ converges with a finite number of iterates. Prove that, however, the Gauss-Seidel is not convergent.

1.2. Now take $d > 4$. Prove that the Gauss-Seidel method is convergent for the matrix $\tilde{A} = L + dD + U$ and determine how many iterates are needed (as a function of d) to reduce the initial error (in sub-infinity norm) in the Gauss-Seidel method in a factor of 10^{-8} .

2. Obtain all the values of α for which the Jacobi method is convergent. Are there values of α for which the Jacobi and Gauss-Seidel methods are simultaneously convergent?

Demostració. Abans de començar, fixem-nos que les matrius L , D i U seran, respectivament:

$$L = \begin{pmatrix} 0 & & \\ -1 & 0 & \\ -\alpha & -2 & 0 \end{pmatrix}, \quad D = I, \quad U = \begin{pmatrix} 0 & -2 & \alpha \\ & 0 & -1 \\ & & 0 \end{pmatrix}$$

Comencem suposant que $\alpha = 2$.

1. Calcular B_J i B_{GS} és senzill. Trobarem els VAPs i calcularem $\rho(B_J)$ i $\rho(B_{GS})$. Per al primer cas, demostrarem que la seqüència convergeix amb un nombre finit d'iterats i, per al segon, que no convergeix en absolut.

$$B_J = -D^{-1}(L + U) = -(L + U) = \begin{pmatrix} 0 & 2 & -2 \\ 1 & 0 & 1 \\ 2 & 2 & 0 \end{pmatrix} \implies p_{B_J}(\lambda) = \begin{vmatrix} -\lambda & 2 & -2 \\ 1 & -\lambda & 1 \\ 2 & 2 & -\lambda \end{vmatrix} = -\lambda^3,$$

ja que el determinant és $-\lambda^3 + 4 - 4 + 2\lambda + 2\lambda - 4\lambda$. Per tant, l'únic VAP de B_J és 0 i $\rho(B_J) = 0$. De fet, això ens indica que B_J és nilpotent. Fent els càlculs pertinents trobaríem que $B_J^3 = 0$ i per tant:

$$\|x^{(k)} - \bar{x}\| = \|e^{(k)}\| = \|B_J^k\| \cdot \|e^{(0)}\|, \quad \|e^{(3)}\| = \|B_J^3\| \cdot \|e^{(0)}\| = 0, \quad \forall x^{(0)}, \quad e^{(0)} = x^{(0)} - \bar{x}.$$

D'altra banda:

$$B_{GS} = -(I + L)^{-1}U = -\begin{pmatrix} 1 & & \\ -1 & 1 & \\ -2 & -2 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 0 & -2 & 2 \\ & 0 & -1 \\ & & 0 \end{pmatrix} = -\begin{pmatrix} 1 & & \\ 1 & 1 & \\ 4 & 2 & 1 \end{pmatrix} \begin{pmatrix} 0 & -2 & 2 \\ & 0 & -1 \\ & & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 & -2 \\ 0 & 2 & -1 \\ 0 & 8 & -6 \end{pmatrix}.$$

Els VAPs de B_{GS} són 0 i $-2 \pm 2\sqrt{2}$. Per tant, $\rho(B_{GS}) = 2 + 2\sqrt{2} > 1$ i Gauss-Seidel divergeix.

Fixem-nos que $\tilde{A}$ és estrictament diagonal dominant; per tant, Gauss-Seidel convergeix⁶. Seguint fixant $\alpha = 2$ i busquem la matriu de Gauss-Seidel; en aquest cas, tenim $D = dI$ i podem calcular $\tilde{B}_{GS}$:

$$\tilde{B}_{GS} = -(D + L)^{-1}U = -\begin{pmatrix} d & & \\ -1 & d & \\ -2 & -2 & d \end{pmatrix}^{-1} \cdot \begin{pmatrix} 0 & -2 & 2 \\ & 0 & -1 \\ & & 0 \end{pmatrix} = \begin{pmatrix} 0 & \frac{2}{d} & -\frac{2}{d} \\ 0 & \frac{2}{d^2} & \frac{d-2}{d^2} \\ 0 & 4\frac{d+1}{d^3} & -2\frac{d+2}{d^3} \end{pmatrix}.$$

Calculem la norma subinfinít per determinar el nombre d'iterats necessaris:

$$\|\tilde{B}_{GS}\|_{\infty} = \max_{1 \leq i \leq n} \sum_{j=1}^n |(\tilde{B}_{GS})_{ij}| = \max \left\{ \frac{4}{d}, \frac{1}{d}, \frac{6d+8}{d^3} \right\} = \frac{4}{d}.$$

Volem $k \in \mathbb{N}$ tal que $(\frac{4}{d})^k < 10^{-8}$, pel que ens surt $k > \frac{8}{\log_{10} d - \log_{10} 4}$. I finalment, mirem si existeixen valors d' α tal que Jacobi i Gauss-Seidel siguin convergents alhora. Calcularem els VAPs de B_J i B_{GS} , $\rho(B_J)$ i $\rho(B_{GS})$ i, en cas de convergir, veurem per a quines α comunes ho fan.

$$B_J = -D^{-1}(L + U) = -(L + U) = \begin{pmatrix} 0 & 2 & -\alpha \\ 1 & 0 & 1 \\ \alpha & 2 & 0 \end{pmatrix} \Rightarrow p_{B_J}(\lambda) = \begin{vmatrix} -\lambda & 2 & -\alpha \\ 1 & -\lambda & 1 \\ \alpha & 2 & -\lambda \end{vmatrix} = -\lambda^3 + (4 - \alpha^2)\lambda,$$

i els VAPs són $\lambda = 0$ i $\lambda = \pm\sqrt{\alpha^2 - 4}$, si $|\alpha| > 2$. Si $|\alpha| < 2$, aleshores $\lambda = \pm i\sqrt{4 - \alpha^2}$. Per tant, Jacobi convergeix si, i només si $\rho(B_J) = \sqrt{|\alpha^2 - 4|} < 1$, si $|\alpha^2 - 4| < 1 \iff \sqrt{3} < |\alpha| < \sqrt{5}$. Fem el mateix per Gauss-Seidel:

$$B_{GS} = -(D + L)^{-1}U = -\begin{pmatrix} 1 & & \\ -1 & 1 & \\ -\alpha & -2 & 1 \end{pmatrix}^{-1} \cdot \begin{pmatrix} 0 & -2 & \alpha \\ & 0 & -1 \\ & & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 & -\alpha \\ 0 & 2 & -\alpha + 1 \\ 0 & 2\alpha + 4 & -\alpha^2 - 2\alpha + 2 \end{pmatrix}.$$

Podem calcular el polinomi característic de B_{GS} :

$$\det(B_{GS} - \lambda I) = \begin{vmatrix} -\lambda & 2 & -\alpha \\ 0 & 2 - \lambda & -\alpha + 1 \\ 0 & 2\alpha + 4 & -\alpha^2 - 2\alpha + 2 - \lambda \end{vmatrix} = -\lambda((2 - \lambda)(-\alpha^2 - 2\alpha + 2 - \lambda) + (\alpha - 1)(2\alpha + 4)).$$

⁶ Recordem que la convergència de Gauss-Seidel està assegurada si o bé A és simètrica i definida positiva, o bé A és estrictament diagonal dominant.

Això queda $p_{B_{GS}}(\lambda) = -\lambda(\lambda^2 + (\alpha + 2\alpha - 4)\lambda - 2\alpha)$. $\lambda_1 = 0$, però λ_2 i λ_3 no són gens senzills de calcular. N'intentem donar una fita inferior. De $\lambda_2\lambda_3 = -2\alpha$ en deduïm que $|\alpha| < \frac{1}{2}$, imposant $|\lambda_2|, |\lambda_3| < 1$ (que $\rho(B_{GS}) < 1$, que convergeixi).

$$|\lambda_{2,3}| = \left| \frac{-(\alpha^2 + 2\alpha - 4) \pm \sqrt{(\alpha^2 + 2\alpha)^2 - 8\alpha^2 - 8\alpha + 16}}{2} \right| > \frac{\frac{11}{4} + \sqrt{8}}{2} > 1 \implies \text{GS no és convergent,}$$

ja que $-(\alpha^2 + 2\alpha - 4) > \frac{11}{4}$ i:

$$(\alpha^2 + 2\alpha)^2 \geq 0, -8 \leq -8\alpha^2 - 8\alpha \leq 4 \implies -(\alpha^2 + 2\alpha - 4) \pm \sqrt{(\alpha^2 + 2\alpha)^2 - 8\alpha^2 - 8\alpha + 16} \geq 8. \blacksquare$$

Exercici 1.23. Let A be an $n \times n$ symmetric positive definite matrix and B_{GS} the iteration matrix of the Gauss-Seidel method corresponding to the system $Ax = b$. We define $P = A - B_{GS}^T A B_{GS}$.

1. Prove that for any vector $v \in \mathbb{C}^n \setminus \{0\}$ we have that $v^H A v > 0$.
2. Prove that P is a symmetric matrix and that $B_{GS} = I - B$ with $B = (D + L)^{-1} A$ (being L the strictly lower triangular of A and D the diagonal part).
3. Prove that $P = B^T (AB^{-1} + (B^T)^{-1} A - A) B$, and, by using this result, that $P = B^T D B$.
4. Prove that P is symmetric positive definite. Compute $u^H P u$, with $u \in \mathbb{C}^n$ an eigenvector of eigenvalue $\lambda \in \mathbb{C}$ of B_{GS} . What can we deduce from this?

2

VALORS I VECTORS PROPIS

Exercici 2.1. Let A be an n -by- n matrix, and let λ be an eigenvalue of A .

1. If B is an n -by- n matrix such that $\lambda \notin \text{Spec}(B)$, prove that $\|(B - \lambda I)^{-1}(A - B)\| \geq 1$ (hint: $\rho(A) \leq \|A\|$).
2. Use the first part to prove the Gerschgorin Theorem (hint: choose B appropriately).

Demostració.

1. Si $\lambda \notin \text{Spec}(B)$, aleshores $0 \notin \text{Spec}(B - \lambda I)$. Per tant, existeix la matriu inversa $(B - \lambda I)^{-1}$. Sigui $(B - \lambda I)^{-1}(A - B)$:

$$(B - \lambda I)^{-1}(A - B) = (B - \lambda I)^{-1}(A - \lambda I + \lambda I - B) = (B - \lambda I)^{-1}(A - \lambda I) - I =: M.$$

Sigui $v \neq 0$ un VEP d' A amb VAP λ , de manera que $Av = \lambda v$. Per tant:

$$Mv = (B - \lambda I)^{-1}(Av - \lambda v) = v \xrightarrow{Av=\lambda v} Mv = -v.$$

Per tant, -1 és un VAP de M . Per tant, $\|M\| \geq \rho(M) \geq 1$.

2. El teorema de Gerschgorin diu el següent:

Teorema 2.2 (de Gerschgorin). *Sigui $A = (a_{ij})_{\substack{1 \leq i \leq n \\ 1 \leq j \leq n}}$ una matriu $n \times n$ amb VAPs $\lambda_1, \dots, \lambda_n$. Aleshores:*

$$\text{Spec}(A) \in \bigcup_{i=1}^n D_i(a_{ii}, r_i), \quad D_i(a_{ii}, r_i) = \{z \in \mathbb{C} \mid |z - a_{ii}| \leq r_i\}, \quad r_i = \sum_{\substack{j=1 \\ i \neq j}}^n |a_{ij}|.$$

Passem a demostrar-lo. Si $\lambda_i = a_{ii}$, $1 \leq i \leq n$, aleshores el resultat és trivial (evidentment, $a_{ii} \in D_i(a_{ii}, r_i)$). Sigui, doncs, $\lambda \notin \text{Spec}(A)$ ($\lambda \neq a_{ii}$) per a tot i . Considerant $\|\cdot\| = \|\cdot\|_\infty$:

$$B = \text{diag}(A) \implies \|(B - \lambda I)^{-1}(A - B)\|_\infty = \sum_{\substack{j=1 \\ j \neq i}}^n \frac{|a_{ij}|}{|a_{ii} - \lambda|} \geq 1 \iff |a_{ii} - \lambda| \leq \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}| = r_i,$$

ja que:

$$(B - \lambda I)^{-1}(A - B) = \begin{pmatrix} \frac{1}{a_{11} - \lambda} & & & \\ & \ddots & & \\ & & \frac{1}{a_{nn} - \lambda} & \\ & & & \ddots \end{pmatrix} \begin{pmatrix} 0 & a_{12} & \cdots & a_{1n} \\ a_{21} & 0 & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & 0 \end{pmatrix} = \begin{pmatrix} 0 & \frac{a_{12}}{a_{11} - \lambda} & \cdots & \frac{a_{1n}}{a_{11} - \lambda} \\ \frac{a_{21}}{a_{22} - \lambda} & 0 & \cdots & \frac{a_{2n}}{a_{22} - \lambda} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{a_{n1}}{a_{nn} - \lambda} & \frac{a_{n2}}{a_{nn} - \lambda} & \cdots & 0 \end{pmatrix}.$$

Per tant, $\lambda \in D_i(a_{ii}, r_i)$. Com podem generalitzar-ho a tot $\lambda \in \text{Spec}(A)$, ens queda que $\text{Spec}(A) \in \bigcup_{i=1}^n D_i(a_{ii}, r_i)$, com volíem. ■

Exercici 2.3. Consider the matrix:

$$A = \begin{pmatrix} -3.0 & 0.1 & 0.1 \\ 0.2 & 0.0 & -0.1 \\ -0.1 & 0.2 & 3.0 \end{pmatrix}$$

1. Use the Gerschgorin Theorem to determine a region $\Omega \subset \mathbb{C}$ such that all the eigenvalues of A lie in Ω .
2. Can you conclude that all eigenvalues of A are real? If so, find an interval for each eigenvalue.

Demostració.

1. Pel teorema de Gerschgorin, tenim que $\text{Spec}(A) \subset \Omega_1 := D(-3, 0.2) \cup D(0, 0.3) \cup D(3, 0.3)$ ⁷. Podem fer el mateix per A^T , ja que A i A^T han de tenir els mateixos VAPs. Però compte, perquè igualment $\text{Spec}(A) \neq \text{Spec}(A^T)$! Aleshores, $\text{Spec}(A^T) \subset \Omega_2 := D(-3, 0.3) \cup D(0, 0.3) \cup D(3, 0.2)$. Per tant, $\Omega := \Omega_1 \cap \Omega_2 = D(-3, 0.2) \cup D(0, 0.3) \cup D(3, 0.2)$.
2. A cada component connexa (és a dir, a cada disc disjunt) hi ha un sol VAP. Si algun fos un VAP no real, el seu conjugat hauria de viure en el mateix disc, però no és el cas. Per tant, $\lambda_1 \in [-3.2, -2.8]$, $\lambda_2 \in [-0.3, 0.3]$ i $\lambda_3 \in [2.8, 3.2]$. ■

⁷ Com és la primera vegada, una petita observació. Estem centrant cada disc en a_{ii} i r_{ii} és $\sum_{j \neq i} |a_{ij}|$. Per exemple, per al segon disc tenim $D(a_{22}, | -0.1 | + | 0.2 |) = D(0, 0.3)$.

Exercici 2.4. Let A be a symmetric n -by- n matrix, and let us denote by λ one of its eigenvalues. If $\|\cdot\|_F$ denotes the Frobenius norm and B is a matrix such that $\|B\|_F = 1$, show that there is an eigenvalue λ_ε of $A + \varepsilon B$ such that $|\lambda_\varepsilon - \lambda| = O(\varepsilon)$. Give an explicit bound on $|\lambda_\varepsilon - \lambda|$ (hint: show that, for any vector $v \in \mathbb{R}^n$, $\|v\|_1 \leq \sqrt{n}\|v\|_2$).

Demostració. Sigui $v \in \mathbb{R}^n$, tal que $\|v\|_1 = |v_1| + \dots + |v_n|$, i $u = (\text{sign}(v_1), \dots, \text{sign}(v_n))$. Com per a tot $u, v \in \mathbb{R}^n$ tenim $|\langle u, v \rangle| \leq \|u\|_2 \cdot \|v\|_2$, aleshores:

$$|\langle u, v \rangle| = \left| \sum_{i=1}^n \text{sign}(v_i) \cdot v_i \right| = \sum_{i=1}^n |v_i| \leq \|u\|_2 \cdot \|v\|_2 = \|v\|_2 \sqrt{\sum_{i=1}^n \text{sign}(v_i)^2} \leq \|v\|_2 \sqrt{n}.$$

Com A és simètrica, aleshores existeix Q una matriu ortogonal tal que $D = Q^T A Q$. Definim $A(\varepsilon) = A + \varepsilon B$:

$$Q^T A(\varepsilon) Q = Q^T A Q + \varepsilon Q^T B Q = D + \varepsilon E \implies \text{Spec}(A + \varepsilon B) = \text{Spec}(D + \varepsilon E).$$

Tal com hem definit E , per la definició de $\|\cdot\|_2$ i per la ortogonalitat de Q :

$$\|E\|_2 \leq \|Q^T\|_2 \cdot \|BQ\|_2 = \|Q^T\|_2 \cdot \|B\|_2 \cdot \|Q\|_2 = \rho(QQ^T) \cdot \|B\|_2 \cdot \rho(Q^T Q) = \|B\|_2.$$

Està clar que $D + \varepsilon E$ és de la forma:

$$D + \varepsilon E = \begin{pmatrix} d_{11} + \varepsilon e_{11} & \varepsilon e_{12} & \cdots & \varepsilon e_{1n} \\ \varepsilon e_{21} & d_{22} + \varepsilon e_{22} & \cdots & \varepsilon e_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \varepsilon e_{n1} & \varepsilon e_{n2} & \cdots & d_{nn} + \varepsilon e_{nn} \end{pmatrix}$$

Per tant, existeix $\lambda_\varepsilon \mid \lambda_\varepsilon \in D(d_{ii} + \varepsilon e_{ii}, r_i)$, on $r_i = \varepsilon \sum_{j=2}^n |e_{ij}|$. Intentem donar ara una cota explícita en $|\lambda_\varepsilon - \lambda|$. Recordem que $e_i = (e_{i1}, \dots, e_{in})$.

$$\begin{aligned} |\lambda_\varepsilon - \lambda| &= |\lambda_\varepsilon - d_{ii}| \leq \underbrace{|\lambda_\varepsilon - \lambda - \varepsilon e_{ii}|}_{r_i} + |\varepsilon| \cdot |e_{ii}| \leq \varepsilon \left(\sum_{j \neq i} |e_{ij}| \right) + \varepsilon |e_{ii}| \\ &= \varepsilon \|e_i\|_1 \leq \varepsilon \sqrt{n} \cdot \|e_i\|_2 \leq \varepsilon \sqrt{n} \cdot \|E\|_F \end{aligned}$$

Vegem que $\|E\|_F = \|B\|_F = 1$. Usem que $\|A\|_F^2 = \text{tr}(A^T A)$ i que la traça és invariant per a permutacions d'ordre del producte:

$$\|Q^T B Q\|_F^2 = \text{tr}(Q^T B^T (Q^T)^T Q^T B Q) = \text{tr}(Q^T B^T B Q) = \text{tr}(Q Q^T B^T B) = \text{tr}(B^T B) = \|B\|_F^2.$$

Per tant, la desigualtat finalment queda:

$$|\lambda_\varepsilon - \lambda| \leq \varepsilon \sqrt{n} \cdot \|E\|_F = \varepsilon \sqrt{n} \|B\|_F = \varepsilon \sqrt{n}.$$

■

Exercici 2.5. *Is it possible to apply the power method to matrix A in Exercise 2.3? And the shifted power method? Explain how to use these two methods to compute all eigenvalues and eigenvectors of A .*

Demostració. Obtenim que $\lambda_1 \approx -3$, $\lambda_2 \approx 0$ i $\lambda_3 \approx 3$. Com $|\lambda_1| \approx |\lambda_3|$, no és convenient aplicar el mètode de la potència (la convergència és lenta). Si fem potència inversa desplaçada amb $\mu = 3$ i $\mu = -3$, és a dir, apliquem el mètode de la potència a $(A - \mu I)^{-1}$, obtenim λ_3 i λ_1 ; si fem potència inversa, obtindrem λ_2 .

$$(A - \lambda I)^{-1}v = \mu v \implies \frac{1}{\mu}v = (A - \lambda I)v \iff Av = \left(\frac{1}{\mu} + \lambda\right)v. \quad \blacksquare$$

Exercici 2.6. *Consider the matrix:*

$$A = \begin{pmatrix} -1 & 1 & 1 \\ 1 & 8 & -1 \\ 1 & -1 & -2 \end{pmatrix}$$

1. *Prove that it has a unique eigenvalue of largest modulus, which is real.*
2. *Use the power method to determine this eigenvalue and its eigenvector.*
3. *Use the trace and determinant of the matrix to obtain all eigenvalues.*

Demostració.

1. Apliquem el teorema de Gerschgorin. Aleshores, $\text{Spec}(A) \subset D(-1, 2) \cup D(8, 2) \cup D(-2, 2)$. Queda una component connexa aïllada, $D(8, 2)$, pel que existeix un únic λ tal que $|\lambda| \approx 8$ i λ sigui real.
2. Prenem $z^{(0)} = (1, 0, 0)^T$ i apliquem el mètode de la potència:

$$\begin{cases} y^{(k+1)} = Az^{(k)} \\ z^{(k+1)} = \frac{y^{(k+1)}}{\|y^{(k+1)}\|} \end{cases} \quad v = \begin{pmatrix} 0.09835 \\ 0.99128 \\ -0.08765 \end{pmatrix}, \quad \sigma_{k+1} = (z^{(k+1)})^T Az^{(k+1)} \implies \lambda = 8.1876.$$

Per tant, tenim $\lambda = 8.1876$, que és real, amb $v = (0.09835, 0.99128, -0.08765)$.

3. Hem d'utilitzar que $\lambda \cdot \lambda_1 \cdot \lambda_2 = \det(A) = 9$ i $\lambda + \lambda_1 + \lambda_2 = \text{tr}(A) = 5$. Aplicant que $\lambda = 8.1876$:

$$\lambda_{1,2}^2 - (5 - \lambda) \cdot \lambda_{1,2} + \frac{9}{\lambda} = 0 \iff \lambda_1 = -0.3933, \quad \lambda_2 = -2.7942517. \quad \blacksquare$$

Exercici 2.7. *Let T be an n -by- n matrix of the form:*

$$T = \begin{pmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{pmatrix},$$

where T_{11} and T_{22} are squared matrices.

1. *Prove that $\text{Spec}(T) = \text{Spec}(T_{11}) \cup \text{Spec}(T_{22})$.*
2. *Explain the relationship between the eigenvectors of T_{11} and T_{22} and the ones of T .*

Demostració.

1. Sigui $\lambda \in \text{Spec}(T)$. Existeix $v \neq 0$ tal que $Tv = \lambda v$, $v = (v_1, v_2)^T$ el VAP corresponent. Podem posar:

$$\begin{cases} T_{11}v_1 + T_{12}v_2 = \lambda v_1 \\ T_{22}v_2 = \lambda v_2 \end{cases}$$

Si $v_2 \neq 0$, aleshores $\lambda \in \text{Spec}(T_{22})$ (v_2 serà un VAP de T_{22} amb VAP $\lambda \in \text{Spec}(T_{22})$). Si $v_2 = 0$, aleshores $T_{11}v_1 = \lambda v_1$ i, per tant, com $v_2 \neq 0$ (perquè si els dos fossin zero, aleshores $v = 0$, contradicció): $\lambda \in \text{Spec}(T_{11})$. Està clar, doncs, que $\lambda \in \text{Spec}(T_{11}) \cup \text{Spec}(T_{22})$. Provem ara la inclusió contrària: suposem que $\lambda \in \text{Spec}(T_{11}) \cup \text{Spec}(T_{22})$.

Suposem que $\lambda \in \text{Spec}(T_{11})$; per tant, existeix $v_1 \neq 0$ tal que $T_{11}v_1 = \lambda v_1$, $v = (v_1, 0)^T$ és el VEP de T (i.e. $Tv = \lambda v$). Així, $\lambda \in \text{Spec}(T)$.

Ara suposem que $\lambda \in \text{Spec}(T_{22})$; per tant, existeix $v_2 \neq 0$ tal que $T_{22}v_2 = \lambda v_2$. Volem trobar $v = (v_1, v_2)^T$ tal que $Tv = \lambda v$:

$$\begin{cases} T_{11}v_1 + T_{12}v_2 = \lambda v_1 \\ T_{22}v_2 = \lambda v_2 \end{cases} \implies (T_{11} - \lambda I)v_1 = -T_{12}v_2.$$

Si $\lambda \notin \text{Spec}(T_{11})$, tenim que $v_1 = -(T_{11} - \lambda I)^{-1}T_{12}v_2$. D'altra banda, si $\lambda \in \text{Spec}(T_{11})$, aleshores $\lambda \in \text{Spec}(T)$. Acabem de provar que $\text{Spec}(T_{11}) \cup \text{Spec}(T_{22}) \subset \text{Spec}(T)$.

2. Si v_1 és VEP de T_{11} i no de T_{22} , $Tv_1 = \lambda v_1$ pel que $Tv = \lambda v$ si $v = (v_1, 0)^T$. En canvi, si v_1 és VAP de T_{11} i de T_{22} , $Tv = \lambda v$, on $v = (v_1, v_2)^T$:

$$(T_{11} - \lambda I)v_1 = -T_{12}v_2 = \lambda v_2 \text{ i } T_{22}v_2 = \lambda v_2.$$

Ara, si v_1 és VAP de T_{22} i no de T_{11} , tenim $Tv_1 = \lambda v_1$ pel que $Tv = \lambda v$ si $v = (v_1, v_2)^T$:

$$v_1 = -(T_{11} - \lambda I)^{-1}T_{12}v_2 \text{ i } T_{22}v_2 = \lambda v_2. \quad \blacksquare$$

Exercici 2.8. If A denotes a n -by- n matrix, let us denote by $A_{k,\alpha}$ the matrix obtained by multiplying row k of A by α and dividing then column k by α (of course, $1 \leq k \leq n$, $\alpha \in \mathbb{R} \setminus \{0\}$).

1. Show that $\text{Spec}(A_{k,\alpha}) = \text{Spec}(A)$. Let us define:

$$A = \begin{pmatrix} 2.14 & -0.10 & 0.00 \\ -0.10 & 4.34 & 0.20 \\ 0.00 & 0.20 & 4.48 \end{pmatrix}.$$

2. Use Gerschgorin Theorem to give an approximation (with an error bound) to the smallest eigenvalue of A .
3. Choosing suitable values of k and α , use the first part to produce a better (that is, with smaller error bound) approximation to the smallest eigenvalue.

4. Discuss if it is possible to use the same technique as in the second part to improve the error bound of the remaining eigenvalues.

Demostració.

1. La matriu $A_{k,\alpha}$ seria de la forma:

$$A_{k,\alpha} = \begin{pmatrix} & & \frac{a_{1k}}{\alpha} & & \\ & & \vdots & & \\ \alpha a_{k1} & \cdots & a_{kk} & \cdots & \alpha a_{kn} \\ & & \vdots & & \\ & & \frac{a_{nk}}{\alpha} & & \end{pmatrix}.$$

Prenent $C_{k,\alpha}$ de la següent manera:

$$C_{k,\alpha} = \begin{pmatrix} c_{11} & & & & 0 \\ & \ddots & & & \\ & & c_{kk} & & \\ & & & \ddots & \\ 0 & & & & c_{nn} \end{pmatrix} = \begin{pmatrix} 1 & & & & 0 \\ & \ddots & & & \\ & & \alpha & & \\ & & & \ddots & \\ 0 & & & & 1 \end{pmatrix}.$$

Trobem que $A_{k,\alpha} = C_{k,\alpha} \cdot A \cdot C_{k,\alpha}^{-1}$; per tant, són matrius semblants i tenen els mateixos VAPs.

2. Aplicarem el teorema de Gerschgorin. $\text{Spec}(A) \subset D(2.14, 0.1) \cup D(4.34, 0.3) \cup D(4.48, 0.2)$. Com A és simètrica, els VAPs seran reals i $\lambda_{\min} = 2.14 \pm 0.1$ (i.e. $\lambda_{\min} \in [2.04, 2.24]$).
3. Siguin $k = 1$ i $\alpha \geq 1$, en virtut dels apartats anteriors. Tenim:

$$A_{1,\alpha} = \begin{pmatrix} 2.14 & -0.1\alpha & 0 \\ -\frac{0.1}{\alpha} & 4.34 & 0.2 \\ 0 & 0.2 & 4.48 \end{pmatrix} \implies A_{1,\alpha}^T = \begin{pmatrix} 2.14 & -\frac{0.1}{\alpha} & 0 \\ -0.1\alpha & 4.34 & 0.2 \\ 0 & 0.2 & 4.48 \end{pmatrix}.$$

Aplicant el teorema de Gerschgorin, $\lambda_1 \in D_1(2.14, \frac{0.1}{\alpha})$, $\lambda_2, \lambda_3 \in D(4.34, 0.2 + 0.1\alpha) \cup D(4.48, 0.2)$.

Per tant, $\lambda_1 \in [2.14 - \frac{0.1}{\alpha}, 2.14 + \frac{0.1}{\alpha}]$ i $\lambda_2, \lambda_3 \in [4.34 - 0.2 - 0.1\alpha, 4.34 + 0.2 + 0.1\alpha] \cup [4.28, 4.68]$.

A partir d'aquí, acotem α :

$$2.14 + \frac{0.1}{\alpha} < 4.34 - 0.2 - 0.1\alpha \iff \alpha^2 - 20\alpha + 1 < 0 \iff \alpha \in (10 - \sqrt{99}, 10 + \sqrt{99}).$$

Triant $\alpha = 19.9$, tenim $\lambda_1 = 2.14 \pm \frac{0.1}{19.9}$.

4. No es pot perquè són disjunts. ■

Exercici 2.9. Let A be a $n \times n$ matrix.

1. Assume that the value $\delta = \min_{i \neq j} \{|a_{ii} - a_{jj}|\}$ is different from zero and that there exist n real values λ_j ($j = 1, \dots, n$) such that $\|Ae_j - \lambda_j e_j\|_\infty < \varepsilon < \frac{\delta}{2n}$, where $\{e_1, \dots, e_n\}$ is the standard basis of $\mathbb{R}^n$.

- I.1. Does A have n different real eigenvalues?
- I.2. Bound the difference between the values $\lambda_1, \dots, \lambda_n$ and the eigenvalues of A .
2. Assume now that A is a symmetric matrix and that we (only) know that the first vector of the standard basis, e_1 , satisfies $\|Ae_1 - \lambda e_1\|_2 < \varepsilon$ for a given value λ . Show that, if ε is small enough, the matrix A has an eigenvalue near λ . Bound the distance between this eigenvalue and λ .

Demostració.

- I. Supposem $\|Ae_j - \lambda_j e_j\|_\infty < \varepsilon$.

$$Ae_j = \begin{pmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{pmatrix}, \quad Ae_j - \lambda_j e_j = \begin{pmatrix} a_{1j} \\ \vdots \\ a_{jj} - \lambda_j \\ a_{nj} \end{pmatrix}.$$

Aleshores, $|a_{ij}| < \varepsilon$ per a $i \neq j$ i $|a_{jj} - \lambda_j| < \varepsilon$. Pel teorema de Gerschgorin, $\text{Spec}(A) \in \bigcup_{i=1}^n D(a_{ii}, \sum_{j \neq i} |a_{ij}|) = \bigcup_{i=1}^n D(a_{ii}, r_i)$ i:

$$r_i < (n-1)\varepsilon < \frac{n-1}{2n} \delta < \frac{\delta}{2}.$$

Com $r_i < \frac{\delta}{2}$ i $\delta = \min_{i \neq j} |a_{ii} - a_{jj}|$, $D_k \cap D_l = \emptyset$, per a qualssevol k, l . Per tant, tenim n VAPs diferents. Pel que fa a la segona part d'aquest apartat, sigui μ_i VAP d' A , tal que $\mu_i \in D(a_{ii}, r_i)$. Aleshores, $|\mu - a_{ii}| < r_i < (n-1)\varepsilon$ i:

$$|\lambda_i - \mu_i| \leq |\lambda_i - a_{ii}| + |a_{ii} - \mu_i| < \varepsilon + (n-1)\varepsilon = n\varepsilon.$$

2. Supposem $\|Ae_1 - \lambda e_1\|_2 < \varepsilon$.

$$Ae_1 = \begin{pmatrix} a_{11} \\ \vdots \\ a_{n1} \end{pmatrix}, \quad Ae_1 - \lambda e_1 = \begin{pmatrix} a_{11} - \lambda \\ \vdots \\ a_{j1} \\ \vdots \\ a_{n1} \end{pmatrix} \implies \|Ae_1 - \lambda e_1\|_2 = \sqrt{(a_{11} - \lambda)^2 + (a_{21}^2 + \dots + a_{n1}^2)} < \varepsilon.$$

Per tant, $|a_{11} - \lambda|, |a_{i1}|, \|a_1\|_2 < \varepsilon$, on:

$$\left(\begin{array}{c|c} a_{11} & a_1^T \\ \hline a_1 & A_1 \end{array} \right).$$

Com A_1 és simètrica, existeix Q_1 ortogonal tal que $D_1 = Q_1^T A_1 Q_1$. Definim Q ortogonal i calculem

$$D = Q^T A Q:$$

$$Q = \left(\begin{array}{c|c} \mathbf{I} & \mathbf{0} \\ \hline \mathbf{0} & Q_1 \end{array} \right) \Rightarrow Q^T A = \left(\begin{array}{c|c} a_{11} & a_1^T \\ \hline Q_1^T a_1 & Q_1^T A_1 \end{array} \right) \Rightarrow Q^T A Q = \left(\begin{array}{c|c} a_{11} & a_1^T \\ \hline Q_1^T a_1 & Q_1^T A_1 Q_1 \end{array} \right) =: \left(\begin{array}{c|c} a_{11} & a_1^T \\ \hline b_1 & D_1 \end{array} \right).$$

. Calculem ara $\|b_1\|_2$ i $\|v\|_1$ i $\|b_1\|_1$ per aplicar el teorema de Gerschgorin sobre $Q^T A Q$.

$$\|b_1\|_2 = \|Q_1^T a_1\|_2 = \rho((Q_1^T a_1)^T Q_1^T a_1) = \rho(a_1^T Q_1 Q_1^T a_1) = \rho(a_1^T a_1) = \|a_1\|_2 < \varepsilon.$$

$$\|b_1\|_1^8 \leq \sqrt{n-1} \|b_1\|_2 < \sqrt{n-1} \varepsilon.$$

Obtenim que $\text{Spec}(Q^T A Q) \in D_1 \cup D_2 \cup \dots \cup D_n$. Cadascun d'aquests discs té com a centre a_{ii} i radi $\|b_i^{(i)}\|$. Ara, $a_{22} = \lambda_2, \dots, a_{nn} = \lambda_n$, però no té per què ser així per a $\lambda_1 = a_{11}$:

- Si $\lambda_j \neq a_{11}$, per a tot $1 \leq j \leq n$. Per tant, existeix ε tal que $D_1 \cap D_j = \emptyset$. És a dir, existeix $\mu \in D_1$ pel que $|\mu - a_{11}| < \sqrt{n-1} \varepsilon$ i:

$$|\lambda - \mu| \leq |\lambda - a_{11}| + |a_{11} - \mu| \leq (1 + \sqrt{n-1}) \varepsilon.$$

- Segui ara $\lambda_\ell = a_{11}$ per a algun $2 \leq \ell \leq n$, pel que $D_\ell \subset D_1$. Aleshores, $D_1 \cap D_j = \emptyset$ i $j \neq 1, \ell$. Per tant, D_1 conté almenys 1 VAP tal que $|\mu - \lambda| < (1 + \sqrt{n-1}) \varepsilon$.

■

Exercici 2.10. Consider the matrix:

$$A = \begin{pmatrix} 7 & 6 \\ 3 & 4 \end{pmatrix}.$$

1. Compute the QR factorization of A .
2. Use the QR iteration to compute the eigenvalues of A , with an error lower than 10^{-6} .

Demostració.

1. Segui $x = (7, 3)^T$ i $\|x\|_2 = \sqrt{7^2 + 3^2} = 7.615773$.

$$v = \frac{x + \|x\|_2 e_1}{x_1 + \|x\|_2} = \begin{pmatrix} 1 \\ 0.2052577 \end{pmatrix}.$$

Definim $P_1 = I - \frac{2vv^T}{v^T v}$, pel que $P_1 A = A - \frac{2}{v^T v} v v^T A = A + v w^T \iff w = \frac{2}{v^T v} A^T v = \beta A^T v$.

Aleshores:

$$\beta = -1.919145, \quad w = \begin{pmatrix} -14.615773 \\ -13.0905547 \end{pmatrix}.$$

Ens queda:

$$P_1 A = \begin{pmatrix} -7.6157731 & -7.0905477 \\ 0 & 1.3106432 \end{pmatrix}, \quad Q = P_1 \Rightarrow$$

⁸ Hem aplicat que $\|v\|_1 \leq \sqrt{n} \|v\|_2$, per a $v \in \mathbb{R}^n$.

2. Pel que fa als VAPs, $A_0 = A$, $A_{k+1} = R_k Q_k$ i $A_k = Q_k P_k$:

$$A_1 = \begin{pmatrix} 9.7931034 & -3.5172414 \\ -0.5172414 & 1.2068965 \end{pmatrix}, \quad A_7 = \begin{pmatrix} 9.9999998 & -3.0000000 \\ -5.39 \cdot 10^{-7} & 1.0000000 \end{pmatrix}.$$

I surten $\lambda_1 = 9.9999998 \pm 5.3999 \cdot 10^{-7}$ i $\lambda_2 = 1.0000000 \pm 5.3999 \cdot 10^{-7}$. ■

Exercici 2.11. Use the QR factorization to solve, in the least squares sense, the overdetermined system:

$$\begin{pmatrix} 2 & 3 \\ 1 & 3 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \\ 4 \end{pmatrix}.$$

Demostració. Sigui $x = (2, 1, 2)^T$. Clarament, $\|x\|_2 = \sqrt{2^2 + 1^2 + 2^2} = 3$,

$$v = \frac{x + \|x\|_2 e_1}{x_1 + \|x\|_2} = \frac{\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} + 3 \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}}{2 + 3} = \frac{1}{5} \begin{pmatrix} 5 \\ 1 \\ 2 \end{pmatrix}.$$

Un altre cop, $P_1 = I - \frac{2}{v^T v} v v^T$ i $\beta = -\frac{2}{v^T v} = -\frac{5}{30}$ i $w = \beta A^T v = -\begin{pmatrix} 5 \\ 8 \end{pmatrix}$. Per tant:

$$P_1 A = A + v w^T = \begin{pmatrix} -3 & -5 \\ 0 & \frac{7}{5} \\ 0 & -\frac{1}{5} \end{pmatrix}.$$

Ens quedem amb $2x = \frac{1}{5} \cdot \begin{pmatrix} 7 \\ -1 \end{pmatrix}$, $\|x\|_2 = \sqrt{2}$. Trobem v :

$$v = \frac{x + \|x\|_2 e_1}{\frac{7}{5} + \sqrt{2}} = \begin{pmatrix} 1 \\ -\frac{1}{\frac{7}{5} + \sqrt{2}} \end{pmatrix} \Rightarrow P_2 = \left(\begin{array}{c|cc} 1 & 0 & 0 \\ 0 & & \\ 0 & & \tilde{P}_2 \end{array} \right), \quad A_1 = \left(\begin{array}{c|c} -3 & -5 \\ 0 & \frac{7}{5} \\ 0 & -\frac{1}{5} \end{array} \right),$$

on $\tilde{P}_2 = I - \frac{2}{v^T v} v v^T$. Calculem $P_2 A_1$ per trobar R :

$$P_2 A_1 = \begin{pmatrix} -3 & -5 \\ 0 & -\sqrt{2} \\ 0 & 0 \end{pmatrix} = R$$

Per tant, $Q = P_1 \cdot P_2$ és ortogonal i:

$$QR \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \\ 4 \end{pmatrix}, \quad R \begin{pmatrix} x \\ y \end{pmatrix} = Q^{-1} \begin{pmatrix} 5 \\ 4 \\ 4 \end{pmatrix} = P_2 P_1 \begin{pmatrix} 5 \\ 4 \\ 4 \end{pmatrix},$$

i trobem $P_1 b$ i $P_2(P_1 b)$:

$$P_1 b = \begin{pmatrix} -\frac{22}{3} \\ \frac{23}{15} \\ -\frac{14}{15} \end{pmatrix}, \quad P_2(P_1 b) = \begin{pmatrix} -\frac{22}{3} \\ \frac{-7\sqrt{2}}{6} \\ -\frac{\sqrt{2}}{2} \end{pmatrix}.$$

Resolent el sistema:

$$\begin{pmatrix} -3 & -5 \\ 0 & -\sqrt{2} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -\frac{22}{3} \\ \frac{-7\sqrt{2}}{6} \\ -\frac{\sqrt{2}}{2} \end{pmatrix} \iff x = \frac{1}{2}, \quad y = \frac{7}{6}. \quad \blacksquare$$

Exercici 2.12. Let $\lambda_1 < \lambda_2 < \dots < \lambda_n$ be real numbers and $\varepsilon > 0$. We consider $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. We want to study the eigenvalues of the matrix $A_\varepsilon = D + \varepsilon B$ where $\text{diag}(B) = 0$ and $\|B\|_F = 1$, for ε small.

1. Prove that, if ε is small enough, for each $j \in \{1, \dots, n\}$ there exist $\lambda_\varepsilon^j \in \text{Spec}(A_\varepsilon)$ and a constant $\kappa_1 > 0$ such that $|\lambda_\varepsilon^j - \lambda_j| \leq \kappa_1 \varepsilon$.
2. Prove that, if ε is small enough, for each $j \in \{1, \dots, n\}$ there exist $\lambda_\varepsilon^j \in \text{Spec}(A_\varepsilon)$ and a constant $\kappa_2 > 0$ such that $|\lambda_\varepsilon^j - \lambda_j| \leq \kappa_2 \varepsilon^{3/2}$.
3. Is it possible to improve the bound of the previous item? That is, are there constants $\kappa_3 > 0$ and $\alpha > 3/2$ such that $|\lambda_\varepsilon^j - \lambda_j| \leq \kappa_3 \varepsilon^\alpha$?

Hint: Multiply or divide rows/columns by suitable values.

Demostració.

1. Pel teorema de Gerschgorin, existeix λ_ε^j VAP d' A_ε tal que $\lambda_\varepsilon^j \in D(\lambda_i, r_j)$, $r_j = \varepsilon \sum_{j \neq i} |b_{ij}|$. Per tant:

$$|\lambda_j - \lambda_\varepsilon^j| \leq r_j = \varepsilon \sum_{j \neq i} |b_{ij}| = \varepsilon \|b_j\|_1 \leq \varepsilon \sqrt{n-1} \|b_j\|_2 \leq \varepsilon \sqrt{n-1} \|B\|_F = \varepsilon \sqrt{n-1},$$

on $b_j = (b_{1j}, \dots, b_{nj}) \in \mathbb{R}^{n-1}$. Obtenim:

$$\lambda_j + \varepsilon \sqrt{n-1} < \lambda_i - \varepsilon \sqrt{n-1} \iff 2\varepsilon \sqrt{n-1} < \lambda_i - \lambda_j \iff \varepsilon < \frac{\lambda_i - \lambda_j}{2\sqrt{n-1}}.$$

2. A_ε és de la forma:

$$(A_\varepsilon)_{i,\sqrt{\varepsilon}} = \begin{pmatrix} & a_{ij}\sqrt{\varepsilon} & & \\ & \vdots & & \\ \frac{a_{ji}}{\sqrt{\varepsilon}} & \cdots & a_{jj} & \cdots & \frac{a_{jn}}{\sqrt{\varepsilon}} \\ & \vdots & & & \\ & a_{nj}\sqrt{\varepsilon} & & & \end{pmatrix}.$$

Tornem a aplicar el teorema de Gerschgorin: existeix $\lambda_\varepsilon^j \in D(a_{jj}, r_j) = D(\lambda_j, r_j)$, tal que $r_j = \varepsilon\sqrt{\varepsilon} \sum_{j \neq i} |b_{ij}|$:

$$|\lambda_\varepsilon - \lambda_j| \leq r_j = \varepsilon\sqrt{\varepsilon} \cdot \sqrt{n-1}$$

3. Si prenem $\beta < 1$ i considerem ε^β , aleshores $(A_\varepsilon)_{j,\varepsilon^\beta}$ ens dona $|\lambda_j - \lambda_\varepsilon^j| \leq \sqrt{n-1}\varepsilon^{1+\beta}$. ■

Exercici 2.13. Let A be a real $n \times n$ with SVD $A = U\Sigma V^T$. Suppose that $\|A^T A - I\|_2 = \varepsilon < 1$.

1. Prove that the singular values of A belong to the interval $[1 - \varepsilon, 1 + \varepsilon]$.
2. Prove that there exists an orthogonal matrix Q such that $\|A - Q\|_2 \leq \varepsilon$.
3. Prove that $1 - \varepsilon \leq \rho(A) \leq 1 + \varepsilon$.
4. We define the disks $D(q_{ii}, r_i)$, such that $r_i = \sum_{j=1, j \neq i}^n |q_{ij}|$, for $1 \leq i \leq n$, and $Q = (q_{ij})_{1 \leq i, j \leq n}$. Prove that if there exists j such that $D(q_{jj}, r_j) \cap D(q_{ii}, r_i) = \emptyset$, for all $i \neq j$, $1 \leq i \leq n$, and $\det A > 0$, then 1 is a simple eigenvalue of Q .

Demostració.

1. Per trobar $\|A^T A - I\|_2$, calculem primer $A^T A$:

$$A^T A = (U\Sigma V^T)^T \cdot (U\Sigma V^T) = V\Sigma U^T U\Sigma V^T = V\Sigma^2 V^T,$$

on $\Sigma^2 = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$.

$$\|A^T A - I\|_2 = \|V\Sigma^2 V^T - VV^T\|_2 = \|V(\Sigma^2 - I)V^T\|_2 = \|\Sigma^2 - I\|_2,$$

on hem aplicat que B és ortonormal pel que:

$$\|BA\|_2 = \rho((BA)^T BA) = \rho(A^T B^T BA) = \rho(A^T A) = \|A\|_2,$$

$$\|AB\|_2 = \sup_{\|x\|_2=1} \|ABx\|_2 = \sup_{\|y\|_2=1} \|Ay\|_2 = \|A\|_2.$$

Per tant, $\|A^T A - I\|_2 = \|\Sigma^2 - I\|_2 = \max(\sigma_i^2 - 1) < \varepsilon$ i:

$$(1 - \varepsilon)^2 \leq 1 - \varepsilon \leq \sigma_i^2 \leq 1 + \varepsilon \leq (1 + \varepsilon)^2 \iff \sigma_i \in [1 - \varepsilon, 1 + \varepsilon].$$

2. Sigui ara $Q = UV^T$. Usant l'apartat anterior, evidentment $\|A - Q\|_2 = \|\Sigma - I\|_2 \leq \varepsilon$.

3. Siguin $\lambda_1, \dots, \lambda_n$ els VAPs d' A . Aleshores:

$$|\lambda_1 \cdots \lambda_n| = |\det(A)| = \sqrt{\det(A^T A)} \iff (1 - \varepsilon)^n \leq \sigma_1 \cdots \sigma_n \leq \rho(A)^n \iff \rho(A) \geq 1 - \varepsilon.$$

Finalment:

$$\rho(A) \leq \|A\|_2 = \sqrt{\rho(A^T A)} \leq \sqrt{\max_i (\sigma_i)^2} < \sqrt{(1 + \varepsilon)^2} = 1 + \varepsilon. \quad \blacksquare$$

3

SISTEMES D'EQUACIONS NO LINEALS

Exercici 3.1. Consider $G : \mathbb{R}^n \longrightarrow \mathbb{R}^n$, $\alpha \in (0, 1)$, $r > 0$ and let $\|\cdot\|$ be a norm of $\mathbb{R}^n$, such that $G(0) = 0$ and:

$$\|G(x) - G(y)\| \leq \alpha \|x - y\| \quad \forall x, y \in B_r := \{z \in \mathbb{R}^n \mid \|z\| \leq r\}$$

1. Let $v \in \mathbb{R}^n$ be a vector. Prove that if $\|v\| \leq (1 - \alpha)r$ then the equation $z = G(z) + v$ has a unique solution in B_r .
2. Take $n = 2$, $\|\cdot\|_\infty$, $v \in \mathbb{R}^2$ and:

$$G \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} \sin(z_1 + z_2) \\ \cos(z_1 - z_2) - 1 \end{pmatrix}.$$

See that the equation $z = G(z) + v$ has a unique solution in $\mathbb{R}^2$.

3. If we take $z^0 = (0, 0)^T$ and $z^i = G(z^{i-1}) + v$, $i = 1, 2, \dots$; say, as a function of $\|v\|_\infty$, how many iterates we have to perform in order to have an approximation of the solution with an error less than 10^{-6} , using the $\|\cdot\|_\infty$ norm.

Demostració.

1. Com G és contractiva, definim $F(z) = G(z) + v$ (volem provar que és contractiva):

$$\|F(x) - F(y)\| = \|G(x) + v - G(y) - v\| = \|G(x) - G(y)\| \leq \alpha \|x - y\| \implies F \text{ és contractiva.}$$

Prenent $x^{(0)} = 0$, tenim que $x^{(1)} = F(x^{(0)}) = F(0) = G(0) + v = v$, pel que $\|x^{(1)} - x^{(0)}\| = \|v\| < (1 - \alpha)r$. Pel teorema del punt fix de Banach F té un únic punt fix $F(\bar{x}) = \bar{x}$ i

$$x^{(i)} = F(x^{(i-1)}), \quad x^{(i)} \rightarrow \bar{x} \text{ i } \|x^{(i)} - \bar{x}\| \leq \frac{\alpha^i}{1 - \alpha} \cdot \|x^{(1)} - x^{(0)}\| = \frac{\alpha^i}{1 - \alpha} \|v\|.$$

2. Usant que $G(0) = 0$,

$$DG = \frac{1}{3} \begin{pmatrix} \cos(z_1 + z_2) & \cos(z_1 + z_2) \\ -\sin(z_1 - z_2) & \sin(z_1 - z_2) \end{pmatrix} \implies \|DG\|_\infty = \frac{1}{3} \max\{2|\cos(z_1 + z_2)|, 2|\sin(z_1 - z_2)|\} \leq \frac{2}{3} < 1.$$

Per tant, podem aplicar el teorema del valor mitjà per dir que G és contractiva amb la constant $\alpha \leq \frac{2}{3}$. En aquestes condicions podem aplicar el primer apartat: existeix $\bar{x} \mid G(\bar{x}) + v = \bar{x}$ en $B_r := \{z \in \mathbb{R}^n \mid \|z\| \leq r\}$.

$$\|x^{(i)} - \bar{x}\|_\infty \leq \frac{\alpha^i}{1 - \alpha} \cdot \|v\|_\infty < \frac{\alpha^i}{1 - \alpha} (1 - \alpha)r = \alpha^i r.$$

Com que $B_r(0) \xrightarrow{r \rightarrow \infty} \mathbb{R}^2$ aleshores existeix una única solució en $\mathbb{R}^2$.

3. Sigui $z^0 = (0, 0)^T$ i $z^i = G(z^{i-1}) + v$. De la mateixa manera que a l'apartat anterior:

$$\begin{aligned} \|z^{(i)} - \bar{x}\| &\leq \frac{\alpha^i}{1 - \alpha} \|v\|_\infty \implies \|z^{(i)} - \bar{x}\| \leq \frac{\left(\frac{2}{3}\right)^i}{1 - \frac{2}{3}} \|v\|_\infty < 10^{-6} \\ \iff i \ln\left(\frac{2}{3}\right) + \ln(3\|v\|_\infty) &< -6 \ln(10) \iff i > \frac{-6 \ln 10 - \ln(3\|v\|_\infty)}{\ln \frac{2}{3}}. \end{aligned}$$

I ja hem acabat. ■

Exercici 3.2. Consider the equations:

$$\begin{cases} x^3 + y^2 + x + y - 2 = 0 \\ xy^2 + x^2 + x - y = 0 \end{cases}$$

that have a solution close to $(0.4, 0.8)$.

1. Compute this solution with an error smaller than 10^{-8} using the Newton method.
2. Compute this solution with a quasi-Newton method, using the Jacobian matrix only at the initial approximation $(0.4, 0.8)$.

Demostració.

1. El mètode de Newton ens diu que:

$$\begin{pmatrix} x_{k+1} \\ y_{k+1} \end{pmatrix} = \begin{pmatrix} x_k \\ y_k \end{pmatrix} - Df^{-1}(x_k, y_k) f(x_k, y_k).$$

Podem calcular la matriu diferencial d' f directament:

$$f = \begin{pmatrix} x^3 + y^2 + x + y - 2 \\ xy^2 + x^2 + x - y \end{pmatrix} \implies Df = \begin{pmatrix} 3x^2 + 1 & 2y + 1 \\ y^2 + 2x + 1 & 2xy - 1 \end{pmatrix}.$$

Partint de $(x^{(0)}, y^{(0)})^T = (0.4, 0.8)^T$, podem calcular el primer iterat:

$$\begin{pmatrix} x^{(1)} \\ y^{(1)} \end{pmatrix} = \begin{pmatrix} 0.4 \\ 0.8 \end{pmatrix} - \begin{pmatrix} 1 - 48 & 1.6 \\ 2.44 & 0.64 \end{pmatrix}^{-1} \cdot \begin{pmatrix} -1.696 \\ 0.016 \end{pmatrix} = \begin{pmatrix} 0.7758 \\ -0.6076 \end{pmatrix}.$$

El criteri de parada és $\|f(x^{(k)}, y^{(k)})\|_\infty < 10^{-8}$ o, equivalentment, anomenant $u^{(k)} = (x^{(k)}, y^{(k)})$,

$\|u^{(k+1)} - u^{(k)}\| < 10^{-8}$. En particular, tenim que $\|u^{(1)} - u^{(0)}\|_\infty = 1.4076$ i $\|f^{(1)}(u_0)\|_\infty = 2.2717$. Calculant iterativament, $(x^{(2)}, y^{(2)}), \dots$, trobem que $(x^{(4)}, y^{(4)}) = (3.987 \cdot 10^{-1}, 8.27 \cdot 10^{-1})$ ja compleix el criteri de parada.

2. Semblant, però avaluant $f^{-1}(x_0, y_0)$:

$$\begin{pmatrix} x^{(k+1)} \\ y^{(k+1)} \end{pmatrix} = \begin{pmatrix} x^{(k+1)} \\ y^{(k+1)} \end{pmatrix} - Df^{-1}(x_0, y_0)f(x_k, y_k).$$

Ens queda:

$$\begin{pmatrix} 0.4 \\ 0.8 \end{pmatrix} = \begin{pmatrix} x^{(0)} \\ y^{(0)} \end{pmatrix} \rightarrow \begin{pmatrix} x^{(1)} \\ y^{(1)} \end{pmatrix} \rightarrow \begin{pmatrix} x^{(2)} \\ y^{(2)} \end{pmatrix} \rightarrow \begin{pmatrix} x^{(3)} \\ y^{(3)} \end{pmatrix} \rightarrow \begin{pmatrix} x^{(4)} \\ y^{(4)} \end{pmatrix} \rightarrow \begin{pmatrix} x^{(5)} \\ y^{(5)} \end{pmatrix}. \quad \blacksquare$$

Exercici 3.3. Consider the system $Ax = b + \varepsilon f(x)$, where A is an $n \times n$ regular matrix such that $a_{ii} \neq 0$, $i = 1, \dots, n$, $b \in \mathbb{R}^n$, $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$, and $\varepsilon \in \mathbb{R}$.

1. Suppose that there exists a constant $K > 0$ such that $\|f(x) - f(y)\|_\infty \leq K\|x - y\|_\infty$, for all $x, y \in \mathbb{R}^n$. If we write $A = L + D + U$, where L is strictly lower triangular, D is diagonal and U is strictly upper triangular, we define the following iterative method to find the solution of $Ax = b + \varepsilon f(x)$:

$$x^{(k+1)} = D^{-1} \left[b + \varepsilon f(x^{(k)}) - (L + U)x^{(k)} \right], \quad k \geq 0.$$

1.1. Prove that if $(x^{(k)})_{k \geq 0} \rightarrow \alpha$ then $x = \alpha$ is the solution of $Ax = b + \varepsilon f(x)$.

1.2. If A is strictly diagonally dominant by rows, prove that there exists $\varepsilon_0 > 0$ such that if $|\varepsilon| < \varepsilon_0$, for any initial condition $x^{(0)} \in \mathbb{R}^n$, the sequence $(x^{(k)})_k$ has a limit and (1) has a unique solution.

Determine ε_0 as a function of K , $\beta = \max_{1 \leq i \leq n} \frac{1}{|a_{ii}|} \sum_{j=1, j \neq i}^n |a_{ij}|$ i $\gamma = \frac{1}{\min_{1 \leq i \leq n} |a_{ii}|}$.

2. Now we take:

$$A = \begin{pmatrix} 3 & 5 \\ 2 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad f(x) = \begin{pmatrix} \sin(x_1 + x_2) \\ \cos(x_2) \end{pmatrix},$$

where $x = (x_1, x_2)$.

2.1. Prove that if $|\varepsilon|$ is small enough, there exists a solution x of $Ax = b + \varepsilon f(x)$ close to the solution when $\varepsilon = 0$.

2.2. Find the Taylor expansion of the solution up to second order in ε .

Demostració.

1. Separem la demostració en els dos subapartats.

1.1. Suposem que si $k \rightarrow \infty$, tenim que $x^{(k)} \rightarrow \alpha$. Per aquesta raó:

$$\begin{aligned} \alpha &= D^{-1}(b + \varepsilon f(x^{(k)}) - (L + U)\alpha) \iff D\alpha = b + \varepsilon f(x^{(k)}) - (L + U)\alpha \\ &\iff (L + D + U)\alpha = b + \varepsilon f(x^{(k)}). \end{aligned}$$

1.2. Definim la funció d'iteració:

$$\Phi(x) = D^{-1}(b + \varepsilon f(x) - (L + U)x).$$

Com A és diagonalment dominant, $\beta = \|-1\| \cdot \|D(L + U)\|_\infty < 1$ i:

$$\begin{aligned} \|\Phi(x) - \Phi(y)\|_\infty &\leq |\varepsilon| \cdot \|D^{-1}(f(x) - f(y))\|_\infty + \|D^{-1}(L + U)(x - y)\|_\infty \\ &\leq |\varepsilon| \cdot \|D^{-1}\|_\infty \|f(x) - f(y)\|_\infty + \|D^{-1}(L + U)\|_\infty \cdot \|x - y\|_\infty. \end{aligned} \quad (3.1)$$

Usant que $\|D^{-1}\|_\infty = \alpha$ i $\beta = \|-D(L + U)\|_\infty$ i que f és contractiva:

$$(3.1) \leq |\varepsilon| \cdot \gamma \cdot K \|x - y\|_\infty + \beta \|x - y\|_\infty = (|\varepsilon| \cdot \gamma K + \beta) \cdot \|x - y\|_\infty < \|x - y\|_\infty,$$

ja que $|\varepsilon| \cdot \gamma K + \beta < 1$. Per tant, $|\varepsilon| < \frac{1-\beta}{\gamma K} = \varepsilon_0$.

2. Dividim els dos subapartats:

2.1. Si $\varepsilon = 0$, aleshores $x_1^0 = -1$ i $x_2^0 = 1$. Definim $F(x_1, x_2, \varepsilon) = Ax - b - \varepsilon f(x)$ i:

$$DF_{x,\varepsilon} = \begin{pmatrix} A & -\varepsilon f'(x) & -f(x) \end{pmatrix} \implies D_x F(-1, 1, 0) = A, \quad \det(A) \neq 0.$$

Apliquem el teorema de la funció implícita: existeix una funció derivable $x = x(\varepsilon)$ a prop de $\varepsilon = 0$ tal que $x_1(0) = -1$, $x_2(0) = 1$ i $F(x_1(\varepsilon), x_2(\varepsilon)) = 0$.

2.2. Trobem l'expansió de Taylor al voltant de la solució sense molta dificultat:

$$x_1(\varepsilon) = x_1(0) + \varepsilon x_1'(0) + \varepsilon^2 x_1''(0) + o(|\varepsilon|^3), \quad x_2(\varepsilon) = x_2(0) + \varepsilon x_2'(0) + \varepsilon^2 x_2''(0) + o(|\varepsilon|^3).$$

Ara, si $v^T = (x_1(\varepsilon), x_2(\varepsilon))$ calculem $Av = b + \varepsilon f(x)$:

$$\begin{cases} 3x_1(\varepsilon) + 5x_2(\varepsilon) = 2 + \varepsilon \sin(x_1(\varepsilon) + x_2(\varepsilon)) \\ 2x_1(\varepsilon) + 3x_2(\varepsilon) = 1 + \varepsilon \cos(x_2(\varepsilon)). \end{cases}$$

Derivant:

$$\begin{cases} 3x_1'(\varepsilon) + 5x_2'(\varepsilon) = \sin(x_1(\varepsilon) + x_2(\varepsilon)) + \varepsilon \cos(x_1(\varepsilon) + x_2(\varepsilon)) \cdot (x_1'(\varepsilon) + x_2'(\varepsilon)). \\ 2x_1'(\varepsilon) + 3x_2'(\varepsilon) = \cos(x_2(\varepsilon)) + \varepsilon(-\sin(x_2(\varepsilon))) \cdot x_2'(\varepsilon). \end{cases} \quad (3.2)$$

Si $\varepsilon = 0$:

$$\begin{cases} 3x_1'(0) + 5x_2'(0) = 1 \cdot 0 + 0 \\ 2x_1'(0) + 3x_2'(0) = 1 \cos 1 + 0 \end{cases} \iff x_1'(0) = 5 \cos 1, \quad x_2'(0) = -3 \cos 1.$$

Ara, derivem un altre cop (3.2):

$$\begin{aligned} 3x_1''(\varepsilon) + 5x_2''(\varepsilon) &= \cos(x_1(\varepsilon) + x_2(\varepsilon)) \cdot (x_1'(\varepsilon) + x_2'(\varepsilon)) + \cos(x_1(\varepsilon) + x_2(\varepsilon)) \cdot (x_1'(\varepsilon) + x_2'(\varepsilon)) \\ &\quad + \varepsilon(-\sin(x_1(\varepsilon) + x_2(\varepsilon))) \cdot (x_1'(\varepsilon) + x_2'(\varepsilon)) \\ &\quad + \cos(x_1(\varepsilon) + x_2(\varepsilon)) \cdot (x_1''(\varepsilon) + x_2''(\varepsilon)). \end{aligned}$$

De la mateixa manera:

$$\begin{aligned} 2x_1''(\varepsilon) + 3x_2''(\varepsilon) &= -\sin(x_2(\varepsilon)) \cdot x_2'(\varepsilon) \\ &\quad + (-\sin(x_2(\varepsilon)) \cdot x_2'(\varepsilon)) - \varepsilon(\cos(x_2(\varepsilon)) \cdot (x_2'(\varepsilon))^2 + \sin(x_2(\varepsilon)) \cdot x_2''(\varepsilon)). \end{aligned}$$

Per a $\varepsilon = 0$:

$$\begin{cases} 3x_1''(0) + 5x_2''(0) = 2 \cdot 2 \cos(1) \\ 2x_1''(0) + 3x_2''(0) = -2 \sin(1)(-3 \cos(1)) \end{cases} \iff \begin{cases} x_1''(0) = 30 \sin(1) \cos(1) - 12 \cos(1) \\ x_2''(0) = -18 \sin(1) \cos(1) + 8 \cos(1) \end{cases}$$

Aplicant Taylor:

$$\begin{cases} x_1(\varepsilon) = -1 + 5 \cos(1)\varepsilon + (30 \sin(1) \cos(1) - 12 \cos(1))\varepsilon^2 + o(|\varepsilon|^3), \\ x_2(\varepsilon) = -1 + 3 \cos(1)\varepsilon + (-18 \sin(1) \cos(1) + 8 \cos(1))\varepsilon^2 + o(|\varepsilon|^3). \end{cases} \quad \blacksquare$$

Exercici 3.4. Consider the equations:

$$\begin{cases} x^3 + y^2 + x + y = \lambda, \\ x y^2 + x^2 + x - y = 0. \end{cases}$$

Note that, for $\lambda = 0$, we have the solution $(x, y) = (0, 0)$.

1. Prove that there exists a value $\delta > 0$ and a function $\lambda \in (-\delta, \delta) \mapsto (\bar{x}(\lambda), \bar{y}(\lambda)) \in \mathbb{R}^2$ such that the values $(\bar{x}(\lambda), \bar{y}(\lambda), \lambda)$ solve the previous equation.
2. Compute $\bar{x}'(\lambda)$ and $\bar{y}'(\lambda)$ for $\lambda = 0$.
3. Use last part to compute an approximation to the solution of the equation for $\lambda = 0.01$.
4. Use the Newton method (you can use the same program as in exercise 2) to solve the equation (with an error smaller than 10^{-8}) using the result of third part as seed. Compare the number of Newton iterations with those needed if the seed is $x^{(0)} = y^{(0)} = 0$.

Demostració.

1. Volem aplicar el teorema de la funció implícita, així que definim $F(x, y)$ de manera que en trobem la

matriu diferencial $DF(x, y)$:

$$F(x, y) = \begin{pmatrix} x^3 + y^2 + x + y - \lambda \\ x y^2 + x^2 + x - y \end{pmatrix} \implies DF(x, y)|_{(0,0,0)} = \begin{pmatrix} 3x^2 + 1 & 2y + 1 \\ y^2 + 2x + 1 & 2xy - 1 \end{pmatrix}|_{(0,0,0)} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

pel que $|\det(DF(x, y)|_{(0,0,0)})| = 2 \neq 0$. Pel teorema de la funció implícita, existeix $(\bar{x}(\lambda), \bar{y}(\lambda))$ a prop de $\lambda = 0$ tal que $F(\bar{x}(\lambda), \bar{y}(\lambda), \lambda) = 0$, $\bar{x}(0) = 0$ i $\bar{y}(0) = 0$.

2. Derivem respecte a λ :

$$\begin{cases} 3\bar{x}(\lambda)^2 \cdot \bar{x}'(\lambda) + 2\bar{y}(\lambda)\bar{y}'(\lambda) + \bar{x}(\lambda) + \bar{y}'(\lambda) - 1 = 0 \\ (\bar{x}'(\lambda) \cdot \bar{y}^2(\lambda) + \bar{x} \cdot 2\bar{y}(\lambda)\bar{y}'(\lambda)) + 2\bar{x}(\lambda) \cdot \bar{x}'(\lambda) - \bar{y}'(\lambda) = 0. \end{cases}$$

Per a $\lambda = 0$:

$$\begin{cases} \bar{x}'(0) + \bar{y}'(0) - 1 = 0 \\ \bar{x}'(0) - \bar{y}'(0) = 0 \end{cases} \iff \bar{x}'(0) = \frac{1}{2} \text{ i } \bar{y}'(0) = \frac{1}{2}.$$

Ara podem fer l'expansió de Taylor per aproximar la solució:

$$\bar{x}(\lambda) = 0 + \frac{1}{2}\lambda + o(|\lambda|^2), \quad \bar{y}(\lambda) = 0 + \frac{1}{2}\lambda + o(|\lambda|^2).$$

Com prenem $\lambda = 0.01$, $\bar{x}(0.01) \approx 0.005$ i $\bar{y}(0.01) \approx 0.005$.

3. Usem el mètode de Newton per a resoldre l'equació amb un error més petit que 10^{-8} , usant el resultat de l'apartat anterior:

$$\begin{pmatrix} x^{(k+1)} \\ y^{(k+1)} \end{pmatrix} = \begin{pmatrix} x^{(k)} \\ y^{(k)} \end{pmatrix} - DF^{-1}(x^{(k)}, y^{(k)}) \cdot F(x^{(k)}, y^{(k)}).$$

Calculant la matriu diferencial $DF(x, y)$ i escrivint el mètode per a $x^{(0)} = y^{(0)} = 0.005$:

$$DF(x, y) = \begin{pmatrix} 3x^2 + 1 & 2y + 1 \\ y^2 + 2x + 1 & x - 1 \end{pmatrix}.$$

I en dos iterats trobem $(x^{(2)}, y^{(2)}) = (4.975 \cdot 10^{-3}, 4.999 \cdot 10^{-3})$. ■

Exercici 3.5. Consider the equation:

$$f(x, \lambda) = x^2 + \sin(x) + \lambda = 0,$$

with $x = \lambda = 0$ as a solution.

1. Use the Implicit Function Theorem to show that the equation $f(x, \lambda) = 0$ defines a function $\lambda \mapsto \bar{x}(\lambda)$ near $x = \lambda = 0$.
2. Compute the first and second derivatives of function $\bar{x}(\lambda)$ at $\lambda = 0$. Use them to compute an approxima-

tion of the solution for $\lambda = 0.01$.

3. Find values (x^*, λ^*) satisfying $f(x^*, \lambda^*) = 0$ and such that the Implicit Function Theorem cannot be applied at (x^*, λ^*) . Describe the solutions of $f(x, \lambda) = 0$ near (x^*, λ^*) and discuss if the function $\lambda \mapsto \bar{x}(\lambda)$ defined in first part exists.
4. Give a value $\lambda_0 > 0$ such that, if $|\lambda| < \lambda_0$ then there exists $x(\lambda)$ such that $f(x(\lambda), \lambda) = 0$ and $x(\lambda) \rightarrow 0$ when $\lambda \rightarrow 0$.

Demostració.

1. Calculem $\frac{\partial f}{\partial x}|_{(0,0)} = 2x + \cos x = 1 \neq 0$, pel que podem aplicar el teorema de la funció implícita i existeix $\bar{x}(\lambda)$ a prop de $\lambda = 0$ tal que $\bar{x}(0) = 0$ i $f(\bar{x}(\lambda), \lambda) = 0$.
2. Calculem les dues primeres derivades de la funció $\bar{x}(\lambda)$ per a trobar una aproximació de la solució per a $\lambda = 0.01$:

$$\frac{\partial f}{\partial x} \cdot \frac{\partial \bar{x}(\lambda)}{\partial \lambda} + \frac{\partial f}{\partial \lambda} = 2x + \cos x + 1 = 0 \xrightarrow{\lambda=0} 1 \cdot \frac{\partial \bar{x}(0)}{\partial \lambda} + 1 = 0 \iff \frac{\partial \bar{x}(0)}{\partial \lambda} = -1.$$

Pel que fa a la segona derivada:

$$2(x')^2 + 2x \cdot x'' - \sin(x) \cdot (x')^2 + \cos(x) \cdot x'' = 0 \xrightarrow{\lambda=0} 2 + x''(0) = 0 \iff x''(0) = -2.$$

Per tant, $x(\lambda) \approx -\lambda - \lambda^2$ i $x(0.01) \approx -0.0101$.

3. Hem de trobar (x^*, λ^*) de manera que $f(x^*, \lambda^*) = 0$ i $\partial_x f(x^*, \lambda^*) = 0$; és a dir,

$$\begin{cases} (x^*)^2 + \sin x^* + \lambda^* = 0 \\ g(x^*) = 2x^* + \cos x^* = 0 \implies g'(x^*) = 2 - \sin x^*. \end{cases} \xrightarrow{\text{Newton}} x_{n+1} = x_n - \frac{2x_n + \cos x_n}{2 - \sin x_n}$$

de manera que $x^* \approx -0.45018361129$ i $\lambda^* \approx -(x^*)^2 - \sin(x^*) \approx 0.23246557516$. I no hi ha més solucions perquè $2x + \cos x$ és creixent. Com $\partial_\lambda f(x^*, \lambda^*) = 1 \neq 0$, podem aplicar el teorema de la funció implícita de manera que existeix $\lambda(x) \mid \lambda(x^*) = \lambda^*$ i $f(x, \lambda(x)) = 0$, x en un entorn de x^* . A més, $\lambda'(x^*) = 0$, ja que derivant:

$$\lambda'(x) = \frac{\partial f}{\partial x}(x^*, \lambda^*) + \frac{\partial f}{\partial \lambda}(x^*, \lambda^*) \frac{\partial \lambda}{\partial x}(x^*) = 0 \implies \lambda'(x^*) = 0.$$

Trobem, a més, que x^* és un màxim de λ , ja que:

$$\lambda''(x^*) = 2 + \sin(x^*) + \lambda''(x^*) = 0 \implies \lambda''(x^*) = \sin(x^*) - 2 < 0.$$

Per tant, si $\lambda > \lambda^*$ no hi ha solució per x ; si $\lambda < \lambda^*$ hi ha dues solucions de manera que $\bar{x}(\lambda)$ no existeix.

4. Teorema de Newton-Kantorovich. Fixem λ , de manera que $f(x, \lambda) = f(x) = x^2 + \sin x + \lambda$ és una

funció d'una variable. Prenem $x_0 = 0$ i tenim:

$$\frac{\partial f}{\partial x}(x_0) = 2x_0 + \cos x_0 = 1 \iff a = \frac{1}{f'(x_0)} = 1.$$

Així doncs:

$$\|x^{(1)} - x^{(0)}\| = \|x^{(1)} - x_0\| = \left| \frac{f(x_0)}{f'(x_0)} \right| = \frac{|\lambda|}{1} = b \text{ i } \left| \frac{\partial^2 f}{\partial x^2}(x) \right| = |2 - \sin x| \leq 3 = c.$$

Per tant, si $|\lambda| \leq \frac{1}{6} = \lambda_0$ obtenim $b = abc = 3|\lambda| \leq \frac{1}{2}$. Estem en condicions d'aplicar el teorema de Newton-Kantorovich: existeix un únic $\bar{x}$ tal que $f(\bar{x}) = 0$ i $\bar{x} = \bar{x}(\lambda) \in [x_0 - 2b, x_0 + 2b] \subset [-2|\lambda|, 2|\lambda|]$. És clar, doncs, que $\bar{x} \xrightarrow{|\lambda| \rightarrow 0} 0$, com volíem. ■

Exercici 3.6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be two C^∞ functions such that $f(1) = g(1) = 1$, $f(3) = g(3) = 3$. Consider the system of equations:

$$\begin{cases} x^2 + xy + y^2 &= f(\lambda) \\ x + y + 1 &= g(\lambda) \end{cases}$$

1. We see that for $\lambda = 1$, the system has the solution $(x, y) = (1, -1)$. Use the Implicit Function Theorem to prove that, for λ close to 1, the system defines a function $\lambda \mapsto (\bar{x}(\lambda), \bar{y}(\lambda))$ in a neighbourhood of $\lambda = 1$. Compute $(\bar{x}'(1), \bar{y}'(1))$ (as a function of derivatives of f and g evaluated at $\lambda = 1$).
2. Now suppose that $f'(3) = -2g'(3) < 0$. For $\lambda = 3$ the system has the solution $(x, y) = (1, 1)$. If $\delta > 0$ is small enough, use the IFT to say which is the correct answer:
 - There exists a solution close to $x = y = 1$ for all λ such that $|\lambda - 3| < \delta$.
 - There exists a solution close to $x = y = 1$ when $3 - \delta < \lambda \leq 3$.
 - There exists a solution close to $x = y = 1$ when $3 \leq \lambda < 3 + \delta$.

Demostració.

1. Definim $F(x, y, \lambda)$ de la següent manera:

$$F(x, y, \lambda) = \begin{pmatrix} x^2 + y^2 + xy - f(\lambda) \\ x + y + 1 - g(\lambda) \end{pmatrix}.$$

Volem aplicar el teorema de la funció implícita en un entorn de la solució per a $\lambda = 1$, $(x, y) = (1, -1)$. Calculem $DF|_{(1, -1, 1)}$ i comprovem que $\det(DF|_{(1, -1, 1)}) \neq 0$:

$$DF(x, y)|_{(1, -1, 1)} = \begin{pmatrix} 2x + y & 2y + x & -f'(\lambda) \\ 1 & 1 & -g'(\lambda) \end{pmatrix}_{|(1, -1, 1)} = \begin{pmatrix} 1 & -1 & -f'(1) \\ 1 & 1 & -g'(1) \end{pmatrix}.$$

És clar, doncs, que:

$$\det(DF(x, y)|_{(1, -1, 1)}) = \det \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = 2 \neq 0.$$

Pel teorema de la funció inversa, existeixen $\bar{x} = \bar{x}(\lambda)$, $\bar{y} = \bar{y}(\lambda)$ tals que $\bar{x}(1) = 1$, $\bar{y}(1) = -1$ i $F(\bar{x}(\lambda), \bar{y}(\lambda), \lambda) = 0$. Si derivem:

$$\begin{cases} 2\bar{x} \cdot \bar{x}' + 2\bar{y} \cdot \bar{y}' + \bar{x}' \cdot \bar{y} + \bar{x} \cdot \bar{y}' - f'(\lambda) = 0 \\ \bar{x}' + \bar{y}' - g'(\lambda) = 0. \end{cases} \xrightarrow{\lambda=1} \begin{cases} \bar{x}'(1) - \bar{y}'(1) = f'(1) \\ \bar{x}'(1) + \bar{y}'(1) = g'(1) \end{cases}$$

Concloem que $\bar{x}'(1) = \frac{f'(1)+g'(1)}{2}$ i $\bar{y}'(1) = \frac{-f'(1)+g'(1)}{2}$.

2. Apliquem el mateix procediment, ara sobre $\lambda = 3$:

$$DF(x, y)|_{(1, -1, 3)} = \begin{pmatrix} 2x + y & 2y + x & -f'(\lambda) \\ 1 & 1 & -g'(\lambda) \end{pmatrix}_{|(1, -1, 3)} = \begin{pmatrix} 1 & -1 & -f'(3) \\ 1 & 1 & -g'(3) \end{pmatrix}.$$

És clar, doncs, que:

$$\det(DF(y, \lambda)|_{(1, -1, 3)}) = \det \begin{pmatrix} 3 & -f'(3) \\ 1 & -g'(3) \end{pmatrix} = f'(3) - 3g'(3) = -5g'(3) < 0.$$

Pel teorema de la funció inversa, existeixen $\bar{y} = \bar{y}(x)$, $\bar{\lambda} = \bar{\lambda}(x)$ tals que x és propera a 1, $\bar{y}(1) = 1$, $\bar{\lambda}(1) = 3$ i $F(x, \bar{y}(x), \bar{\lambda}(x)) = 0$. El sistema queda:

$$\begin{cases} 2x + \bar{y} + x\bar{y}' + 2\bar{y}\bar{y}' = f'(\bar{\lambda}) \cdot \bar{\lambda}' \\ 1 + \bar{y}' = g'(\bar{\lambda}) \cdot \bar{\lambda}' \end{cases} \xrightarrow{x=1} \begin{cases} 3\bar{y}'(1) + 3 = f'(3) \cdot \bar{\lambda}'(1) \\ \bar{y}'(1) + 1 = g'(3) \cdot \bar{\lambda}'(1) \end{cases} \iff \begin{cases} \bar{y}'(1) = \frac{3g'(3)-f'(3)}{f'(3)-3g'(3)} = 1 \\ \bar{\lambda}'(1) = 0. \end{cases}$$

Derivem i prenem $x = 1$:

$$\begin{cases} 2 + 3\bar{y}''(1) = f'(3)\bar{\lambda}''(1) \\ \bar{y}''(1) = g'(3)\bar{\lambda}''(1) \end{cases} \implies \bar{\lambda}''(1) = -\frac{2}{5g'(3)} < 0,$$

pel que $\bar{\lambda}(x)$ té un màxim local per $x = 1$ (i.e. $\lambda \in (3 - \delta, 3]$). ■

Exercici 3.7. Consider the system of equations:

$$\begin{cases} x + \mu \cos(x + y) = \mu^2, \\ x + \frac{1}{2}\mu y + \mu x^2 + y^2 = 0 \end{cases}$$

1. If $\mu = 1$ the system has the solution $(x, y) = 0$. Prove that for μ close to 1, the equations define a function $(\bar{x}(\mu), \bar{y}(\mu))$ in a suitable neighbourhood of $\mu = 1$. Compute $(\bar{x}'(1), \bar{y}'(1))$ and $(\bar{x}''(1), \bar{y}''(1))$.

2. For $\mu = 0$ we also have a solution at $(x, y) = (0, 0)$. If $\delta > 0$ is small enough, use the Implicit Function Theorem to find which option is correct:

- There exists a solution near $x = y = 0$, for all μ such that $|\mu| < \delta$.
- There exists a solution near $x = y = 0$ when $-\delta < \mu \leq 0$.
- There exists a solution near $x = y = 0$ when $0 \leq \mu < \delta$.

Demostració.

1. Sigui:

$$F(x, y, \mu) = \begin{pmatrix} x + \mu \cos(x + y) - \mu^2 \\ x + \frac{1}{2}\mu x y + \mu x + y^2 \end{pmatrix} \implies DF(0, 0, 1) = \begin{pmatrix} 1 & 0 & -1 \\ 1 & \frac{1}{2} & 0 \end{pmatrix},$$

i si prenem $DF_{x,y}$, és clar que $\det(DF(x, y)|_{(0,0,1)}) \neq 0$. Pel teorema de la funció implícita, existeixen $\bar{x} = \bar{x}(\mu)$ i $\bar{y} = \bar{y}(\mu)$ tals que $\bar{x}(1) = 0$, $\bar{y}(1) = 0$ i $F(\bar{x}(\mu), \bar{y}(\mu), \mu) = 0$. Derivem respecte μ i substituïm $\mu = 1$:

$$\bar{x}' + \cos(\bar{x} + \bar{y}) - \mu \sin(\bar{x} + \bar{y}) \cdot (\bar{x}' + \bar{y}') - 2\mu = 0 \xrightarrow{\mu=1} \begin{cases} \bar{x}'(1) - 1 = 0 \\ \bar{x}'(1) + \frac{1}{2}\bar{y}'(1) = -2 \end{cases}$$

Tornant a derivar, obtenim:

$$\begin{aligned} \bar{x}'' - \sin(\bar{x} + \bar{y})(\bar{x}' + \bar{y}') - \sin(\bar{x} + \bar{y})(\bar{x}' + \bar{y}') - \mu \cos(\bar{x} + \bar{y})^2 - \mu \sin(\bar{x} + \bar{y})(\bar{x}'' + \bar{y}'') - 2 = 0 \\ \iff \bar{x}'' + \frac{1}{2}\bar{y}' + \frac{1}{2}\bar{y}' + 2\bar{x}\bar{x}' + 2\mu(\bar{x})^2 + 2\mu\bar{x}\bar{x}'' + 2(\bar{y})^2 + 2\bar{y}\bar{y}'' = 0. \end{aligned}$$

Fent $\mu = 1$, obtenim:

$$\begin{cases} \bar{x}''(1) - (1 - 2)^2 - 2 = 0 \\ \bar{x}''(1) + \frac{1}{2}(2 \cdot (-2)) + 2 + \frac{1}{2}\bar{y}''(1) + 2(-2)^2 = 0 \end{cases} \iff \begin{cases} \bar{x}''(1) = 3 \\ \bar{y}''(1) = -22. \end{cases}$$

2. Podem calcular el jacobià:

$$DF(x, y, \mu)|_{(0,0,0)} = \begin{pmatrix} 1 - \mu \sin(x + y) & -\mu \sin(x + y) & \cos(x + y) - 2\mu \\ 1 + 2\mu x & \frac{1}{2}\mu + 2y & \frac{1}{2}y + x^2 \end{pmatrix}|_{(0,0,0)} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Com $\text{rang}(DF|_{(0,0,0)}) = 2$, ja que $\det(DF_{x,\mu}(0, 0, 0)) = 1 \neq 0$, podem aplicar el teorema de la funció implícita. Existeixen $x = x(y)$ i $\mu = \mu(y)$ tals que y és a prop de zero i $x(0) = \mu(0) = 0$ i $F(x(y), y, \mu(y)) = 0$. Derivant un cop:

$$\begin{cases} x' + \mu' \cos(x + y) - \mu \sin(x + y)(x' + 1) - 2\mu\mu' = 0, \\ x' + \frac{1}{2}\mu' y + \frac{1}{2}\mu + \mu' x^2 + 2\mu \cdot x x' + 2y = 0. \end{cases} \xrightarrow{(x,y,\mu)=(0,0,0)} \begin{cases} x'(0) + \mu'(0) = 0 \\ x'(0) = 0 \end{cases}$$

Tornant a derivar, comprovem les característiques dels extrems que hem trobat.

$$\begin{cases} x'' + \mu'' \cos(x + y) - \mu \sin(x + y)(x' + 1) \\ - ((\mu' \sin(x + y) + \mu \cos(x + y)(x' + 1)) \cdot (x' + 1) + \mu \sin(x + y)x'') - 2(\mu')^2 - 2\mu\mu'' = 0 \\ x'' + \frac{1}{2}\mu''y + \frac{1}{2}\mu' + \frac{1}{2}\mu' + \mu''x^2 + 2\mu'xx' + 2\mu'xx' + 2\mu'xx' + 2\mu'(x')^2 + xx'' + 2 = 0 \end{cases}$$

Avaluant en $(0, 0, 0)$ i usant $x'(0) = \mu'(0) = 0$, obtenim que $x''(0) + \mu''(0) = 0$ i $x''(0) + 2 = 0$; per tant, $x''(0) = -2 < 0$ i $\mu''(0) = 2 > 0$. Així, μ té un mínim en $y = 0$ i x té un màxim en $y = 0$. ■

Exercici 3.8. Consider the system of equations:

$$\begin{cases} x^4 + ax^2 + bx + c = 0, \\ 4x^3 + 2ax + b + 2 = 0, \\ 36x^4 + 12ax^2 + 4x + a^2 = 0. \end{cases}$$

1. It is immediate to see that the system has the solution $(x, a, b, c) = (0, 0, -2, 0)$. Prove that for a close to zero, the equation defines infinitely differentiable functions $x(a), b(a), c(a)$, such that $x(0) = c(0) = 0$ and $b(0) = -2$.
2. Compute the Taylor expansion of $x(a), b(a), c(a)$ up to second order about $a = 0$.
3. Which of the functions $x(a), b(a)$, and $c(a)$ do have a maximum or a minimum at $a = 0$?

Demostració.

1. Definim F i calculem el jacobià DF al voltant de la solució:

$$F(x, a, b, c) = \begin{pmatrix} x^4 + ax^2 + bx + c \\ 4x^3 + 2ax + b + 2 \\ 36x^4 + 12ax^2 + 4x + a^2 \end{pmatrix} \Rightarrow DF|_{(0,0,-2,0)} = \begin{pmatrix} 4x^3 + 2ax + b & x^2 & x & 1 \\ 12x^2 + 2a & 2x & 1 & 0 \\ 144x^3 + 24ax + 4 & 12x^2 + 2a & 0 & 0 \end{pmatrix}_{|(0,0,-2,0)},$$

pel que:

$$DF|_{(0,0,-2,0)} = \begin{pmatrix} -2 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 4 & 0 & 0 & 0 \end{pmatrix} \text{ i } \det(DF_{x,b,c}(0,0,-2,0)) = \begin{vmatrix} -2 & 0 & 1 \\ 0 & 1 & 0 \\ 4 & 0 & 0 \end{vmatrix} \neq 0.$$

Pel teorema de la funció inversa existeix $\delta > 0$ prou petit tal que $x = x(a), b = b(a)$ i $c = c(a)$ són funcions $[-\delta, \delta] \rightarrow \mathbb{R}$ i tals que $x(0) = 0, b(0) = -2, c(0) = 0$ i $F(x(a), a, b(a), c(a)) = 0$.

2. Per trobar l'expressió de Taylor a segon ordre en $a = 0$ haurem de trobar $\varphi'(0), \varphi''(0), \varphi \in \{x, b, c\}$.

$$\begin{cases} 4x^3x' + x^2 + a2xx' + b'x + bx' + c' = 0 \\ 12x^2x' + 2x + 2ax' + b' = 0 \\ 144x^3x' + 12x^2 + 24a \cdot xx' + 4x' + 2a = 0 \end{cases} \xrightarrow{x(0)=0, b(0)=-2, c(0)=0} \begin{cases} -2x'(0) + c'(0) = 0 \\ b'(0) = 0 \\ 4x'(0) = 0 \end{cases}$$

Per tant, $x'(0) = b'(0) = c'(0) = 0$. Derivant novament:

$$\begin{cases} 12x^2(x')^2 + 4x^3x'' + 2xx' + 2xx' + 2a((x')^2 + xx'') + b''x + b'x' + b'x' + bx'' + c'' = 0 \\ 12 \cdot 2x(x')^2 + 12x^2x'' + 2x' + 2ax'' + b'' = 0 \\ 144 \cdot 3x^2 \cdot (x')^2 + 144x^3x'' + 12 \cdot 2xx' + 24xx' + 24a((x')^2 + xx'') + 4x'' + 2 = 0. \end{cases}$$

Substituint $a = 0$, $x(0) = c(0) = 0$ i $b(0) = -2$, $x'(0) = b'(0) = c'(0) = 0$:

$$\begin{cases} -2x''(0) + c''(0) = 0 \\ b''(0) = 0 \\ 4x''(0) + 2 = 0 \end{cases} \iff \begin{cases} x''(0) = -\frac{1}{2} \\ b''(0) = 0 \\ c''(0) = -1 \end{cases} \iff \begin{cases} x(a) = \frac{1}{2} \left(-\frac{1}{2}\right) a^2 + o(|a|^3), \\ b(a) = -2 + o(|a|^3), \\ c(a) = -\frac{1}{2}a^2 + o(|a|^3). \end{cases}$$

3. Molt senzill al disposar ja de les derivades en $a = 0$ calculades. Tenim $x'(0) = b'(0) = c'(0) = 0$ i $x''(0), c''(0) < 0$, pel que en ambdós casos (x i c) $a = 0$ és un màxim local. Però $b''(0) = 0$, pel que caldria tornar a derivar per afirmar que b té un punt d'inflexió en $a = 0$.

$$\begin{cases} 2x'''(0) + c'''(0) = 0 \\ 6x''(0) + b'''(0) = 0 \\ 4x'''(0) = 0 \end{cases} \implies b'''(0) = 3 > 0. \quad \blacksquare$$

Exercici 3.9. The equations

$$\begin{cases} x^2 + y^2 = 1 \\ x^2 + z^2 = 1 \end{cases}$$

have the solution $(x, y, z) = (0, 1, 1)$.

1. Is there a curve of solutions going through the point $(x, y, z) = (0, 1, 1)$?
2. If so, study if some of the variables (x , y and z) have a maximum/minimum when going through $(x, y, z) = (0, 1, 1)$.

Demostració.

1. Estudiem F i DF al voltant de $(0, 1, 1)$.

$$F(x, y, z) = \begin{pmatrix} x^2 + y^2 - 1 \\ x^2 + z^2 - 1 \end{pmatrix} \implies DF(x, y, z)|_{(0,1,1)} = \begin{pmatrix} 2x & 2y & 0 \\ 2x & 0 & 2z \end{pmatrix}|_{(0,1,1)} = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Com $\det(D_{y,z}(0, 1, 1)) \neq 0$, podem aplicar el teorema de la funció implícita. En efecte, existeixen $y = y(x)$ i $z = z(x)$, $y, z : [-\delta, \delta] \rightarrow \mathbb{R}$ (per a δ prou petit), tals que $y(0) = z(0) = 1$ i $F(x, y(x), z(x)) = 0$.

2. Derivem per trobar si alguna de les variables té algun màxim o mínim a prop de $(x, y, z) = (0, 1, 1)$:

$$\begin{cases} 2x + 2y y' = 0 \\ 2x + 2z z' = 0 \end{cases} \xrightarrow{x=0, y(0)=z(0)=1} \begin{cases} 2y'(0) = 0 \iff y'(0) = 0 \\ 2z'(0) = 0 \iff z'(0) = 0 \end{cases}$$

Tornant a derivar:

$$\begin{cases} 2 + 2(y')^2 + 2y y'' = 0 \\ 2 + 2(z')^2 + 2z z'' = 0 \end{cases} \xrightarrow[y'(0)=z'(0)=0]{x=0, y(0)=z(0)=1} y''(0) = z''(0) = -1 < 0.$$

Per tant, y, z tenen un màxim en comú en $x = 0$. ■

Exercici 3.10. The equations

$$x^2 + y^2 = 2\lambda$$

$$x^2 + z^2 = 2\lambda$$

$$y^2 + z^2 = 2$$

have, for $\lambda = 1$, the solution $(x, y, z) = (1, 1, 1)$. Discuss the existence of solutions for λ close to 1. If there exist a curve of solutions, compute its Taylor polynomial (with respect to λ) of degree 2.

Demostració. Definim F i calculem el jacobià DF al voltant de la solució:

$$F(x, y, z, \lambda) = \begin{pmatrix} x^2 + y^2 - 2\lambda \\ x^2 + z^2 - 2\lambda \\ y^2 + z^2 - 2 \end{pmatrix} \implies DF|_{(1,1,1)} = \begin{pmatrix} 2x & 2y & 0 & -2 \\ 2x & 0 & 2z & -2 \\ 0 & 2y & 2z & 0 \end{pmatrix}_{|(1,1,1)} = \begin{pmatrix} 2 & 2 & 0 & -2 \\ 2 & 0 & 2 & -2 \\ 0 & 2 & 2 & 0 \end{pmatrix}.$$

Com $\det_{x,y,z} DF(1, 1, 1, 1) \neq 0$, podem aplicar el teorema de la funció inversa. Existeixen $x = x(\lambda)$, $y = y(\lambda)$ i $z = z(\lambda)$ a prop de $\lambda = 1$ tals que $x(1) = y(1) = z(1) = 1$ i $F(x(\lambda), y(\lambda), z(\lambda), \lambda) = 0$. Derivant:

$$\begin{cases} 2xx' + 2yy' = 2 \\ 2xx' + 2zz' = 2 \\ 2yy' + 2zz' = 0 \end{cases} \xrightarrow{x(1)=y(1)=z(1)=1} \begin{cases} x'(1) = 1 \\ y'(1) = 0 \\ z'(1) = 0 \end{cases}$$

Tornant a derivar, per a $\lambda = 1$:

$$\begin{cases} 2 + 2x'' + 2y'' = 0 \\ 2 + 2x'' + 2z'' = 0 \\ 2y'' + 2z'' = 0 \end{cases} \xrightarrow[x'(1)=1, y'(1)=z'(1)=0]{x(1)=y(1)=z(1)=1} \begin{cases} x''(1) = -1 \\ y''(1) = 0 \\ z''(1) = 0 \end{cases}$$

Per tant, els polinomis de Taylor són els següents:

$$\begin{cases} x(\lambda) = 1 + (\lambda - 1) - \frac{1}{2}(\lambda - 1)^2 + o(|\lambda - 1|^3) \\ y(\lambda) = 1 + o(|\lambda - 1|^3) \\ z(\lambda) = 1 + o(|\lambda - 1|^3) \end{cases} \quad \blacksquare$$

Exercici 3.11. Assume that $f, g : \mathbb{R}^3 \rightarrow \mathbb{R}$ are C^1 functions, and consider the equations $f(x, y, z) = 0$, $g(x, y, z) = 0$.

1. Assume that there exists a point (x_0, y_0, z_0) such that $f(x_0, y_0, z_0) = g(x_0, y_0, z_0) = 0$ and that the vectors $u = \nabla f(x_0, y_0, z_0)$ and $v = \nabla g(x_0, y_0, z_0)$ are different from zero. Moreover, if u and v are not parallel, do the surfaces $f = 0$ and $g = 0$ intersect on a curve?
2. Consider:

$$\begin{cases} x^4 + y^4 - 2 = 0 \\ xz + y - 2 = 0 \end{cases}$$

and note that $(x, y, z) = (1, 1, 1)$ satisfies both equations. Study the existence of a curve of solutions near $(x, y, z) = (1, 1, 1)$. Discuss if one of the variables goes through a maximum or minimum at $(x, y, z) = (1, 1, 1)$ when we move along this curve.

Demostració.

1. Si $u \nparallel v$, aleshores $\text{rank}(u, v) = 2$. Per tant, existeix un menor $M_{2 \times 2}$ tals que $\det M \neq 0$:

$$F(x, y, z) = \begin{pmatrix} f(x, y, z) \\ g(x, y, z) \end{pmatrix} \implies DF|_{(x_0, y_0, z_0)} = \begin{pmatrix} \nabla f(x, y, z) \\ \nabla g(x, y, z) \end{pmatrix}_{|(x_0, y_0, z_0)} = (u, v)^T.$$

Si M correspon a y, z , pel teorema de la funció implícita sobre (x_0, y_0, z_0) existeix $y = y(x)$, $z = z(x)$ tal que $y(x_0) = y_0$ i $z(x_0) = z_0$, $f(x, y(x), z(x)) = 0$. Procedim de la mateixa manera si M correspon a x, y o x, z .

2. Siguin $f(x, y, z) = x^4 + y^4 - 2$ i $g(x, y, z) = xz + y - 2$.

$$u = \nabla f|_{(1,1,1)} = (4x^3, 4y^3, 0)|_{(1,1,1)} = (4, 4, 0), \quad v = \nabla g|_{(1,1,1)} = (z, 1, x)|_{(1,1,1)} = (1, 1, 1).$$

Com $u \nparallel v$, podem aplicar el primer apartat:

$$DF|_{(1,1,1)} = \begin{pmatrix} 4 & 4 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \quad M = D_{y,z}F = \begin{vmatrix} 4 & 0 \\ 1 & 1 \end{vmatrix} = 4 \neq 0.$$

Per tant, pel teorema de la funció implícita existeixen $y = y(x)$ i $z = z(x)$ tals que $y(1) = z(1) = 1$ i

derivem:

$$\begin{cases} 4x^3 + 4y^3 \cdot y' = 0 \\ z + xz' + y' = 0 \end{cases} \xrightarrow[x=1]{y(1)=z(1)=1} \begin{cases} 4 + 4y'(1) = 0 \iff y'(1) = -1 \\ 1 + z'(1) + y'(1) = 0 \iff z'(1) = 0 \end{cases}$$

Tornem a derivar per a obtenir la segona derivada i assegurar la naturalesa dels extrems:

$$\begin{cases} 12x^2 + 12y^2(y')^2 + 4y^3 \cdot y'' = 0 \\ z' + z' + xz'' + y'' = 0 \end{cases} \xrightarrow{x=1} \begin{cases} 12 + 12 + 4y''(1) = 0 \iff y''(1) = -6 \\ 1 \cdot z''(1) + y''(1) = 0 \iff z''(1) = 6 > 0 \end{cases}$$

Per tant, $z(x)$ té un mínim en $x = 1$. ■

Exercici 3.12.

1. The equations:

$$\begin{cases} x^4 + y^4 = 2, \\ (x-1)^2 + (y-1)^2 = \lambda \end{cases}$$

have the solution $x = y = 1, \lambda = 0$. Study the existence of a curve of solutions going through this point and, using a Taylor polynomial of suitable degree (larger than 1), give an approximation to the solution (or solutions) for $\lambda = 0.01$.

2. Let $f : (x, y) \in \mathbb{R}^2 \mapsto \mathbb{R}$ be a C^1 map, and assume we know a point $(x_0, y_0) \in \mathbb{R}^2$ such that $f(x_0, y_0) = 0$. If $\nabla f(x_0, y_0) \neq 0$ use the Implicit Function Theorem to show, if h is small enough, that the following equations admit two solutions.

$$\begin{cases} f(x, y) = 0, \\ (x - x_0)^2 + (y - y_0)^2 = h^2. \end{cases}$$

Demostració.

1. Com als exercicis anteriors, definim F i plantegem DF , al punt on tenim una solució:

$$F(x, y, \lambda) = \begin{pmatrix} x^4 + y^4 - 2 \\ (x-1)^2 + (y-1)^2 - \lambda \end{pmatrix} \implies DF|_{(1,1,0)} = \begin{pmatrix} 4x^3 & 4y^3 & 0 \\ 2(x-1) & 2(y-1) & -1 \end{pmatrix}_{|(1,1,0)} = \begin{pmatrix} 4 & 4 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Com $\det D_{y,\lambda}(1, 1, 0) = -4 \neq 0$, pel teorema de la funció implícita existeix $(x, y(x), \lambda(x))$ per $x = 1$ tal que $y(1) = 1, \lambda(1) = 0$ i $F(x, y(x), \lambda(x)) = 0$. Derivant:

$$\begin{cases} 4x^3 + 4y^3 \cdot y' = 0 \\ 2(x-1) + 2(y-1) \cdot y' = \lambda' \end{cases} \xrightarrow{x=1} \begin{cases} 4 + 4 \cdot y'(1) = 0 \iff y'(1) = -1 \\ \lambda'(1) = 0 \iff \lambda'(1) = 0 \end{cases}$$

Tornant a derivar, trobem la caracterització dels extrems:

$$\begin{cases} 12x^2 + 12y^2(y')^2 + 4y^3 \cdot y'' = 0 \\ 2 + 2y' \cdot y' + 2(y-1) \cdot y'' = \lambda'' \end{cases} \xrightarrow{x=1} \begin{cases} 12 + 12 + 4y''(1) = 0 \iff y''(1) = -6, \\ 2 + 2 \cdot 1 + 0 = \lambda''(1) \iff \lambda''(1) = 4. \end{cases}$$

Aplicant Taylor sobre les funcions implícites $y(x)$ i $\lambda(x)$:

$$y(x) = 1 + (-1)(x-1) + \frac{-6}{2}(x-1)^2 + o(|x-1|^3), \quad \lambda(x) = 0 + 0 + \frac{4}{2}(x-1)^2 + o(|x-1|^3).$$

Per tant, $\lambda \approx 2(x-1)^2$ i:

$$0.01 = 2(x-1)^2 \implies x_{1,2} \approx 1 \pm \sqrt{0.005} \implies x_1 \approx 1.0707107, \quad x_2 \approx 0.9292893.$$

De la mateixa manera, $y_1 \approx 0.9142893$ i $y_2 \approx 1.0557107$.

2. Definim F :

$$F(x, y, \lambda) = \begin{pmatrix} f(x, y) \\ (x-x_0)^2 - (y-y_0)^2 - \lambda \end{pmatrix}$$

Per tant, DF queda de la següent forma:

$$DF|_{(x_0, y_0, 0)} = \begin{pmatrix} \frac{\partial}{\partial x} f(x, y) & \frac{\partial}{\partial y} f(x, y) & 0 \\ 2(x-x_0) & 2(y-y_0) & -1 \end{pmatrix}_{|(x_0, y_0, 0)} = \begin{pmatrix} \frac{\partial}{\partial x} f(x_0, y_0) & \frac{\partial}{\partial y} f(x_0, y_0) & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Suposant $\frac{\partial f}{\partial y}(x_0, y_0) \neq 0$, és clar que $\det D_{y, \lambda} F|_{(x_0, y_0, 0)} \neq 0$. Pel teorema de la funció implícita, existeixen $y(x)$ i $\lambda(x)$ a prop d' $x = x_0$ tal que $y(x_0) = y_0$ i $\lambda(x_0) = 0$. Derivem i $x = x_0$, $\lambda'(x_0) = 0$:

$$y'(x_0) = \frac{-\frac{\partial}{\partial x} f(x_0, y_0)}{\frac{\partial f}{\partial y}(x_0, y_0)}.$$

Derivem i $x = x_0$:

$$\lambda''(x_0) = 2 + 2 \left(\frac{\frac{\partial}{\partial x} f(x_0, y_0)}{\frac{\partial f}{\partial y}(x_0, y_0)} \right)^2 > 0.$$

Per tant, $\lambda(x)$ té un mínim local a $x = x_0$. Podem desenvolupar Taylor:

$$\lambda(x) = \frac{1}{2} \lambda''(x_0) \cdot (x-x_0)^2 + o(|x-x_0|^2).$$

Per tant, $\lambda = h^2 > 0$ i existeixen dues solucions. ■

Exercici 3.13. *We consider the system of equations*

$$\begin{aligned}\lambda x^2 + y^2 &= 2y - \sin(\lambda - 1) \\ x^3 + \lambda y^3 &= y\end{aligned}$$

which has the solution $(x, y) = (0, 0)$ for $\lambda = 1$.

1. Say which of the following answers is correct.

- There exist C^∞ functions $x(\lambda)$ and $y(\lambda)$, for $|1 - \lambda|$ small enough such that $(x(\lambda), y(\lambda), \lambda)$ is a solution of the system.
 - There exist C^∞ functions $x(y)$ and $\lambda(y)$, for $|y|$ small enough such that $(x(y), y, \lambda(y))$ is a solution of the system.
 - There exist C^∞ functions $y(x)$ and $\lambda(x)$, for $|x|$ small enough such that $(x, y(x), \lambda(x))$ is a solution of the system.
2. Suppose that we are following the curve of solutions. Study if some of the variables has a local maximum or minimum at $(x, y, \lambda) = (0, 0, 1)$.
3. Which of the answers of first part are true if we ask for the existence of continuous functions instead of C^∞ functions?

Demostració.

1. Sigui $F(x, y) = (\lambda x^2 + y^2 - 2y + \sin(\lambda - 1), x^3 + \lambda y^3 - y)$, de manera que:

$$DF(x, y) = \begin{pmatrix} 2\lambda x & 2y - 2 & x^2 + \cos(\lambda - 1) \\ 3x^2 & 3\lambda y^2 - 1 & 0 \end{pmatrix} \implies DF(x, y)|_{(0,0,1)} = \begin{pmatrix} 0 & -2 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

Com la matriu no és quadrada, calculem $DF_{x,y}$, $DF_{x,\lambda}$, $DF_{y,\lambda}$ i mirem quan el determinant dona diferent de zero (volem aplicar el TFI):

$$DF_{x,y}(0, 0) = \begin{pmatrix} 0 & -2 \\ 0 & -1 \end{pmatrix} \implies \det(DF_{x,y}(0, 0)) = 0.$$

$$DF_{x,\lambda}(0, 1) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \implies \det(DF_{x,\lambda}(0, 1)) = 0.$$

$$DF_{y,\lambda}(0, 1) = \begin{pmatrix} -2 & 1 \\ -1 & 0 \end{pmatrix} \implies \det(DF_{y,\lambda}(0, 1)) \neq 0.$$

Per tant, existeixen $y(x)$ i $\lambda(x)$ per $|x|$ prou petit (x en un entorn de zero) tals que $(x, y(x), \lambda(x))$ és solució del sistema. A més, $y(0) = 0$ i $\lambda(0) = 1$. Posant $y = y(x)$ i $\lambda = \lambda(x)$, podem derivar les

equacions del sistema inicial:

$$\begin{cases} \lambda' x^2 + 2\lambda x + 2y y' - 2y' + \cos(\lambda - 1) \cdot \lambda' = 0 \\ 3x^2 + \lambda' y^3 + \lambda \cdot 3y^2 \cdot y' - y' = 0 \end{cases} \iff \begin{cases} -2y'(0) + \lambda'(0) = 0 \\ -y'(0) = 0 \end{cases}$$

Per tant, $y'(0) = \lambda'(0) = 0$. Caldria tornar a derivar per comprovar els extrems relatius i els punts d'inflexió, respectivament.

$$\begin{cases} \lambda'' x^2 + \lambda' 2x + \lambda' 2x + 2\lambda + 2(y')^2 + 2y y'' - 2y'' - \sin(\lambda - 1) \cdot (\lambda')^2 + \cos(\lambda - 1) \lambda'' = 0 \\ 6x + \lambda'' y^3 + \lambda' 3y^2 y' + \lambda' 3y^2 y' + \lambda 6y y' + \lambda 3y^2 y'' - y'' = 0 \end{cases}$$

$$\xrightarrow[y'(0)=\lambda'(0)=0]{x=0, y(0)=0, \lambda(0)=1} \begin{cases} y''(0) = 0 \\ \lambda''(0) = 2. \end{cases}$$

Per tant, λ té un màxim en un entorn de $x = 0$.

2. Veiem que $y'' = 0$ pel que existeix la inversa de $y(x)$, $f^{-1}(y)$. Per tant, $x(y) = f^{-1}(y)$ i $\lambda(x) = g(x) = g(f^{-1}(y))$; per l'apartat anterior, tenim un màxim a λ en $x = 0$. No podem trobar la inversa i, per tant, no existeixen les corbes del primer apartat. ■

4

APROXIMACIÓ

Exercici 4.1. Let E be a real vector space and $\langle \cdot, \cdot \rangle$ a scalar product on E . Let $\{\varphi_0, \varphi_1, \dots, \varphi_n\}$ be an orthogonal system. Prove that, for any $(c_0, \dots, c_n) \in \mathbb{R}^{n+1}$,

$$\left\| \sum_{i=0}^n c_i \varphi_i \right\|^2 = \sum_{i=0}^n c_i^2 \|\varphi_i\|^2$$

Demostració. Ho veiem per inducció sobre n . Efectivament, per $n = 1$ tenim:

$$\left\| \sum_{i=0}^1 c_i \varphi_i \right\|^2 = \|c_0 \varphi_0 + c_1 \varphi_1\|^2 \stackrel{P}{=} \|c_0 \varphi_0\|^2 + \|c_1 \varphi_1\|^2 = c_0^2 \|\varphi_0\|^2 + c_1^2 \|\varphi_1\|^2.$$

Ho suposem cert per a $n = k$ i ho veiem per $n = k + 1$:

$$\begin{aligned} \left\| \sum_{i=0}^{k+1} c_i \varphi_i \right\|^2 &= \left\| \sum_{i=0}^k c_i \varphi_i + c_{k+1} \varphi_{k+1} \right\|^2 \stackrel{H}{=} \left\| \sum_{i=0}^k c_i \varphi_i \right\|^2 + \|c_{k+1} \varphi_{k+1}\|^2 \\ &\stackrel{HI}{=} \sum_{i=0}^k c_i^2 \|\varphi_i\|^2 + c_{k+1}^2 \|\varphi_{k+1}\|^2 = \sum_{i=0}^{k+1} c_i^2 \|\varphi_i\|^2. \end{aligned}$$

I amb això ja hem acabat. ■

Exercici 4.2. Let E be a real vector space and $\langle \cdot, \cdot \rangle$ a scalar product on E . Prove that two elements f, g of E verify $\langle f, g \rangle = 0$ if and only if $\|\alpha f + g\| \geq \|g\|$ for all $\alpha \in \mathbb{R}$.

Definició (Millor aproximació). g^* és la millor aproximació de g si, i només si $\|g - g^*\| < \|g - u\|$ per a tot $u \in E^*$. Sigui $u = -\alpha f$, aleshores $\|\alpha + g\| \geq \|g - 0\| = \|g\|$.

Demostració. Definim $E^* = \langle f \rangle$ i trobem la millor aproximació $g^* \in E^*$ de $g \in E$. Pel teorema de la projecció ortogonal, $g^* \in E^*$ és la millor aproximació si, i només si $\langle g - g^*, f \rangle = 0$ per a tot $f \in E^*$. Per tant, $\langle f, g \rangle = 0 \iff \langle g^*, f \rangle = 0 \iff g^* = 0$. ■

Exercici 4.3. Let E be the real linear space $C^0([0, 1])$ with the scalar product:

$$\langle f, g \rangle = \int_0^1 f(x)g(x) dx.$$

Let E^* be the linear subspace of E generated by the functions $1, e^x$ and e^{-x} .

1. Derive an orthonormal basis of E^* .
2. Compute the best approximation $f^* \in E^*$ to $f(x) = x^2$.
3. Compute $\|f - f^*\|$.

Demostració.

1. Utilitzarem l'ortonormalització de Gram-Schmidt per trobar una base ortonormal $\langle \Psi_0, \Psi_1, \Psi_2 \rangle$. Sigui la base inicial $\varphi_0(x) = 1, \varphi_1(x) = e^x$ i $\varphi_2(x) = e^{-x}$.
 - És clar que $\Psi_0(x) = \varphi_0(x) = 1$.
 - Ara, $\Psi_1(x) = \varphi_1(x) - a\Psi_0(x)$ tal que $\langle \Psi_0, \Psi_1 \rangle = 0$. Aleshores:

$$0 = \langle \Psi_0, \Psi_1 \rangle = \langle \varphi_1(x) - a\Psi_0(x), \Psi_0(x) \rangle = \langle \varphi_1(x), \Psi_0(x) \rangle - a\langle \Psi_0(x), \Psi_0(x) \rangle.$$

Per tant:

$$a = \frac{\langle \varphi_1, \Psi_0 \rangle}{\langle \Psi_0, \Psi_0 \rangle} = \langle \varphi_1, \Psi_0 \rangle = e - 1, \text{ ja que } \begin{cases} \langle \Psi_0, \Psi_0 \rangle = \int_0^1 1 \cdot 1 dx = [x]_0^1 = 1 \\ \langle \Psi_0, \varphi_1 \rangle = \int_0^1 1 \cdot e^x dx = [e^x]_0^1 = e - 1 \end{cases}.$$

Una vegada trobada a , és clar que $\Psi_1(x) = \varphi_1(x) - a\Psi_0(x) = e^x - (e - 1)1 = 1 - e + e^x$.

- Sigui ara $\Psi_2(x) = \varphi_2(x) - a\Psi_1(x) - b\Psi_0(x)$ tal que $\langle \Psi_0, \Psi_2 \rangle = \langle \Psi_1, \Psi_2 \rangle = 0$. Aleshores:

$$\begin{cases} 0 = \langle \Psi_2, \Psi_1 \rangle = \langle \varphi_2, \Psi_1 \rangle - a\langle \Psi_1, \Psi_1 \rangle - b\langle \Psi_0, \Psi_1 \rangle \\ 0 = \langle \Psi_2, \Psi_0 \rangle = \langle \varphi_2, \Psi_0 \rangle - a\langle \Psi_1, \Psi_0 \rangle - b\langle \Psi_0, \Psi_0 \rangle \end{cases}$$

Fent els càlculs pertinents trobem a i b :

$$\begin{cases} \langle \varphi_2, \Psi_1 \rangle = \int_0^1 e^{-x} (1 - e + e^x) dx = 3 - e - e^{-1}, \\ \langle \Psi_1, \Psi_1 \rangle = \|\Psi_1\|^2 = \int_0^1 (1 - e + e^x)^2 dx = \frac{1}{2}(1 - e) \cdot (e - 3), \\ \langle \varphi_2, \Psi_0 \rangle = \int_0^1 e^{-x} \cdot 1 dx = 1 - e^{-1} \end{cases} \Rightarrow \begin{cases} a = \frac{\langle \varphi_2, \Psi_1 \rangle}{\langle \Psi_1, \Psi_1 \rangle} = \frac{3 - e - e^{-1}}{\frac{1}{2}(e-1)(3-e)} \\ b = \frac{\langle \varphi_2, \Psi_0 \rangle}{\langle \Psi_0, \Psi_0 \rangle} = 1 - e^{-1} \end{cases}$$

Per tant, $\Psi_2(x) = e^{-x} - a(e^x - e + 1) - (1 - e^{-1})1$ i obtenim:

$$\langle \Psi_2, \Psi_2 \rangle = \|\Psi_2\|^2 = \frac{22 - 16e + \frac{9}{2}(e^2 - e^{-2}) - \frac{3}{2}(e^2 - e^{-2})}{6 - 8e + 2e^2}.$$

Així doncs, ens queda la base ortonormal d' E^* $\{\Psi_0, \frac{\Psi_1}{\|\Psi_1\|}, \frac{\Psi_2}{\|\Psi_2\|}\} = \{\Psi_0, \tilde{\Psi}_1, \tilde{\Psi}_2\}$.

2. Sigui ara $f^* = \langle f, \Psi_0 \rangle \Psi_0 + \langle f, \tilde{\Psi}_1 \rangle \tilde{\Psi}_1 + \langle f, \tilde{\Psi}_2 \rangle \tilde{\Psi}_2$ la millor aproximació. Aleshores:

$$\begin{aligned} \langle f, \Psi_0 \rangle &= \int_0^1 x^2 \cdot 1 dx = \frac{1}{3}. \\ \langle f, \tilde{\Psi}_1 \rangle &= \left\langle f, \frac{\Psi_1}{\|\Psi_1\|} \right\rangle = \frac{1}{\|\Psi_1\|} \int_0^1 x^2 (e^x + 1 - e) dx = \frac{1}{\|\Psi_1\|} \frac{2e - 5}{3}. \\ \langle f, \tilde{\Psi}_2 \rangle &= \frac{1}{\|\Psi_2\|} \langle f, e^{-x} - a\Psi_1 - b \rangle = \frac{1}{\|\Psi_2\|} (\langle f, e^{-x} \rangle - a\langle f, \Psi_1 \rangle - b\langle f, 1 \rangle). \end{aligned}$$

3. Per a calcular $\|f - f^*\|$:

$$\|f - f^*\|^2 = \|f\|^2 - \|f^*\|^2 = \int_0^1 x^2 \cdot x^2 dx - \left(\langle f, \Psi_0 \rangle^2 + \langle f, \tilde{\Psi}_1 \rangle^2 + \langle f, \tilde{\Psi}_2 \rangle^2 \right).$$

Utilitzant els càlculs de l'apartat anterior, ja hem acabat. ■

Exercici 4.4. Let E be the space of real functions, defined on $[0, 1]$, that are of C^1 class, and define:

$$\langle f, g \rangle = f(0)g(0) + \int_0^1 f'(x)g'(x) dx. \quad (4.1)$$

1. Show that $\langle \cdot, \cdot \rangle$ is a scalar product. Is $(f, g) \mapsto \int_0^1 f'(x)g'(x) dx$ a scalar product?
2. Using the scalar product (4.1), derive an orthogonal basis of $\mathbb{R}_2[x]$.
3. Find the best approximation to $f(x) = \cos x$ by an element of $\mathbb{R}_2[x]$.
4. Find the best approximation to $f(x) = \cos x$ by an element of $\{p \in \mathbb{R}_2[x] \mid p(0) = 1\}$.

Demostració.

1. $\langle \cdot, \cdot \rangle$ és producte escalar. En efecte, $\langle f, g \rangle = \langle g, f \rangle$, $\langle f + g, h \rangle = \langle f, h \rangle + \langle g, h \rangle$ i $\langle \alpha f, g \rangle = \alpha \langle f, g \rangle$.
A més:

$$\langle f, f \rangle = f(0)^2 + \int_0^1 (f'(x))^2 dx \geq 0, \quad \langle f, f \rangle = 0 \iff \begin{cases} f(0)^2 = 0 \iff f(0) = 0 \\ \int_0^1 (f'(x))^2 dx = 0 \iff f'(x) = 0 \end{cases}$$

Pel que $f \equiv 0$. Considerem ara $\langle f, g \rangle = \int_0^1 f'(x)g'(x) dx$. Totes les propietats de producte escalar són evidents, però:

$$\langle f, f \rangle = \int_0^1 (f'(x))^2 dx = 0 \iff f'(x) = 0 \iff f(x) = C, C \in \mathbb{R};$$

com $C \neq 0$ també ho compleix, no és producte escalar.

2. Sigui la base canònica $\{1, x, x^2\}$. Volem veure que definits $\varphi_0 = 1, \varphi_1 = x + a, \varphi_2 = x^2 + bx + c$, per a quins a, b $\{\varphi_0, \varphi_1, \varphi_2\}$ és una base. Imposem que $\langle \varphi_0, \varphi_1 \rangle = \langle \varphi_0, \varphi_2 \rangle = \langle \varphi_1, \varphi_2 \rangle = 0$ i resolent ens queda:

$$\langle \varphi_0, \varphi_1 \rangle = a, \langle \varphi_0, \varphi_2 \rangle = c, \langle \varphi_1, \varphi_2 \rangle = a \cdot c + \int_0^1 1 \cdot (2x + b) dx = [x^2 + bx]_0^1 = 1 + b.$$

Per tant, obtenim que $a = c = 0$ i $b = -1$. Per tant, $\varphi_0(x) = 1, \varphi_1(x) = x$ i $\varphi_2(x) = x^2 - x$.

3. Volem trobar la millor aproximació $f^* = c_0\varphi_0 + c_1\varphi_1 + c_2\varphi_2$ on $c_i = \frac{\langle f, \varphi_i \rangle}{\langle \varphi_i, \varphi_i \rangle}$. Simplement caldria fer els càlculs i trobar el valor corresponent dels c_i :

$$\begin{aligned} \langle \varphi_0, \varphi_0 \rangle &= 1 & \langle \varphi_1, \varphi_1 \rangle &= 1 & \langle \varphi_2, \varphi_2 \rangle &= \frac{1}{3} \\ \langle f, \varphi_0 \rangle &= 1 & \langle f, \varphi_1 \rangle &= \cos(1) - 1 & \langle f, \varphi_2 \rangle &= \cos(1) + 1 - 2 \sin(1) \end{aligned}$$

pel que $c_0 = 1, c_1 = \cos(1) - 1$ i $c_2 = 3(\cos(1) + 1 - 2 \sin(1))$, així que f^* queda fixada.

4. La millor aproximació de $f(x) = \cos x$ en $\{p \in \mathbb{R}_2[x] \mid p(0) = 1\}$ és la de l'apartat anterior, ja que $f^*(0) = 1$. ■

Exercici 4.5. Let us define the set $E = \{f \in C_{\mathbb{R}}^1([-1, 1]) \mid f(-x) = -f(x) \forall x \in [-1, 1]\}$, and let us consider the map from $E \times E$ to $\mathbb{R}$ given by

$$(f, g) \in E \times E \mapsto \langle f, g \rangle = \int_{-1}^1 f'(x)g'(x) dx$$

1. Is E a vector space? Is $\langle \cdot, \cdot \rangle$ a scalar product on $C_{\mathbb{R}}^1([-1, 1])$? Is $\langle \cdot, \cdot \rangle$ a scalar product on E ? Let us define $E_3 = \{p \in E \mid p \text{ is a third degree polynomial}\}$. Is E_3 a vector space? If so, find a basis.
2. Let us define $f_0(x) = \sin x$. Compute the element of E_3 closest to f_0 according to the norm induced by the scalar product $\langle \cdot, \cdot \rangle$.
3. Let us define $F = \{f \in E \mid f'(0) = 1\}$ and $F_3 = \{p \in F \mid \deg(p) = 3\}$. Compute the element of F_3 closest to $f_0(x) = \sin x$. Is it possible to use the same scheme to find the element of F_3 closest to $f_0(x) = \cos x$?

Demostració.

1. És evident que és espai vectorial, ja que per a tot $f, g \in E$ i $\alpha, \beta \in \mathbb{R}$ tenim $\alpha f, \beta g \in E$. $\langle f, g \rangle = \langle g, f \rangle$, $\langle f + g, h \rangle = \langle f, h \rangle + \langle g, h \rangle$ i, finalment, que $\langle \alpha f, g \rangle = \alpha \langle f, g \rangle$. Vegem que $\langle f, f \rangle > 0$ si $f \neq 0$. En efecte, $\langle f, f \rangle = \int_{-1}^1 (f'(x))^2 dx \geq 0$ i si $\langle f, f \rangle = 0$, aleshores $f'(x) = 0$ per a tot $x \in [-1, 1]$. Així doncs, $f \equiv c \in \mathbb{R}$ i com $c = f(-x) = -f(x) = -c$, aleshores $c = 0$.

$E_3 = E \cap \mathbb{R}_3[x] = \{p \in E \mid \deg(p) = 3\}$ és un espai vectorial, sigui $p \in E_3$; llavors, $p(x) = ax^3 + bx^2 + cx + d$ per a certs a, b, c, d . Com $p \in E$:

$$p(-x) = -ax^3 + bx^2 - cx + d = -p(x) = -ax^3 - bx^2 - cx - d \implies p(x) = ax^3 + bx.$$

Per tant, prenem la base $\{x, x^3\}$.

2. La millor aproximació és $f^* = c_0x + c_1x^3$, on:

$$\begin{pmatrix} \langle x, x \rangle & \langle x, x^3 \rangle \\ \langle x, x^3 \rangle & \langle x^3, x^3 \rangle \end{pmatrix} = \begin{pmatrix} \langle f, x \rangle \\ \langle f, x^3 \rangle \end{pmatrix} \implies \begin{cases} \langle f, x \rangle = 2 \sin(1) & \langle x, x \rangle = \langle x, x^3 \rangle = \langle x^3, x \rangle = 2 \\ \langle f, x^3 \rangle = 12 \cos(1) - 6 \sin(1) & \langle x^3, x^3 \rangle = \frac{18}{5} \end{cases}$$

Resolent el sistema, trobem $c_0 \approx 0.647611$, $c_1 \approx 0.193859$.

3. Sigui $F = \{f \in E \mid f'(0) = 1\}$ i $F_3 = \{p \in F \mid p \text{ és un polinomi de grau } 3\}$. És clar que $p \in F_3 \iff p(x) = ax^3 + x$; òbviament, F_3 no és un espai vectorial. Trobar l'element més proper a f_0 vol dir trobar $p_0 \in F_3$ tal que $\|f - p_0\| \leq \|f - p\|$, per a tot $p \in F_3$. Això és el mateix que trobar un monomi de la forma $q_0 = ax^3$ tal que $\|g - q_0\| \leq \|g - q\|$, per a tot $q(x) = ax^3$ on $g(x) = \sin x - x$. Com $\sin x - x \in E$, la millor aproximació $p^* = x + ax^3$ satisfà:

$$a = \frac{\langle g, x^3 \rangle}{\langle x^3, x^3 \rangle} = \frac{12 \cos 1 - 6 \sin 1 - 2}{\frac{18}{5}} \approx -0.1569995.$$

No podem seguir la mateixa estratègia per a $\cos x$ ja que $\cos x - x \notin E$. ■

Exercici 4.6. Define $E = \{f \in C^1([-1, 1]) \mid \int_{-1}^1 f(x) dx = 0\}$ and consider the map from $E \times E$ to $\mathbb{R}$ given by:

$$(f, g) \in E \times E \mapsto \langle f, g \rangle = \int_{-1}^1 f'(x) g'(x) dx.$$

1. Is E a vector space? Is $\langle \cdot \rangle$ a scalar product for E ?
2. Define $E_4 = \{p \in E \mid p \in \mathbb{R}_4[x]\}$. Prove that E_4 is a linear subspace of E and find a basis.
3. Find an orthogonal basis of monic polynomials of E_4 .
4. Let $\|\cdot\|$ be the norm associated with the given scalar product. Find between all the monic polynomials of degree 4 belonging to E , the one that achieves the minimum norm.

Demostració.

1. És evident que $\langle f, g \rangle = \langle g, f \rangle$, $\langle f + g, h \rangle = \langle f, h \rangle + \langle g, h \rangle$ i, finalment, que $\langle \alpha f, g \rangle = \alpha \langle f, g \rangle$. Vegem que $\langle f, f \rangle > 0$ si $f \neq 0$. En efecte,

$$\langle f, f \rangle = 0 \implies f'(x) = 0 \implies f(x) = C \text{ en } [-1, 1] \xrightarrow{\int_{-1}^1 f(x) dx = 0} C = 0.$$

2. Sigui $E_4 = \{p \in E \mid p \in \mathbb{R}_4[x]\}$ i sigui $p_4(x) = ax^4 + bx^3 + cx^2 + dx + e$. Per tant:

$$\int_{-1}^1 (ax^4 + bx^3 + cx^2 + dx + e) dx = 0 \iff \frac{a}{5} \cdot 2 + \frac{c}{3} \cdot 2 + 2e = 0 \iff e = -\frac{1}{5}a - \frac{1}{3}c.$$

Així, p és de la forma $p(x) = a\left(x^4 - \frac{1}{5}\right) + bx^3 + c\left(x^2 - \frac{1}{3}\right) + dx$. La base trobada serà $\varphi_1(x) = x$, $\varphi_2(x) = x^2 - \frac{1}{3}$, $\varphi_3(x) = x^3$ i $\varphi_4(x) = x^4 - \frac{1}{5}$.

3. Tenim $\Phi_1(x) = \varphi_1(x)$ i $\Psi_i = \varphi_i - \sum_{j=1}^{i-1} \alpha_j \Psi_j$, $\Psi_i \perp \Psi_1$ per a tot $j > 1$. Aleshores, $\Psi_1 = x$, $\Psi_2 = x^2 - \frac{1}{3}$, $\Psi_3 = x^3 - x$, $\Psi_4 = x^4 - \frac{6}{5}x^2 + \frac{1}{5}$.
4. Podem escriure qualsevol polinomi com a combinació de Ψ_i i que sigui mònic. Per tant, $x^4 + ax^3 + bx^2 + cx + d = \Psi_4 + E_3$, on $E_3 = \lambda_1 \Psi_1 + \lambda_2 \Psi_2 + \lambda_3 \Psi_3 = \mathbb{R}_3[x] \cap E$. Hem de buscar la millor aproximació g^* de Ψ_4 a E_3 , i.e. $\|\Psi_4 - g^*\| \leq \|\Psi_4 - p_3(x)\|$ i, pel teorema de la projecció ortogonal $\langle \Psi_4 - g^*, \Psi_i \rangle = 0$. Pel segon exercici de la llista, aleshores $g^* = 0$. Per tant, $\Psi_4(x)$ és el polinomi mònic amb la norma mínima. ■

Exercici 4.7. Consider the vector space $C^0([0, 1])$ endowed with the scalar product $\langle f, g \rangle = \int_0^1 f(x)g(x) dx$. Let f be an element of $C^0([0, 1])$ such that $\int_0^1 xf(x) dx = \frac{1}{2}$ and $\int_0^1 x^2f(x) dx = \frac{1}{3}$.

1. Suppose that $\int_0^1 f(x) dx = 0$, and define $M_2 = \{p \in \mathbb{R}_2[x] \mid \int_0^1 p(x) dx = 0\}$. Find the best approximation to f by an element of M_2 .
2. Suppose that $\int_0^1 f(x) dx = 1$, and define $D_2 = \{p \in \mathbb{R}_2[x] \mid \int_0^1 p(x) dx = 1\}$. Find the best approximation to f by an element of D_2 .

Demostració.

1. M_2 és espai vectorial. Observem que $\dim \mathbb{R}_2[x] = 3$ i tenim la base canònica $\{1, x, x^2\}$. Anem a veure que $\dim M_2 < 3$. En efecte, $p = ax^2 + bx + c \in M_2$ si, i només si:

$$\int_0^1 (ax^2 + bx + c) dx = 0 \iff \frac{1}{3}a + \frac{1}{2}b + c = 0 \iff 2a + 3b + 6c = 0.$$

Tenim una equació i, per tant, dos graus de llibertat. Posant $a = 0$, $b = 2$, $c = -1$, $\varphi_1(x) = 2x - 1$ i prenent $a = 3$, $b = 0$, $c = -1$, $\varphi_2(x) = 3x^2 - 1$. $\{\varphi_1, \varphi_2\}$ són linealment independents (i en particular una base d' M_2) perquè:

$$A\varphi_1 + B\varphi_2 = 3Bx^2 + 2Ax - (A + B) = 0 \iff A = B = 0.$$

Trobem la millor aproximació f^* a f .

$$f^* = c_1\varphi_1 + c_2\varphi_2 = 18(2x-1) - 10(3x^2-1) \quad \begin{cases} c_1\langle\varphi_1, \varphi_1\rangle + c_2\langle\varphi_1, \varphi_2\rangle = \langle f, \varphi_1\rangle \\ c_1\langle\varphi_2, \varphi_1\rangle + c_2\langle\varphi_2, \varphi_2\rangle = \langle f, \varphi_2\rangle \end{cases} \implies \begin{cases} c_1 = 18, \\ c_2 = -10. \end{cases}$$

En particular,

$$\begin{aligned}\langle f, \varphi_1 \rangle &= \int_0^1 (2x - 1)f(x) dx = \int_0^1 2xf(x) dx - \int_0^1 f(x) dx = \int_0^1 2xf(x) dx = 1, \\ \langle f, \varphi_1 \rangle &= 3 \int_0^1 x^2 f(x) dx - \int_0^1 2xf(x) dx = 1.\end{aligned}$$

2. D_2 no és espai vectorial. Busquem f^* tal que $\|f - f^*\|$ és mínim.

$$\|f - f^*\| = \|g - g^*\| := \|(f - 1) - (f^* - 1)\|, \quad g, g^* \in \mathcal{M}_2.$$

Fent com al primer apartat, $g^* = c_1 \varphi_1 + c_2 \varphi_2$ i per tant, $c_1 = c_2 = 0$ i $g^* = 0 \implies f^* = 1$. ■

Exercici 4.8. Compute the monic orthogonal polynomials of degrees 0, 1, 2, 3 and 4, corresponding to the scalar product:

$$\langle f, g \rangle = \int_{-1}^1 (1 + x^2) f(x) g(x) dx$$

Demostració. Utilitzem l'algorisme que hem vist a classe:

$$\begin{cases} p_0(x) = 1, & p_{-1}(x) = (x - \delta_{i+1})p_i(x) - \gamma_{i+1}^2 p_{i-1}, \quad i \geq 0. \\ \delta_{i+1} = \frac{\langle x p_i, p_i \rangle}{\langle p_i, p_i \rangle}, & \gamma_{i+1}^2 = \begin{cases} 0, & \text{si } i = 0 \\ \frac{\langle p_i, p_i \rangle}{\langle p_{i-1}, p_{i-1} \rangle}, & \text{si } i \geq 1. \end{cases} \end{cases}$$

Per tant, per a $n = 0$ tenim $p_0(x) = 1$ i per $n = 1$ $p_1(x) = x$, ja que:

$$p_1(x) = (x - \delta_1)p_0(x) - \gamma_1^2 p_{-1}(x) = x - \delta_1, \quad \delta_1 = \frac{\langle x p_0, p_0 \rangle}{\langle p_0, p_0 \rangle} = \frac{\langle x, 1 \rangle}{\langle 1, 1 \rangle} = \frac{\int_{-1}^1 (1 + x^2) x dx}{\int_{-1}^1 (1 + x^2) \cdot 1 dx} = \frac{0}{\frac{8}{3}} = 0.$$

Per $p_2(x) = (x - \delta_2)p_1(x) - \gamma_2^2 p_0(x) = (x - \delta_2)x - \gamma_2^2 \cdot 1$. Trobem δ_2 i γ_2^2 ;

$$\delta_2 = \frac{\langle x^2, x \rangle}{\langle x, x \rangle} = \frac{0}{\frac{16}{15}} = 0, \quad \gamma_2^2 = \frac{\langle x, x \rangle}{\langle 1, 1 \rangle} = \frac{\frac{16}{15}}{\frac{8}{3}} = \frac{2}{5}.$$

Per tant, $p_2(x) = x^2 - \frac{2}{5}$. Per $n = 3$ tenim $p_3(x) = (x - \delta_3)p_2(x) - \gamma_3^2 p_1(x)$ i:

$$\delta_3 = \frac{\langle x(x^2 - \frac{2}{5}), (x^2 - \frac{2}{5}) \rangle}{\langle x^2 - \frac{2}{5}, x^2 - \frac{2}{5} \rangle} = 0, \quad \gamma_3^2 = \frac{\langle p_2, p_2 \rangle}{\langle p_1, p_1 \rangle} = \frac{\frac{136}{525}}{\frac{16}{15}} = \frac{17}{70}.$$

Pel que $p_3(x) = x(x^2 - \frac{2}{5}) - \frac{17}{70}x = x^3 - \frac{2}{14}x$. ■

Exercici 4.9. Consider $E = C^0([-1, 1])$ with the scalar product:

$$\langle f, g \rangle = \int_{-1}^1 \frac{f(x)g(x)}{\sqrt{1-x^2}} dx$$

Let $f \in E$ defined by $f(x) = \sqrt{1-x^2}$.

1. Compute the best approximation to f by a 4th degree polynomial p_4 .
2. Compute $\|f - p_4\|$, where $\|\cdot\|$ is the norm associated to the previous scalar product.

Demostració.

1. Hem de trobar la millor aproximació p_4 de f per un polinomi de grau 4. Els polinomis de Tchebixev $T_n = \cos(n \arccos x)$ són ortogonals:

$$\langle T_i, T_j \rangle = \begin{cases} 0 & \text{si } i \neq j \\ \frac{\pi}{2} & \text{si } i = j \neq 0 \\ \pi & \text{si } i = j = 0 \end{cases}$$

Seguint la recursivitat $T_0 = 1$, $T_1 = x$, $T_{n+1} = 2xT_n(x) - T_{n-1}(x)$, trobem que:

$$T_0 = 1, \quad T_1 = x, \quad T_2 = 2x^2 - 1, \quad T_3 = 4x^3 - 3x, \quad T_4 = 8x^4 - 8x^2 + 1.$$

Ara,

$$p_4(x) = \sum_{i=0}^n c_i T_i(x), \quad c_i = \frac{\langle f, T_i \rangle}{\langle T_i, T_i \rangle} \implies p_4(x) = \frac{2}{\pi} T_0 - \frac{4}{3\pi} T_2 - \frac{4}{15} T_4.$$

2. Per Pitàgores, $\|f - p_4\|^2 = \|f\|^2 - \|p_4\|^2$ i obtenim $\|f\|^2 = \langle f, f \rangle = \int_{-1}^1 \sqrt{1-x^2} dx = \frac{\pi}{2}$, usant el canvi de variables $x = \cos t$.

$$\begin{aligned} \langle p_4, p_4 \rangle &= \left(\frac{2}{\pi}\right)^2 \langle T_0, T_0 \rangle + \left(\frac{4}{3\pi}\right)^2 \langle T_2, T_2 \rangle + \left(\frac{4}{15}\right)^2 \langle T_4, T_4 \rangle \\ &= \left(\frac{2}{\pi}\right)^2 \pi + \left(\frac{4}{3\pi}\right)^2 \frac{\pi}{2} + \left(\frac{4}{15}\right)^2 \frac{\pi}{2} = \frac{1108}{225\pi}. \end{aligned}$$

Pel que $\|f - p_4\| = \sqrt{\frac{\pi}{2} - \frac{1108}{225\pi}} = 0.0574$. ■

Exercici 4.10. Let M_n be the set of monic polynomials of degree n , and let us consider the space of real polynomials endowed with the norm:

$$\|f\|_2 = \left[\int_{-1}^1 |f(x)|^2 dx \right]^{\frac{1}{2}}$$

1. If $n = 2$, compute the element of M_n of minimum norm.
2. Compute the element of M_n of minimum norm for an arbitrary value of n (hint: use Legendre polynomials).

Let $A_2 \subset \mathbb{R}_2[x]$ such that $\int_{-1}^1 p(x) dx = 2$.

3. Compute the element of A_2 of minimum norm.
4. Compute the best approximation of x^2 in A_2 .

Demostració.

1. Sigui $M_n = \{\text{polinomis mònics de grau } n\}$. M_2 no és espai vectorial, sigui $p_2(x) \in M_2 \iff p_2(x) = x^2 + ax + b$. Volem minimitzar $\|p_2(x)\|_2 = \|x^2 - (-ax - b)\|_2$, on $-ax - b \in \mathbb{R}_1[x]$. Hem de trobar la millor aproximació de x^2 des de $\mathbb{R}_1[x]$. Prenem com a $\mathbb{R}_1[x]$ -base $\{\varphi_0(x) = 1, \varphi_1(x) = x\}$, pel que sigui $p^*(x) = c_0 + c_1x$. Llavors, les equacions normals són:

$$\begin{pmatrix} \langle 1, 1 \rangle & \langle 1, x \rangle \\ \langle x, 1 \rangle & \langle x, x \rangle \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} = \begin{pmatrix} \langle 1, x^2 \rangle \\ \langle x, x^2 \rangle \end{pmatrix} \iff \begin{pmatrix} 2 & 0 \\ 0 & \frac{2}{3} \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} \\ 0 \end{pmatrix}.$$

D'aquí, obtenim $c_0 = \frac{1}{3}$, $c_1 = 0$ i, per tant, $p^*(x) = \frac{1}{3}$. Així, $p(x) = x^2 - \frac{1}{3}$.

2. Ara volem trobar $p_n(x) \in M_n$ amb la norma mínima, pel que hem de trobar la millor aproximació de 1 en $\mathbb{R}_2[x]$. Sigui $p_n \in M_n$, de manera que:

$$p_n = \hat{L}_n + c_{n-1}L_{n-1} + c_{n-2}L_{n-2} + \dots + c_0L_0, \quad L_n = \frac{L_n}{a_n} \quad (L_n = a_nx^n + \dots + a_1x + a_0).$$

$L_{n-1}, \dots, L_0$ són ortogonals i els polinomis de Legendre definits per la recursió $L_0 = 1$, $L_n = \frac{1}{2^n n!} \frac{d^n}{dx^n} ((x^2 - 1)^n)$. Qualsevol $q \in M_n$ es pot escriure com $q(x) = \hat{L}_n(x) + \sum_{i=0}^{n-1} c_i L_i$ per a coeficients c_i adequats.

$$\|p_n\|_2^2 = \|\hat{L}_n\|_2^2 + (c_{n-1})^2 \|L_{n-1}\|_2^2 + \dots + (c_0)^2 \|L_0\|_2^2.$$

Això és mínim quan $c_i = 0$ per a tot $i < n$, pel que $p_n = \hat{L}_n$ (i.e. $c_{n-1} = \dots = c_0 = 0$) té norma mínima.

3. A_2 no és espai vectorial, però $E_2 = \{q \in \mathbb{R}_2[x] \mid \int_{-1}^1 q(x) dx = 0\}$ sí ho és. Així doncs, si $p \in A_2$, aleshores $1 - p \in E_2$; recíprocament, si $q \in E_2$ aleshores $1 - q \in A_2$.

$$\|p\|_2 = \|1 - (1 - p)\|_2 = \int_{-1}^1 (1 - p) dx = \int_{-1}^1 dx - \int_{-1}^1 p(x) dx = 2 - 2 = 0.$$

Ara volem $p \in A_2$ amb norma mínima i tenim $p = 1 - (1 - p)$, però $1 - p \in E_2$. Per tant, minimitzar la norma equival a trobar la millor aproximació de 1 en E_2 . Una base de E_2 és:

$$\int_{-1}^1 (ax^2 + bx + c) dx = \frac{2}{3}a + 2c = 0 \iff a + 3c = 0 \begin{cases} q_1(x) = x & (b = 1, a = c = 0) \\ q_2(x) = 3x^2 - 1 & (a = 3, b = 0, c = -1) \\ q^* = c_0 q_1(x) + c_1 q_2(x) \end{cases}$$

Resolem les equacions normals (els polinomis són ortogonals):

$$\begin{pmatrix} \langle \varphi_1, \varphi_1 \rangle & \langle \varphi_1, \varphi_2 \rangle \\ \langle \varphi_2, \varphi_1 \rangle & \langle \varphi_2, \varphi_2 \rangle \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} = \begin{pmatrix} \langle \varphi_1, 1 \rangle \\ \langle \varphi_2, 1 \rangle \end{pmatrix} \iff \begin{pmatrix} \frac{2}{3} & 0 \\ 0 & \frac{8}{5} \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Troben $c_0 = c_1 = 0$, pel que $q^* = 0$. Per tant, $p = 1$ és el polinomi que minimitza la norma.

4. Se'ns demana calcular la millor aproximació de x^2 en A_2 . Està clar que $\int_{-1}^1 x^2 dx = \frac{2}{3}$, pel que $x^2 \in A_2$. Per a tot $p^* \in A_2$ podem escriure:

$$\|x^2 - p^*\|_2 = \|x^2 - (1 - (1 - p^*))\|_2 = \|(x^2 - 1) - (p^* - 1)\|_2.$$

Si posem $q^* = p^* - 1 \in E_2$, aleshores $q^* = c_1\varphi_1(x) + c_2\varphi_2(x)$. Tenim les equacions normals:

$$\begin{pmatrix} \langle \varphi_1, \varphi_1 \rangle & \langle \varphi_1, \varphi_2 \rangle \\ \langle \varphi_2, \varphi_1 \rangle & \langle \varphi_2, \varphi_2 \rangle \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} \langle \varphi_1, x^2 \rangle \\ \langle \varphi_2, x^2 \rangle \end{pmatrix} \iff \begin{pmatrix} \frac{2}{3} & 0 \\ 0 & \frac{8}{5} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{8}{5} \end{pmatrix}.$$

D'on obtenim que $c_0 = 0$ i $c_1 = \frac{1}{3}$. Per tant, $q^* = \frac{1}{3}(3x^2 - 1) = x^2 - \frac{1}{3}$ i $p^* = 1 + q^* = x^2 + \frac{2}{3}$. ■

Exercici 4.II. Consider $E = C^0([-1, 1])$ with the scalar product $\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx$, and let us denote by $\|\cdot\|$ the associated norm. Let us define M as $\{p \in \mathbb{R}_2[x] \mid p(0) = 1\}$.

1. Obtain the element of M of minimum norm. Write the value of this minimum norm.
2. Compute the element of M which is closest to $f(x) = e^x$.
3. If $\overline{M} = \{p \in \mathbb{R}_2[x] \mid p(0) = p'(0) = 1\}$, compute the element of $\overline{M}$ which is closest to $f(x) = e^x$.

Demostració.

1. Sigui $M = \{p \in \mathbb{R}_2[x] \mid p(0) = 1\}$. Trobem $p_2 \in M$ amb la norma mínima. M no és espai vectorial, però $M' = \{p \in \mathbb{R}_2[x] \mid p(0) = 0\}$ sí ho és.
2. Trobem el $p \in M$ més proper a $f(x) = e^x$. Sigui $p^* \in M$:

$$\|f(x) - p^*(x)\| \leq \|f(x) - p(x)\| \implies \|f(x) - 1 - (p^*(x) - 1)\| \leq \|f(x) - 1 - (p(x) - 1)\|.$$

Clarament, $p - 1, p^* - 1 \in M'$. Hem de trobar la millor aproximació $f(x) - 1$ per un element de M' . Podem posar $q^* \in M'$ com $q^* = c_0x + c_1x^2$ (ja que $q^*(0) = 0$), amb les equacions normals $c_0 = \frac{1}{3e}$ i $c_1 = \frac{5}{2}e - \frac{25}{2e} - \frac{5}{3}$. Per tant, $p^* = 1 + q^* = 1 + c_0x + c_1x^2$.

3. Sigui $\overline{M} = \{p \in \mathbb{R}_2[x] \mid p(0) = p'(0) = 1\}$. Trobem $p \in \overline{M}$ més proper a $f(x) = e^x$. És clar que $p \in \overline{M}$ si, i només si $p(x) = 1 + x + ax^2$.

$$\|e^x - (1 + x + ax^2)\| = \|e^x - 1 - x - ax^2\| \quad \begin{cases} g(x) = e^x - 1 - x \text{ i } g(0) = g'(0) = 0 \\ h(x) = ax^2 \text{ i } h(0) = h'(0) = 0 \end{cases}$$

Hem de trobar $p^*(x) = a^*x^2$ la millor aproximació de $e^x - 1 - x$:

$$a^* = \frac{\langle e^x - 1 - x, x^2 \rangle}{\langle x^2, x^2 \rangle} = \frac{5}{2}e - \frac{25}{2e} - \frac{5}{3} \implies p(x) = 1 + x + a^*x^2. \quad \blacksquare$$

Exercici 4.12. Consider the discrete scalar product defined on $\mathbb{R}_m[x]$,

$$\langle f, g \rangle = \sum_{i=0}^m f(x_i)g(x_i), \text{ where } x_i = -1 + \frac{2i}{m}.$$

1. Derive, for $m = 4$, the orthogonal polynomials of degrees 0, 1 and 2.
2. Obtain the least-squares approximation polynomials of degrees 1 and 2 to the table:

x	-1	-0.5	0	0.5	1
$f(x)$	1.1	0.3	0.1	0.2	0.9

Demostració.

1. Obtenim els polinomis ortogonals de grau 0, 1, 2. $x_0 = -1, x_1 = -0.5, x_2 = 0, x_3 = 0.5, x_4 = 1$.

$$\begin{cases} p_0(x) = 1 \\ p_{i+1}(x) = (x - \delta_{i+1})p_i(x) - \gamma_{i+1}^2 p_{i-1}, \quad i > 0 \end{cases} \quad \delta_{i+1} = \frac{\langle x p_i, p_i \rangle}{\langle p_i, p_i \rangle}, \quad \gamma_{i+1}^2 = \begin{cases} 0, & \text{si } i = 0 \\ \frac{\langle p_i, p_i \rangle}{\langle p_{i-1}, p_{i-1} \rangle}, & \text{si } i \geq 1 \end{cases}$$

Per a $n = 0$, $p_0(x) = 1$. Per a $n = 1$, $p_1(x) = (x - \delta_1)p_0(x) = x - \delta_1$.

$$\delta_1 = \frac{\langle x, 1 \rangle}{\langle 1, 1 \rangle} = \frac{1 - 0.5 + 0 + 0.5 + 1}{1 + 1 + 1 + 1 + 1} = 0 \implies p_1(x) = x.$$

Per a $n = 2$, $p_2(x) = (x - \delta_2)p_1(x) - \gamma_2^2 p_0(x) = (x - \delta_2)x - \gamma_2^2$ i:

$$\begin{aligned} \delta_2 &= \frac{\langle x^2, x \rangle}{\langle x, x \rangle} = \frac{(-1)^3 + (-0.5)^3 + 0^3 + 0.5^3 + 1^3}{(-1)^2 + (-0.5)^2 + 0^2 + (0.5)^2 + 1^2} = 0, \\ \gamma_2^2 &= \frac{\langle x, x \rangle}{\langle 1, 1 \rangle} = \frac{(-1)^2 + (-0.5)^2 + 0^2 + 0.5^2 + 1^2}{1 + 1 + 1 + 1 + 1} = \frac{1}{2} \implies p_2(x) = x^2 - \frac{1}{2}. \end{aligned}$$

2. Sigui $m = 4$. Obtenim els polinomis d'aproximació de grau 1 i 2 mínima quadràtica a la taula:

x	-1	-0.5	0	0.5	1
$f(x)$	1.1	0.3	0.1	0.2	0.9

Hem de trobar q, q^* tals que $\|f - q^*\| < \|f - q\|$. Per grau 1, $q^* = c_0 p_0(x) + c_1 p_1(x)$, $c_i = \frac{\langle f, p_i \rangle}{\langle p_i, p_i \rangle}$:

$$\begin{aligned} c_0 &= \frac{\langle f, p_0' \rangle}{\langle p_0, p_0 \rangle} = \frac{1.1 + 0.3 + 0.1 + 0.2 + 0.9}{1 + 1 + 1 + 1 + 1} = 0.52 \\ c_1 &= \frac{\langle f, p_1' \rangle}{\langle p_1, p_1 \rangle} = \frac{1.1(-1) + 0.3(-0.5) + 0 + 0.2 \cdot 0.5 + 0.9 \cdot 1}{(-1)^2 + (-0.5)^2 + 0^2 + 0.5^2 + 1^2} = -0.1. \end{aligned}$$

Per tant, $q^* = 0.52 - 0.1x$. Per a grau $n = 2$: $p^* = c_0 p_0(x) + c_1 p_1(x) + c_2 p_2(x)$, $c_i = \frac{\langle f, p_i \rangle}{\langle p_i, p_i \rangle}$:

$$c_2 = \frac{\langle f, p_2 \rangle}{\langle p_2, p_2 \rangle} = \frac{1.1(x_1^2 - \frac{1}{2}) + 0.3(x_2^2 - \frac{1}{2}) + 0.1(x_3^2 - \frac{1}{2}) + 0.2(x_4^2 - \frac{1}{2}) + 0.9(x_5^2 - \frac{1}{2})}{(x_1^2 - \frac{1}{2})^2 + (x_2^2 - \frac{1}{2})^2 + (x_3^2 - \frac{1}{2})^2 + (x_4^2 - \frac{1}{2})^2} = \frac{33}{35}.$$

Per tant, $p^* = 0.52 - 0.1x + \frac{33}{35}(x^2 - \frac{1}{2})$. ■

Exercici 4.13. We define $E = C^1([-1, 1])$ endowed with the scalar product:

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx + \int_{-1}^1 f'(x)g'(x) dx$$

1. Prove that $\langle \cdot, \cdot \rangle$ is a scalar product.
2. Prove that the monic orthogonal polynomials of even degree are even functions and those of odd degree are odd functions.
3. Find the monic orthogonal polynomials of degrees 0, 1, 2 and 3.
4. Find the best approximation to $f(x) = x^7$ by a forth degree polynomial.

Demostració.

1. És simètric, bilineal:

$$\langle f, f \rangle = \int_{-1}^1 f^2 dx + \int_{-1}^1 (f'(x))^2 dx \geq 0.$$

Aleshores, $\langle f, f \rangle = 0 \iff f(x) = 0, f'(x) = 0 \iff f \equiv 0$.

2. p_n el polinomi ortogonal mònic. Sigui $p_n(x) = P_n(x) + S_n(x)$, amb $\deg(P_n) \leq n$ i $\deg(S_n) \leq n$:

$$\|p_n\|^2 = \langle P_n + S_n, P_n + S_n \rangle = \langle P_n, P_n \rangle + 2\langle P_n, S_n \rangle + \langle S_n, S_n \rangle.$$

Dels fets que $P_n S_n$ té grau senar, P'_n és senar o 0, S'_n és parell i $P'_n S'_n$ és senar o 0, aleshores $\langle P_n, S_n \rangle = 0$. A més a més, $\|p_n\|^2 = \langle P_n, P_n \rangle + \langle S_n, S_n \rangle$. Si $n = 2k$, P_n és mònic i $\|P_n\|^2 \leq \|p_n\|^2$. Per la propietat minimal dels polinòmics mònic, $\langle S_n, S_n \rangle = 0 \iff S_n = 0$ i per tant p_n és una funció parella.

3. $p_0 = 1$, $p_1(x) = x + a = x$, $p_2 = x^2 + ax + b = x^2 + b$, pel que $\langle p_0, p_2 \rangle = 0$ i $b = -\frac{1}{3}$. Així, $p_2(x) = x^2 - \frac{1}{3}$. Finalment, $p_3(x) = x^3 + bx$, trobem $b = -\frac{2}{10}$ i $\langle p_1, p_3 \rangle = 0$, pel que $p_3(x) = x^3 - \frac{2}{10}x$.
4. $f(x) = x^7$ té grau senar i $\langle f, p_0 \rangle = \langle f, p_2 \rangle = 0$. Per tant:

$$f^* = c_1 p_1(x) + c_3 p_3(x), \quad c_1 = \frac{\langle f, p_1 \rangle}{\langle p_1, p_1 \rangle} = \frac{5}{6}, \quad c_3 = \frac{\langle f, p_3 \rangle}{\langle p_3, p_3 \rangle} = \frac{8225}{4983}. \quad \blacksquare$$

Exercici 4.14. Let E be the space of polynomials with real coefficients, endowed with the inner product:

$$\langle p, q \rangle = \sum_{j \geq 0} a_j b_j + \int_{-1}^1 p(x)q(x) dx$$

being $p, q \in E$ such that:

$$p(x) = \sum_{j \geq 0} a_j x^j, \quad q(x) = \sum_{j \geq 0} b_j x^j.$$

1. Determine the monic orthogonal polynomials of degrees 0, 1, 2, 3.
2. Let $E^* \subset \mathbb{R}_3[x]$ be such that if $p \in E^*$ then $p'(0) = 0$. Is it a linear subspace of E ? In such a case, find an orthogonal basis of E^* .
3. Find the function from E^* which best approximates $q(x) = x^5 + x$ using the induced norm by the inner product.
4. If we consider the inner product $(p, q) = \sum_{j \geq 0} a_j b_j$ determine if there exist some interval $[a, b]$ and some continuous and positive weight function w such that $(p, q) = \int_a^b p(x)q(x)w(x) dx$ for all $p, q \in E$.

Demostració (sense cap tipus de rigor ni de detall).

1. $p_0(x) = 1, p_1(x) = x, p_2(x) = x^2 - \frac{2}{9}$ i $p_3(x) = x^3 - \frac{6}{25}x$.
2. $p \in E^* \iff p(x) = ax^3 + bx^2 + c$ amb base $\varphi_0 = 1, \varphi_1 = x^2, \varphi_2 = x^3$ i base ortogonal $\Psi_0 = 1, \Psi_1 = x^2 - \frac{2}{9}$ i $\Psi_2 = x^3$.
3. $q^* = c_0 \Psi_0 + c_1 \Psi_1 + c_2 \Psi_2$ i $c_i = \frac{\langle q, \Psi_i \rangle}{\langle \Psi_i, \Psi_i \rangle}, c_0 = 0, c_1 = 0$ i $c_2 = \frac{196}{405}$.
4. $p_0 = 1, q = x^2, \sum_i a_i b_i = 0$ i $\int_a^b x^2 w(x) dx > 0$. ■